

4800 - Master

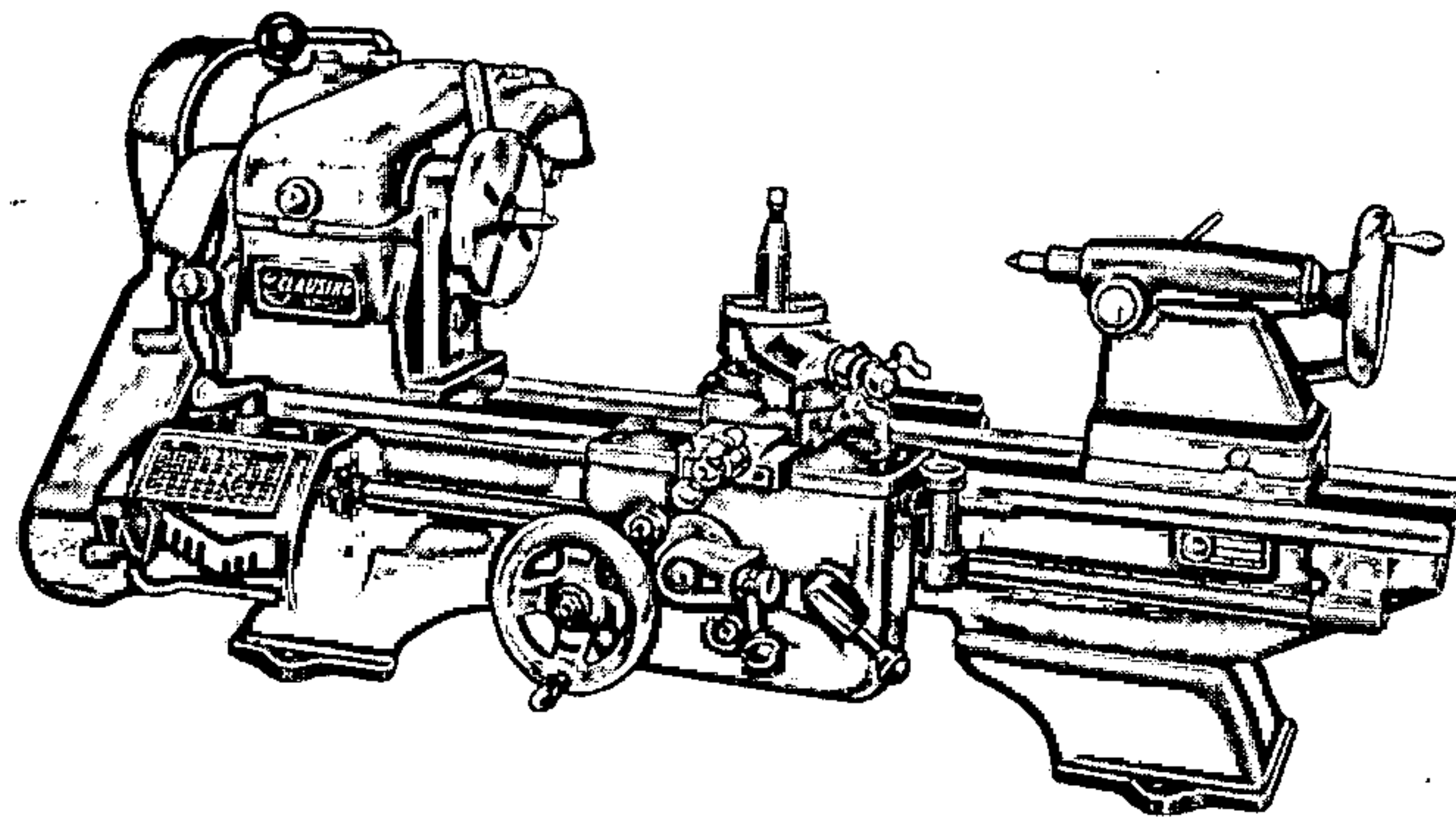
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LATHE BULLETIN

4800 - X

APRIL, 1957

OPERATING MANUAL and PARTS LIST



No. 4800 SERIES
12-inch LATHES



Do Not Operate The Lathe Until . . .

- it is properly mounted and leveled. Clausing lathes pass rigid inspection and operating tests before shipment - to maintain its built-in accuracy it must be properly installed.
- you are acquainted with the lathe and understand all the controls and their functions.
- you have oiled the lathe.
- you have carefully read all the instructions.

Then operate the lathe in back gear - get the "feel" of the controls - set up different threads and feeds - engage the power feeds - get acquainted with the lathe before starting a job - doing that will save time and produce better work.

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INSTRUCTIONS FOR ORDERING REPAIR PARTS

IMPORTANT: The following information must be furnished on all repair part orders:

1. Model Number and Serial Number of your lathe. This is found on the plate attached to the bed.
2. Part Number and Name of part.
3. Quantity required.

Parts shown coded are standard parts and should be purchased locally.

Parts price will be quoted on request.

We reserve the right to make changes in design and specifications without notice.

Instructions For Mounting and Leveling The Lathe

IT IS YOUR RESPONSIBILITY to properly level the lathe - it is the first essential for accurate work and long service life. Satisfactory performance is impossible if the lathe bed is out of level as little as one thousandth of an inch.

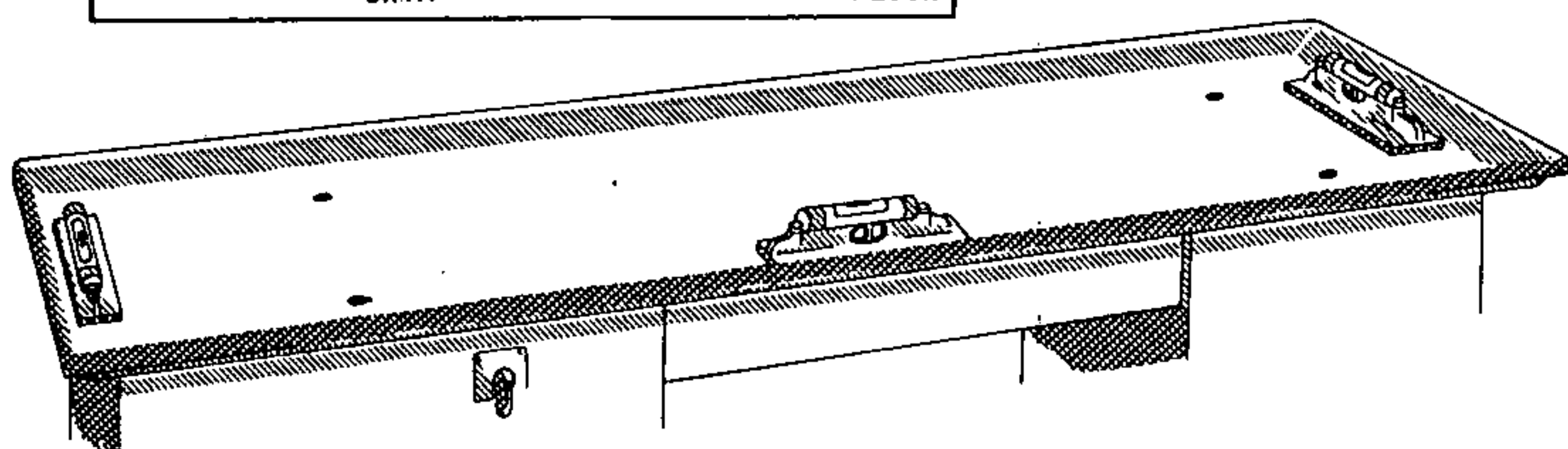
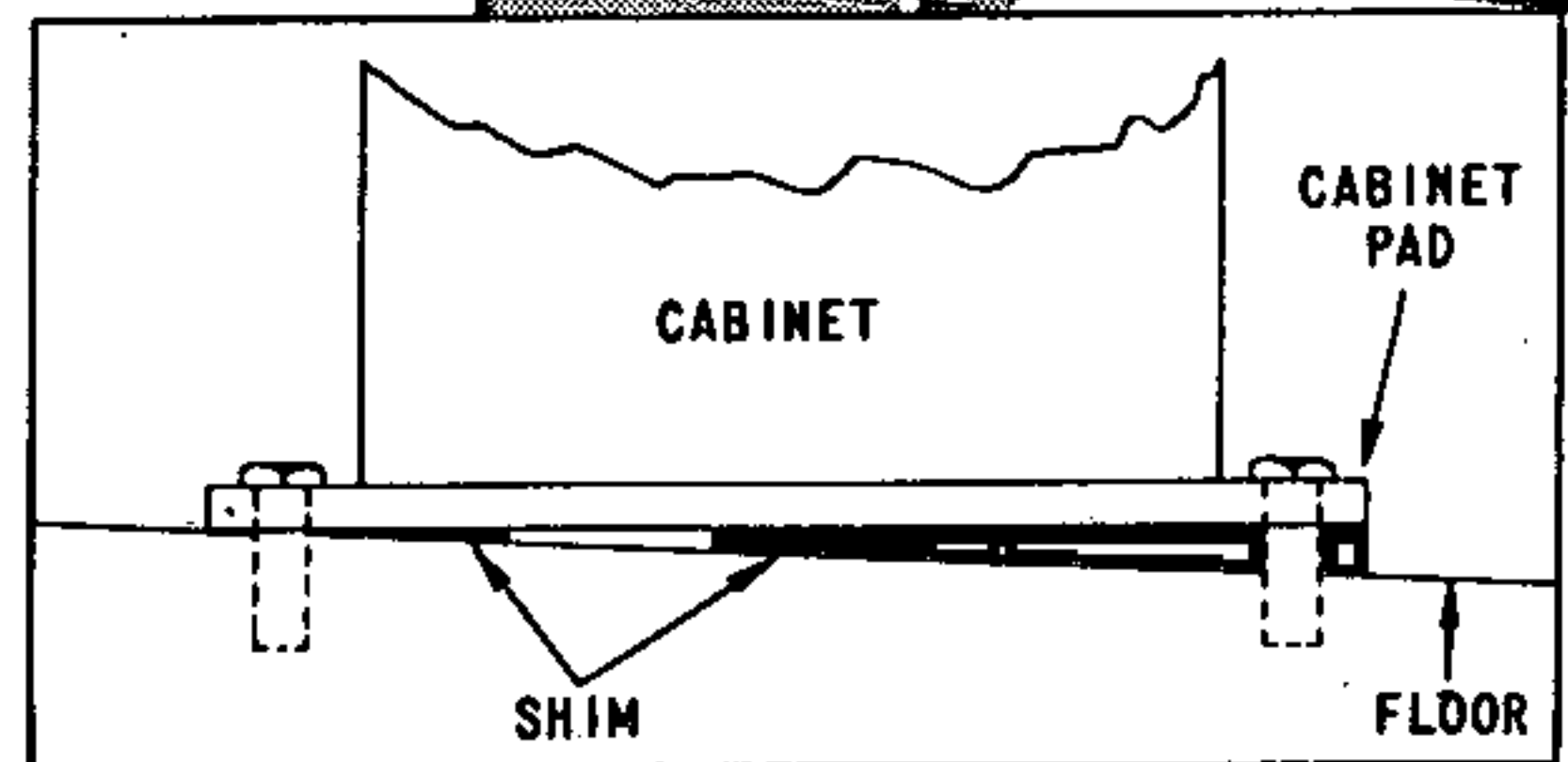
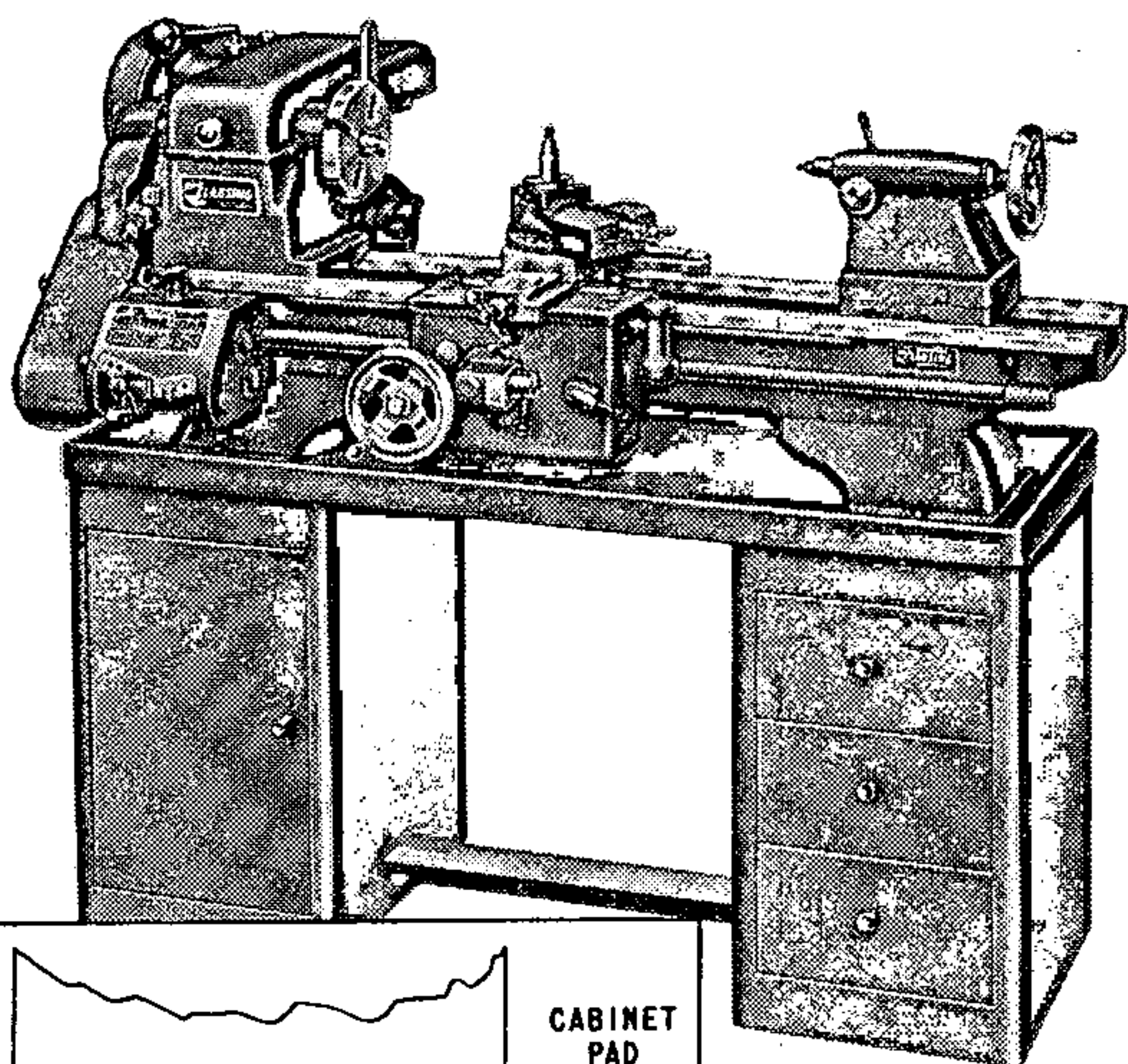
TO DO IT RIGHT-

- First bolt cabinet or bench to the floor
- Next level the cabinet or bench
- Then level the lathe

METAL CABINET INSTALLATION

1. Mount cabinet on a concrete floor or base whenever possible - if a wood floor is used, it should be well braced, capable of absorbing vibration and withstanding the load. Make sure cabinet rests solidly on the floor.

2. Fasten cabinet to concrete floor by marking location of mounting holes and drilling holes large enough to receive expansion bolts, or set studs or bolts in melted lead. Use lag screws or bolts to fasten cabinet to a wood floor.



1 (Above) Lathe mounted on steel cabinet. (Center) Use shims as shown to level cabinet. (Below) Positions for checking level readings on cabinet or bench top.

Cabinet must be bolted to floor, otherwise vibration will result.

3. Level the cabinet - use a precision machinists level. Place shims as required between cabinet pads and floor to accurately level the top. Shims should be of hardwood or metal and bear under at least 75% of the cabinet pad as shown in insert, Figure 1.

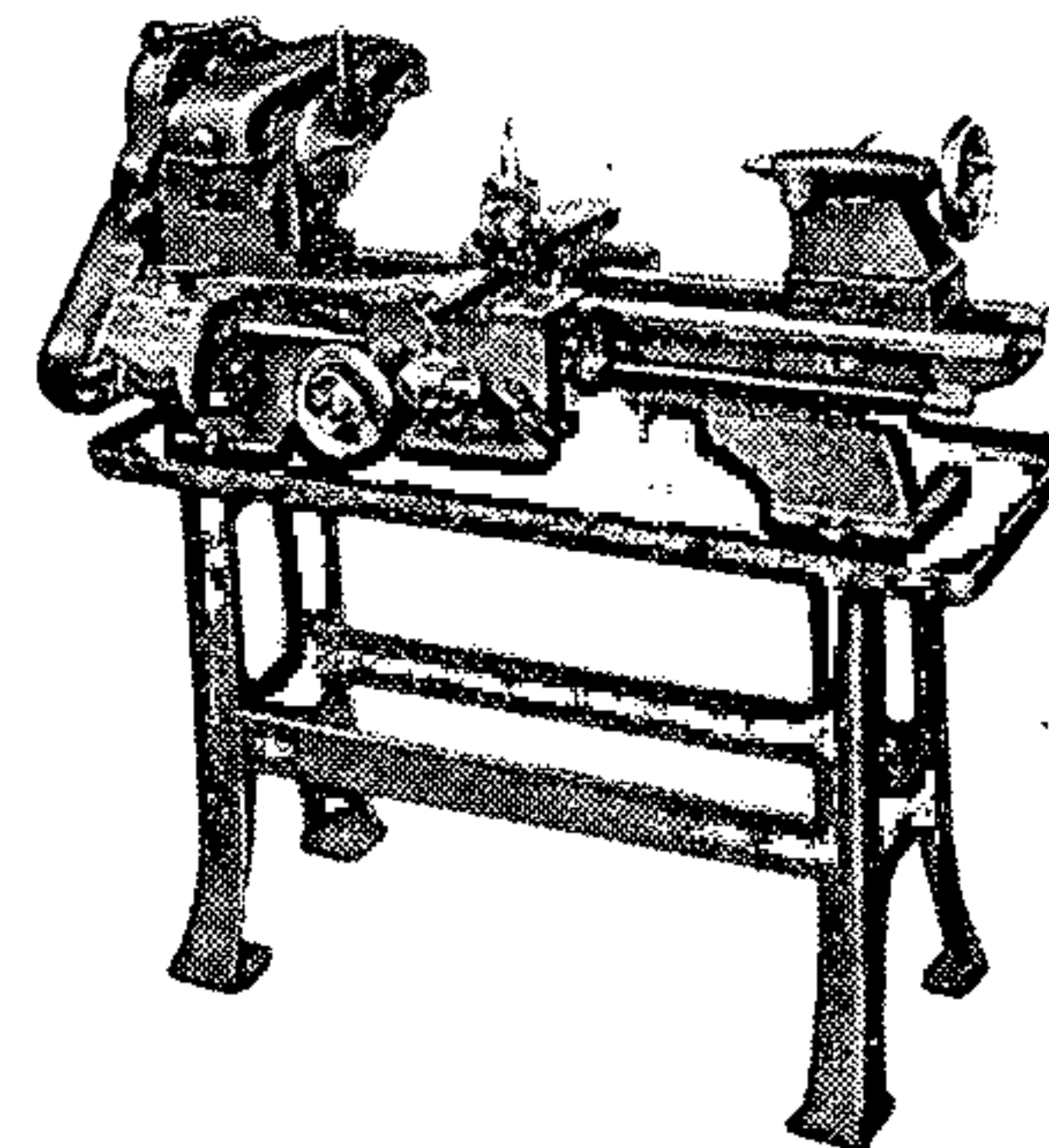
Don't use the lathe bedways to level the cabinet. Place level on cabinet top in the positions shown in sketch, Figure 1.

4. Mount the lathe. Bolt lathe to bench using 3/8" dia. bolts - they may be inserted from either top or underneath side. Do not tighten bolts securely. Lathe may now be leveled - see instructions, page 5.

BE SURE YOU HAVE FOLLOWED THESE INSTRUCTIONS COMPLETELY BEFORE LEVELING THE LATHE.

LATHE BENCH INSTALLATION

1. Mount bench on a concrete floor or base whenever possible - if a wood floor is used, it should be well braced, capable of absorbing vibration and withstanding the load. Make sure bench rests solidly on the floor.



2 Lathe mounted on lathe bench with chip pan

2. Fasten bench to concrete floor by marking location of mounting holes and drilling holes large enough to receive expansion bolts, or set studs or bolts in melted lead. Use lag screws or bolts to fasten bench to a wood floor.

3. Level the bench - use a precision machinists level. Place shims as required between bench legs and floor to accurately level the top.

4. Remove leveling screw that's wired to lathe shift lever and place in bed foot underneath quick-change gear box.

5. Mount the lathe. Bolt lathe to bench using 3/8" dia. bolts - they may be inserted from either top or underneath side. Do not tighten bolts securely. Lathe may now be leveled - see instructions, page 5.

BE SURE YOU HAVE FOLLOWED THESE INSTRUCTIONS COMPLETELY BEFORE LEVELING THE LATHE.

WOOD BENCH REQUIREMENTS and INSTALLATION

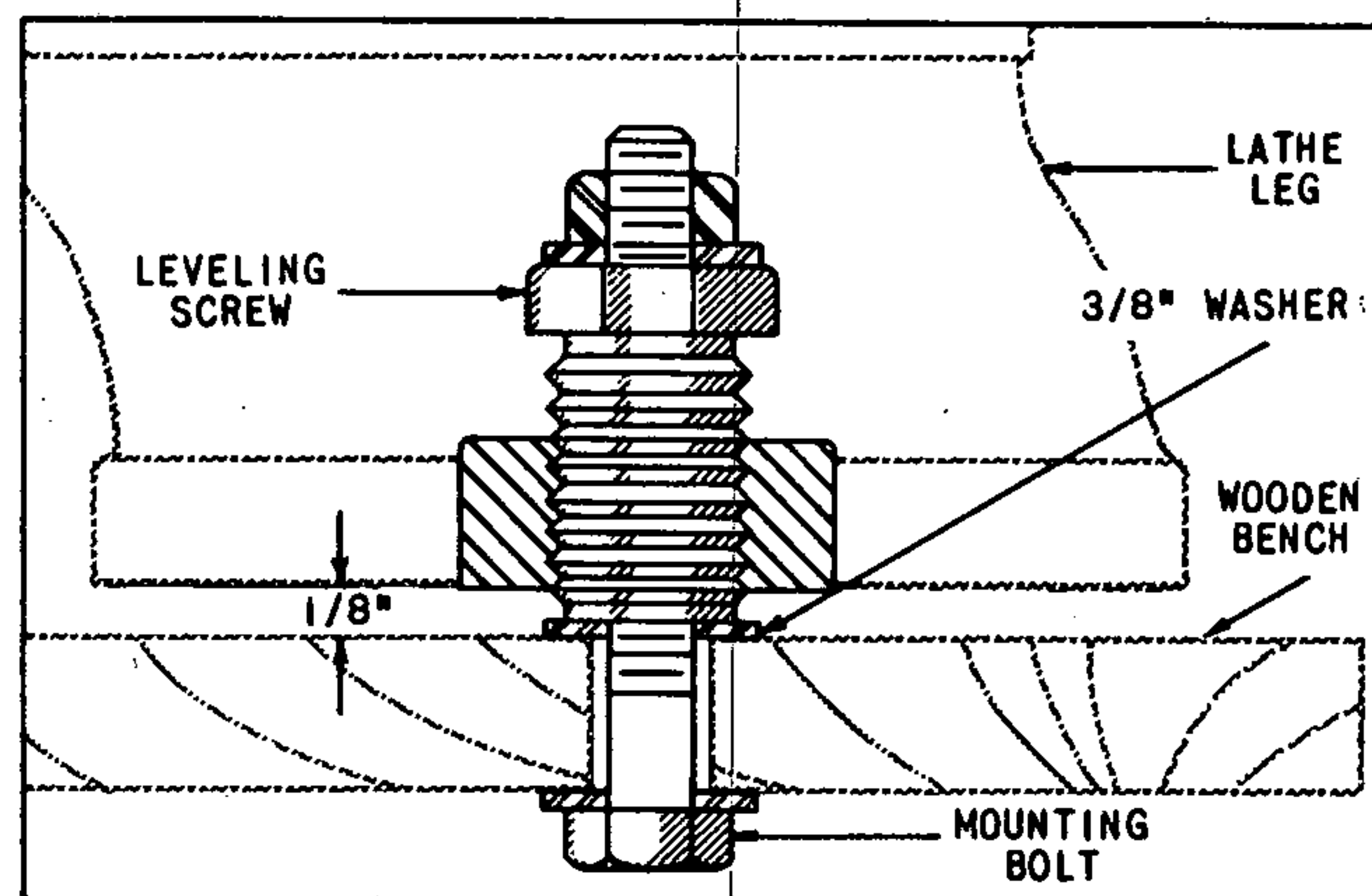
1. Bench top must be semi-hard or hard wood at least 1 5/8" thick, cleated or well doweled to form a rigid table. DO NOT USE SOFTWOODS OR BOARDS NOT CLEATED TOGETHER.

2. Legs should be of heavy construction - preferably 4" x 4" lumber, provided with lugs for bolting bench to floor. overall height of bench should be about 28".

3. Mount bench on a concrete floor or base if possible - if a wood floor is used, it should be well braced, capable of absorbing vibration and withstanding the load. Make sure stand rests solidly on the floor.

4. Fasten stand to concrete by marking location of mounting holes and drilling holes large enough to receive expansion bolts, or set studs or bolts in melted lead. Use lag screws or bolts to fasten bench to a wood floor.

5. Level bench before mounting lathe - use a precision machinists level. Place shims as required between leg pads and floor to accurately level the top.

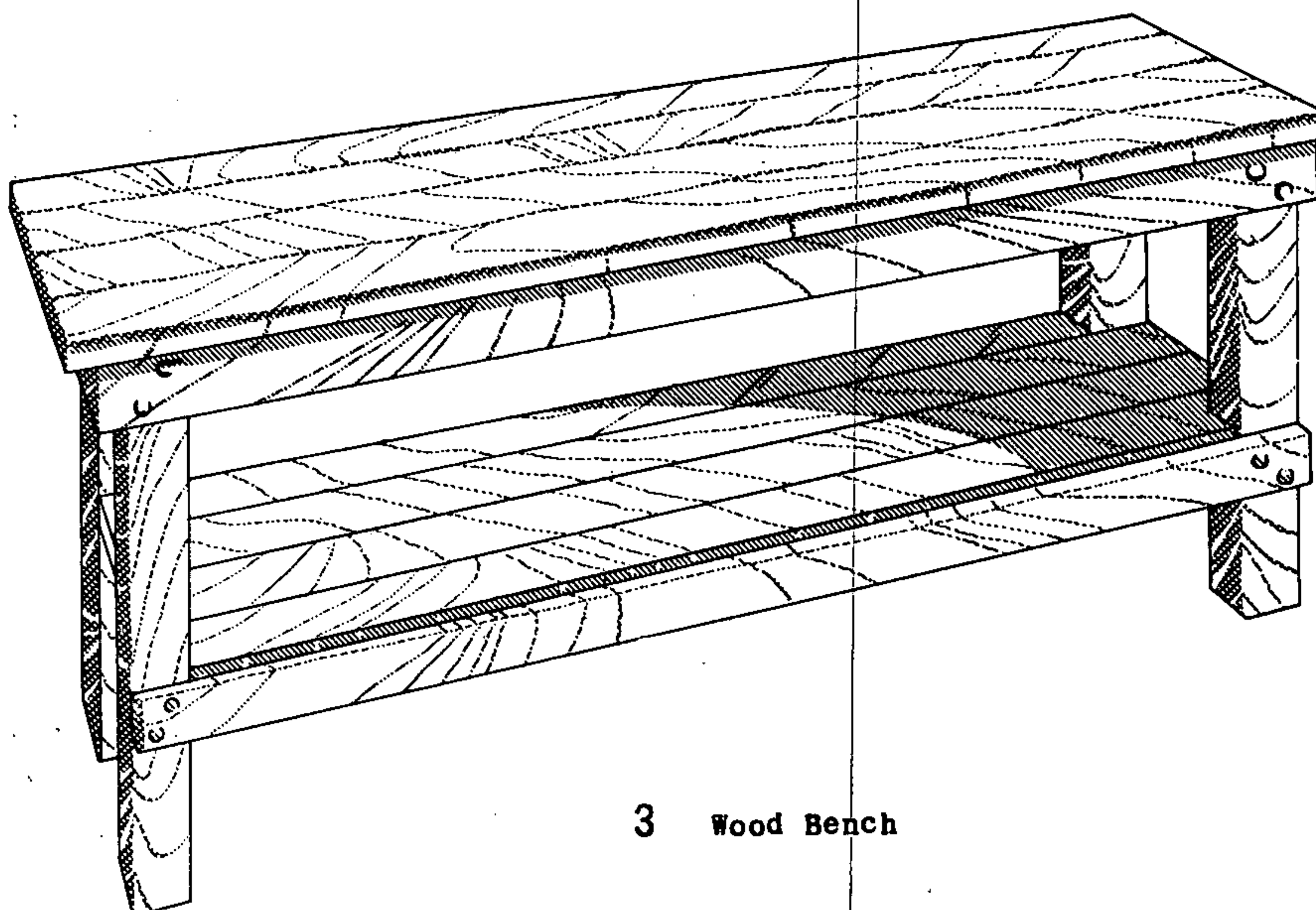


4 Make sure a metal washer is placed between leveling screw and top when using a wood bench.

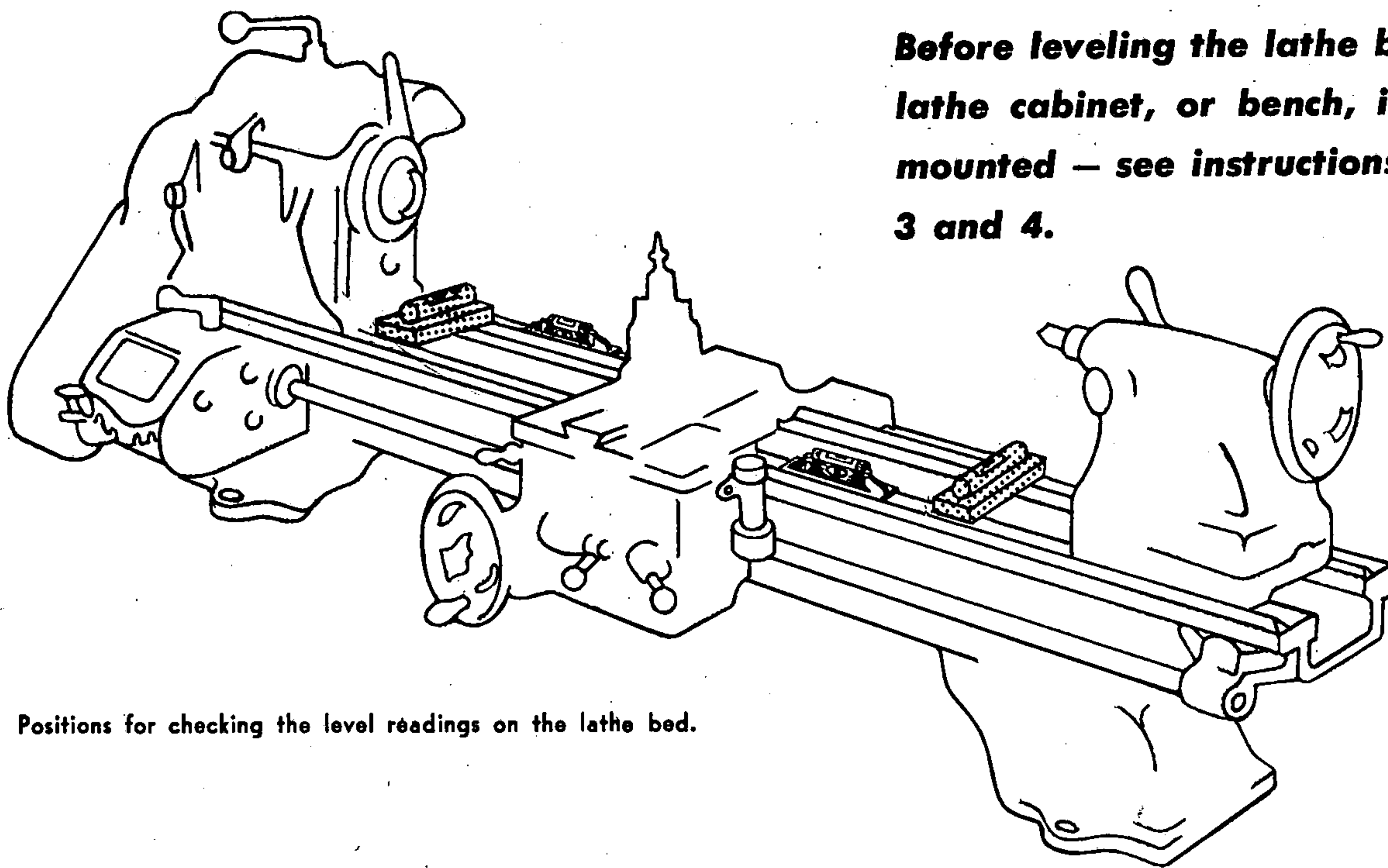
6. Remove leveling screw that's wired to lathe shift lever and place in bed foot underneath quick-change gear box.

7. Mount the lathe. Mark and drill four 7/16" dia. holes in bench top under corresponding holes in lathe legs. Bolt lathe to top using 3/8" dia. bolts, placing a 3/8" washer between lathe leg and bench top - see Figure 4. Bolts may be inserted from either top or underneath side. Do not tighten bolts securely.

BE SURE YOU HAVE FOLLOWED THESE INSTRUCTIONS COMPLETELY BEFORE LEVELING THE LATHE.



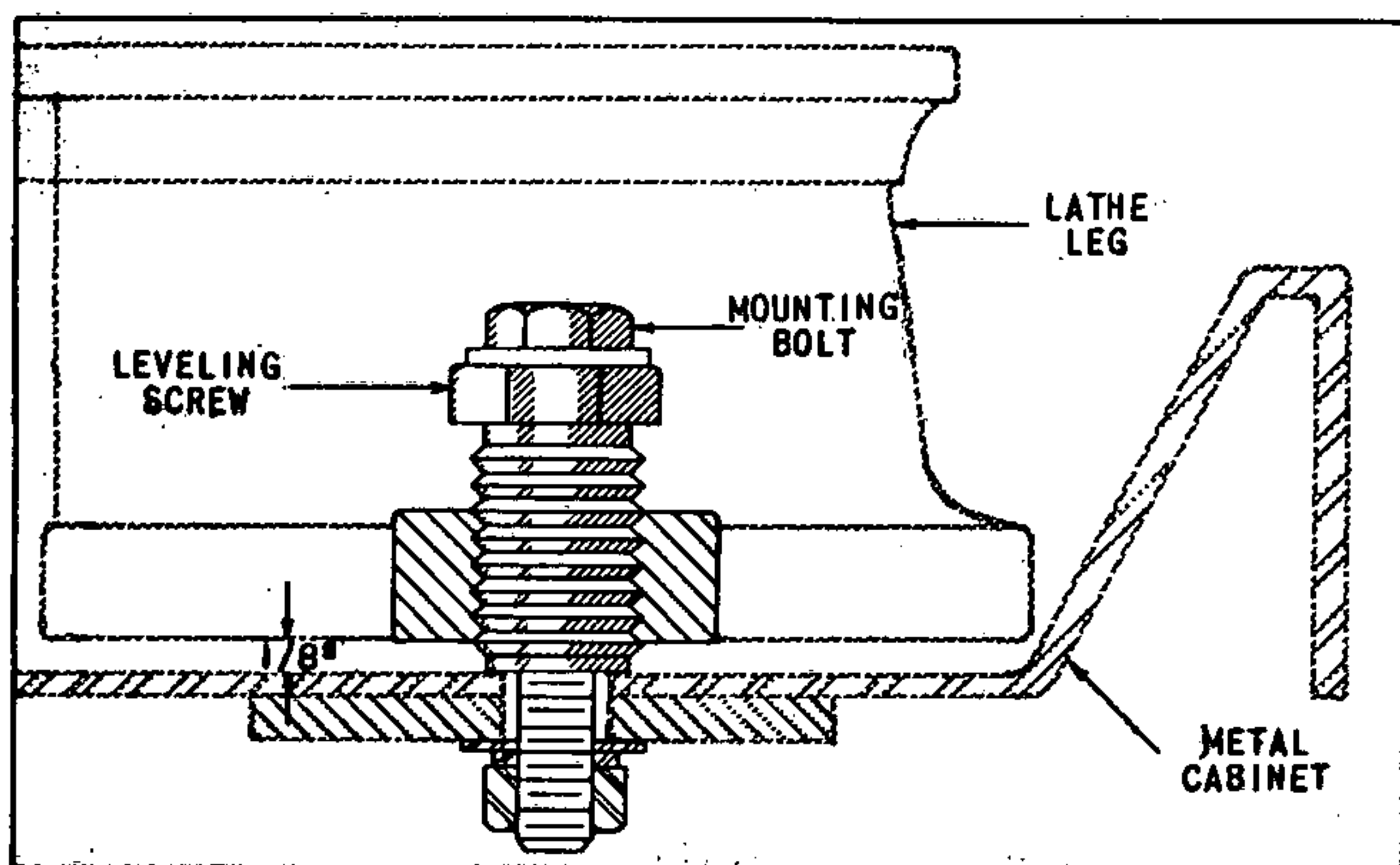
3 Wood Bench



5. Positions for checking the level readings on the lathe bed.

Leveling the Lathe

1. Screw down the four leveling screws, raising the lathe so that lathe legs at no point touch cabinet top — approximately $\frac{1}{8}$ " clearance is sufficient — see fig. 6.
2. Use a precision machinist's spirit level to level the lathe. A **VERY SENSITIVE LEVEL SHOULD BE USED**. Level should be at least 6" long and should show a distinct bubble movement when a .003" shim is placed under one end of the level.
3. Lathe bed must be leveled longitudinally to within .002" per foot before leveling the ends.
4. Both ends of the lathe bed — the headstock end and the tailstock end — must be checked with the level



6. The four leveling screws furnished with lathe will quickly and accurately level the lathe.

Before leveling the lathe be sure the lathe cabinet, or bench, is properly mounted — see instructions on pages 3 and 4.

placed at right angles to the lathe bed — see fig. 5. Use an accurate square to align the level. Place the level on top of the V-ways — about three inches in front of headstock and in front of tailstock.

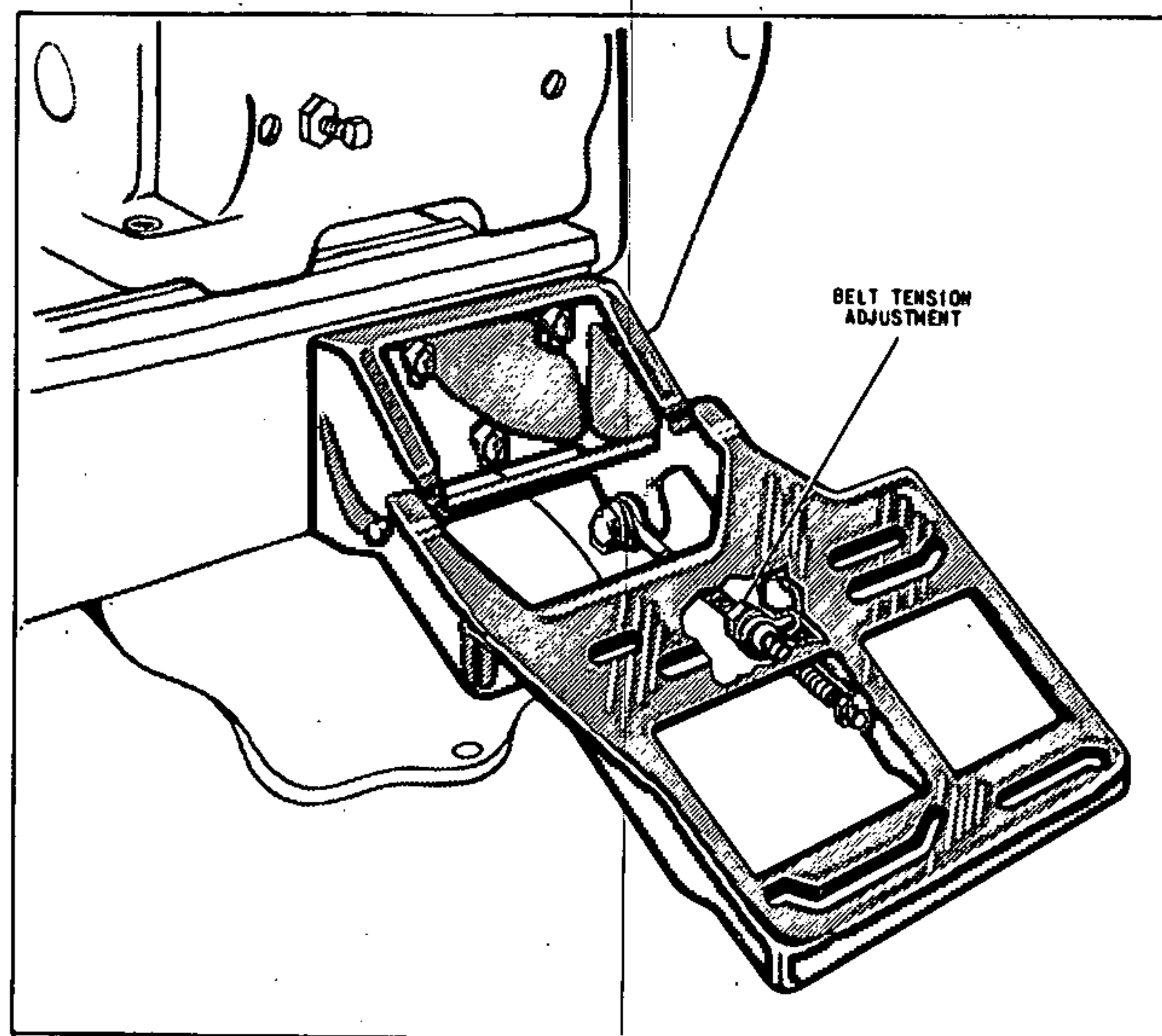
Note: Clean bed ways thoroughly before leveling lathe — make sure level does not rest on a burr or dent caused by mishandling of the machine.

5. Level reading in both positions must be identical. Compensate variations of bubble readings by turning the leveling screws until lathe is level — see fig. 6.
6. Now tighten the four mounting bolts securely.
7. Again check the level of the lathe with the level. Tightening the bolts may have pulled lathe bed out of level. If further adjustment is necessary, hold bolt head with wrench while adjusting leveling screws.

CHECK THE LEVEL OF THE LATHE AT FREQUENT INTERVALS to assure accurate turning. If the lathe is not properly leveled, it will twist the bed resulting in misalignment of the headstock and tailstock with the ways, causing lathe to chatter - turn taper - uneven wear - bore taper - face convex or concave - ruin spindle bearings and make carriage bind.

MOUNTING THE MOTOR

1. Mount motor base assembly to the lathe bed with the three screws furnished.
2. Slide pulley on motor with large step next to motor.
3. Mount motor on motor base and fasten in place with the four bolts furnished.
4. Place belt over pulleys and shift motor until pulleys are aligned and belt is straight, then tighten motor mounting bolts.
5. Adjust spring to obtain proper belt tension - see Figure 7. Belt should be just tight enough to prevent slipping.



7 Motor base mounted to lathe bed. Adjust spring for proper belt tension.

Oiling The Lathe

Apron To fill oil reservoir in apron remove pipe plug on side of apron and use hole for supplying oil. Fill apron to level indicator on side of apron. Maintain this oil level at all times. Use S.A.E. No. 20 machine oil.

Use the lubrication chart furnished with the lathe as a guide for locating the oiling positions.

Gear Train Lubrication points can be reached through holes in guard. Fill oil cup on quadrant and on reverse handle - add oil daily. Oil holes on ends of reverse gear studs and sliding gear handle require daily lubrication. Fill zerk fitting monthly with a light grease for quadrant gear lubrication.

Gear Box Apply oil frequently thru oil cups on top of gear box for general lubrication of all moving parts. Gear box bearings are sealed-for-life ball bearings and do not require lubrication.

Lubricate the following points in headstock and countershaft by raising hood.

Headstock Lubricate the spindle bearings thru the two oil cups on both ends of headstock. Lubricate back gear bearings thru oil hole in back gear shaft quill. To oil spindle pulley bearing, remove set screw in spindle drive pulley. Oil these bearings once a week.

Countershaft All the bearings in the countershaft are lubricated for life and permanently sealed against dust and dirt and need no further attention.

Other parts to oil occasionally are:

1. Right lead screw bearing
2. Tailstock ram
3. Tailstock Screw
4. Carriage handwheel shaft
5. Leadscrew
6. Carriage and compound dovetail ways
7. Lathe bed ways
8. Felt wipers on carriage saddle
9. Rim of threading dial.

KEEP YOUR LATHE CLEAN - Oil and dirt form an abrasive compound which can easily damage carefully fitted bearing surfaces. Wipe the bed and all polished parts with a clean oily rag at frequent intervals. Use a brush to clean spindle threads, gear teeth, lead screw threads, etc.

Maintenance and Controls

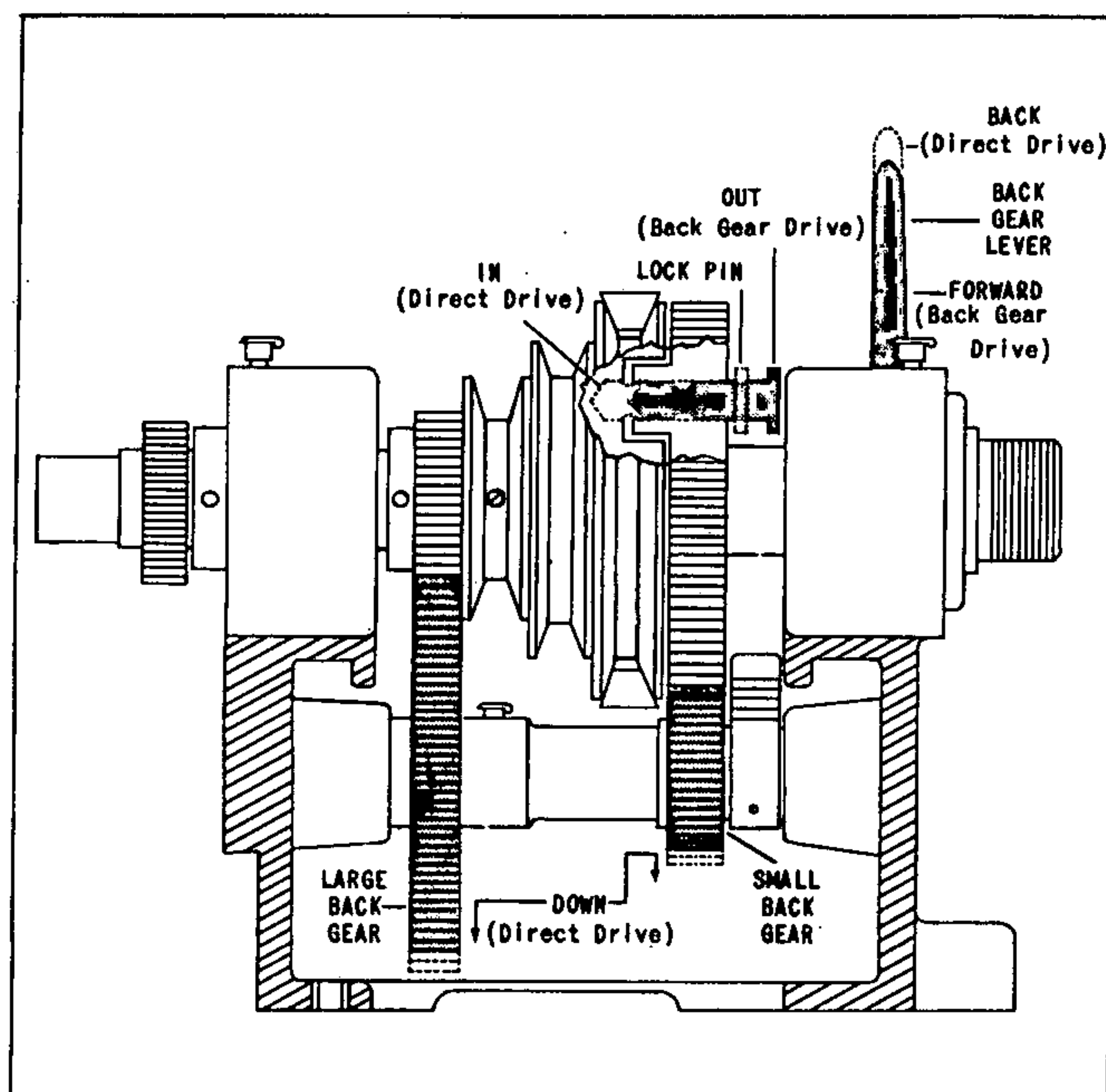
DO NOT OPERATE THE LATHE - until you are thoroughly familiar with all the controls and their functions (read carefully the instructions MAINTENANCE AND CONTROLS). Then operate the lathe in back gear - get the "feel" of the controls - set up different threads and feeds - engage the power feeds - get acquainted with the lathe before you start a job - it will save time and produce better work.

SPINDLE DRIVE, See Figure 8

FOR DIRECT DRIVE - turn pulley until lock pin can be reached and engage lock pin with the pulley, locking pulley to the bull gear and spindle. Place back gear lever in the disengaged position, or away from headstock. Lathe is now in direct drive.

FOR BACK GEAR DRIVE - pull out lock pin, disengaging bull gear from pulley. Shift back-gear lever to the forward position, meshing back gears with spindle gears - it may be necessary to rotate spindle pulley to mesh gears.

CAUTION - Always stop motor before changing from one drive to another.

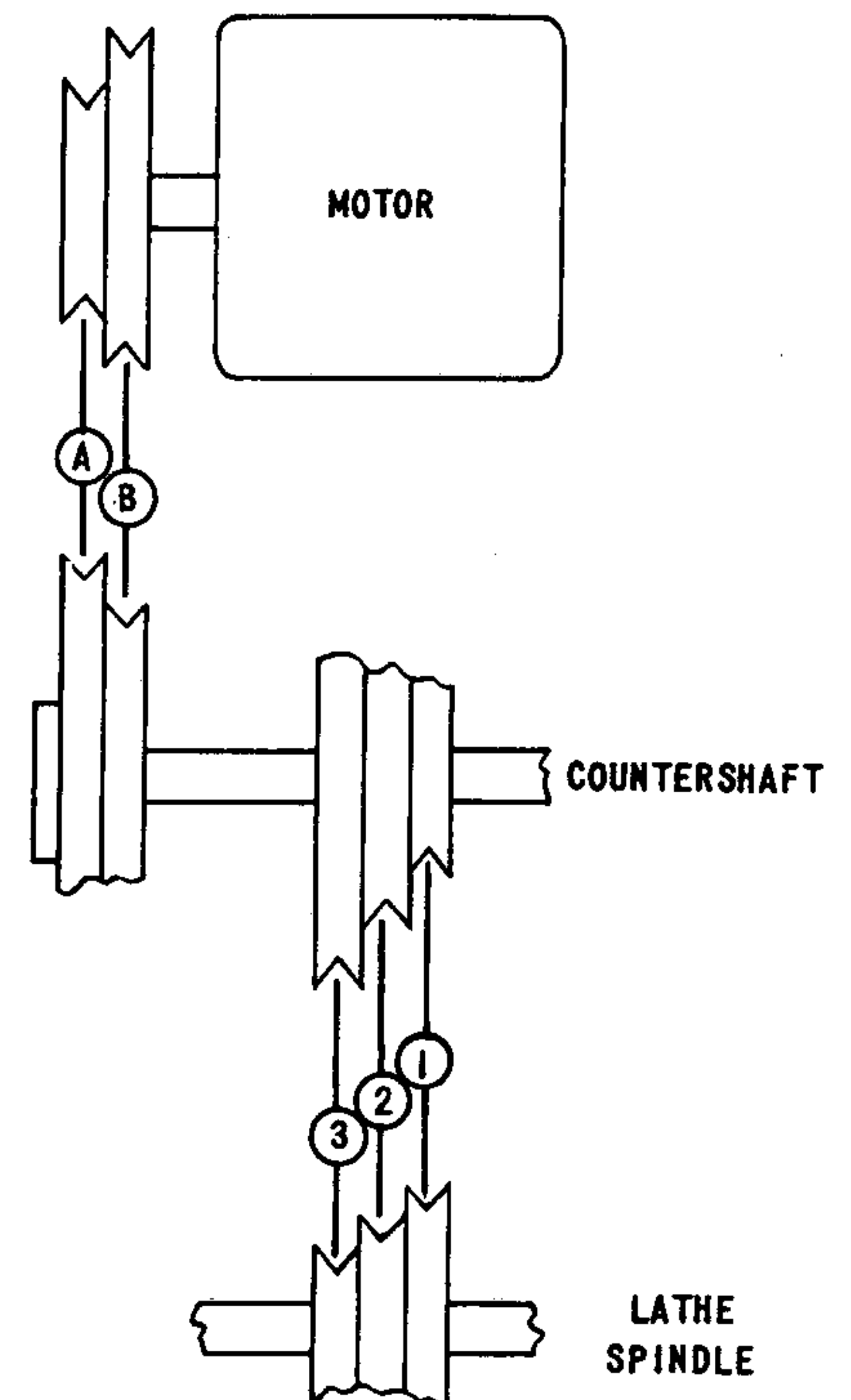


8 Position of back-gears, lever and lock pin for spindle drive in "back-gear" (shaded) and in "direct drive" (brokenline).

LATHE COUNTERSHAFT

Countershaft has friction clutch and brake for instant starting and stopping of the spindle without stopping the motor. Moving clutch lever to left engages spindle drive - to the right disengages it. Moving lever to extreme right tightens the brake shoe stopping the spindle.

Speed changes are made by raising the guard which slackens the drive belt for easy belt changes. The chart, Figure 9, lists the speeds available and shows how they are obtained.



BACK GEAR DRIVE				DIRECT CONE DRIVE			
MOTOR BELT POSITION	SPINDLE BELT POS.			MOTOR BELT POSITION	SPINDLE BELT POS.		
	1	2	3		1	2	3
A	32	56	128	A	192	338	768
B	89	180	362	B	536	1080	2170

9 Spindle Speed Chart

Maintenance and Controls

DO NOT OPERATE THE LATHE — until you are thoroughly familiar with all the controls and their functions (read carefully the instructions **MAINTENANCE AND CONTROLS**). Then operate the lathe in back gear — get the "feel" of the controls — set up different threads and feeds — engage the power feeds — get acquainted with the lathe before you start a job — it will save time and produce better work.

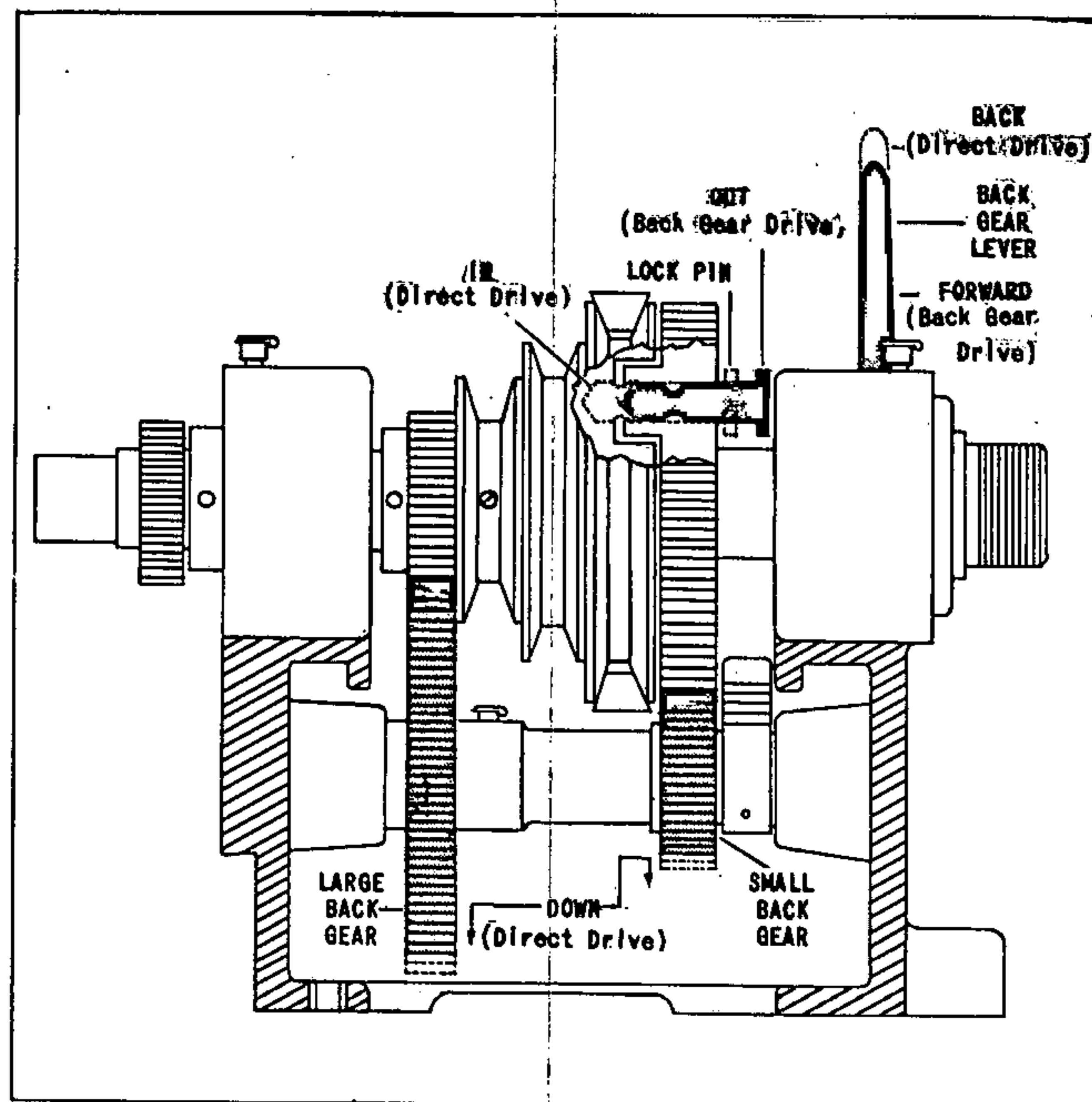
SPINDLE DRIVE

(See Figure 8)

FOR DIRECT DRIVE — turn pulley until lock pin can be reached and engage lock pin with the pulley, locking pulley to the bull gear and spindle. Place back gear lever in the disengaged position, or away from headstock. Lathe is now in direct drive.

FOR BACK GEAR DRIVE — pull out lock pin, disengaging bull gear from pulley. Shift back-gear lever to the forward position, meshing back gears with spindle gears — it may be necessary to rotate spindle pulley to mesh gears.

CAUTION — Always stop motor before changing from one drive to the other.

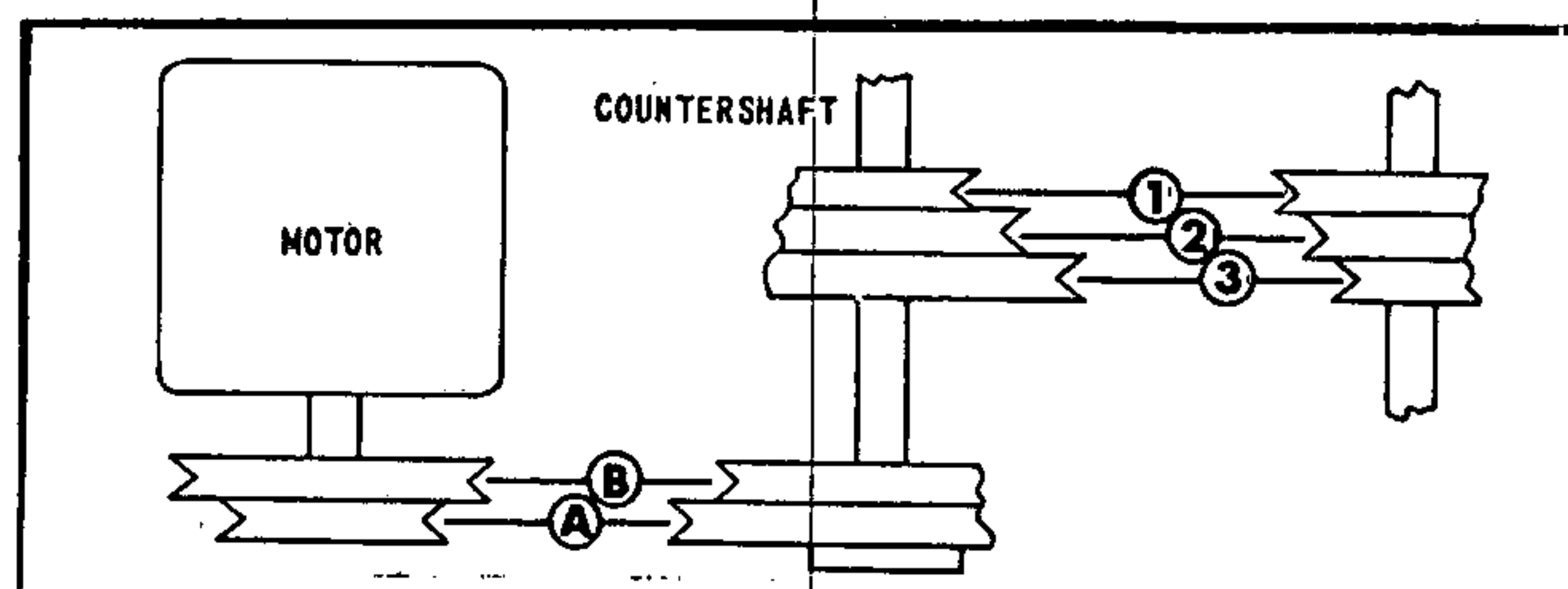


8. Position of back gears, lever and lock pin to drive spindle in back-gear (shaded) and in direct drive (broken line).

HORIZONTAL COUNTERSHAFT

Countershaft has friction clutch and brake for instant starting and stopping of the spindle without stopping the motor. Moving clutch lever to left engages spindle drive — to the right disengages it. Moving lever to extreme right tightens the brake shoe stopping the spindle.

Speed changes are made by raising the spindle-countershaft guard which slackens the drive belt for easy belt changes. Raise guard by pushing in and upward on the guard lock knob. The chart, Figure 9, lists the speeds available and shows how they are obtained.

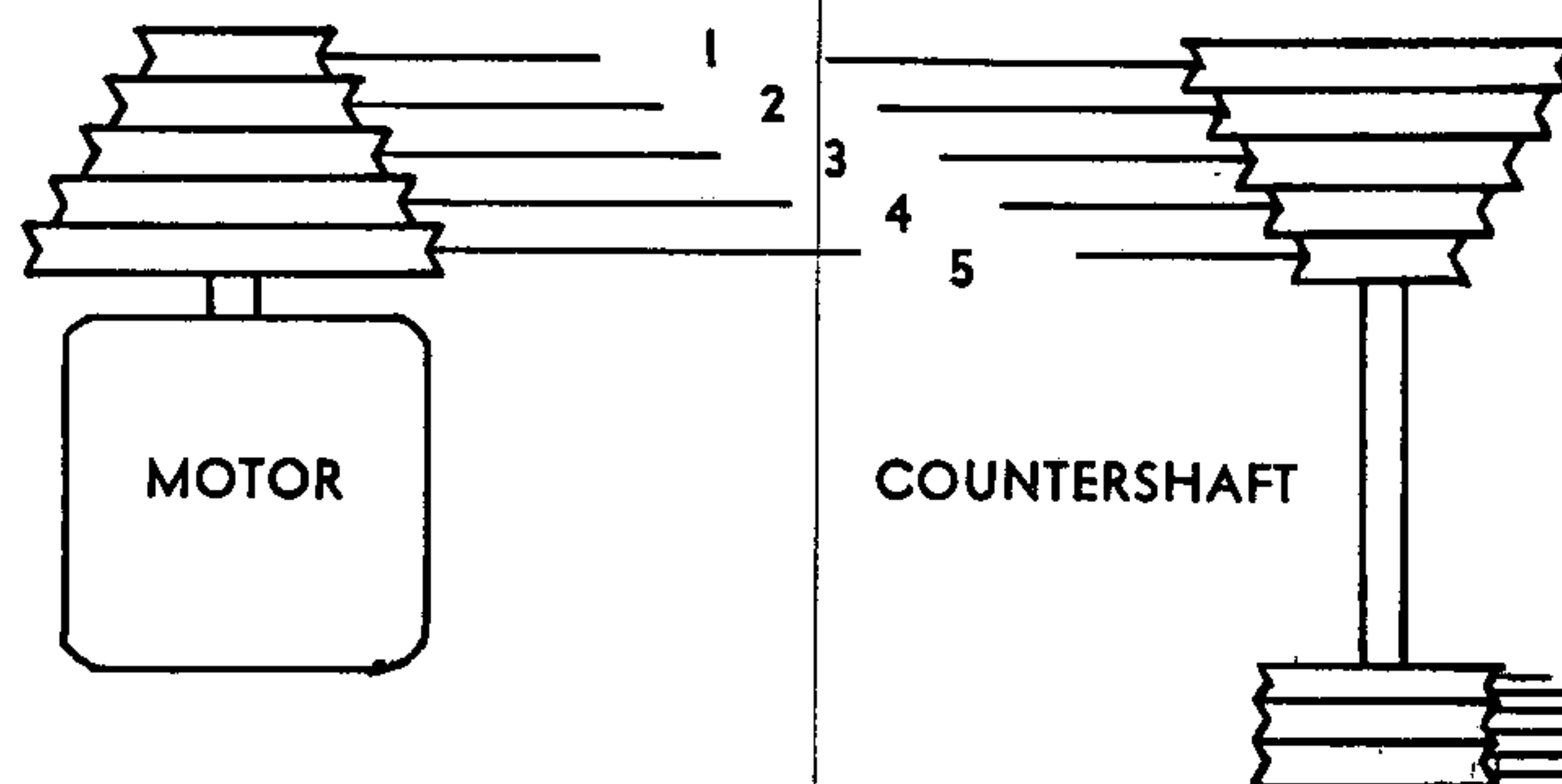


BACK GEAR DRIVE				DIRECT CONE DRIVE			
Motor Belt Position	Spindle Belt Pos.			Motor Belt Position	Spindle Belt Pos.		
	1	2	3		1	2	3
A	32	56	128	A	192	338	768
B	89	180	362	B	536	1080	2170

9. Spindle speed chart, listing speeds available, and diagram showing how they can be obtained.

UNDERNEATH DRIVE

CHANGING SPEEDS — Raise belt tension lever to loosen belts from the pulleys — lever is accessible through cabinet door. Shift belt to position desired and then lower lever to tension belt. The chart, shown below, lists the speeds available and how they are obtained.



SPINDLE SPEEDS PER RPM									
BACK GEAR DRIVE					DIRECT DRIVE				
BELT POSITIONS					BELT POSITIONS				
1	2	3	4	5	1	2	3	4	5
35	65	100	160	270	210	375	650	995	1600

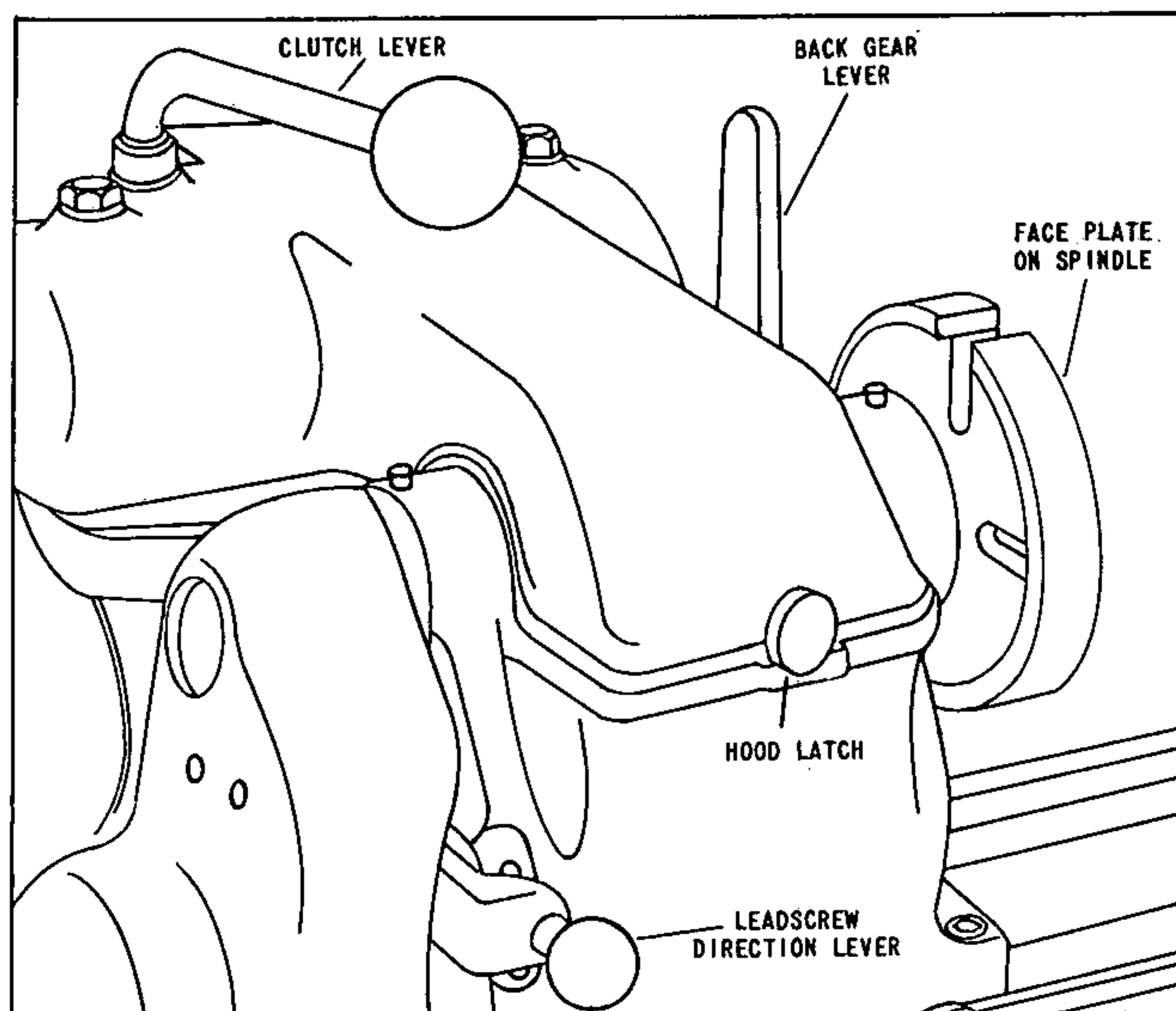
9A Spindle Speed Chart for underneath drive lathes.

ADJUSTING SPINDLE BELT — The spindle drive belts may be adjusted by loosening the two screws on the hanger and moving the hanger on the two vertical supporting rods.

ADJUSTING MOTOR BELT — Motor belt may be adjusted by moving the collars located beneath the motor base tension springs.

FEED REVERSE LEVER

The feed reverse lever, or lead screw direction lever, is located on left side of headstock, Figure 10. Lever has three positions. Center position is neutral and disengages gear train. Upper position moves carriage toward tailstock. Lower position moves carriage toward headstock. *This lever should not be moved while lathe is operating at high speeds - it may strip the gears or result in serious damage to the lathe.* It is possible to quickly reverse lead screw at lower speeds if desired.



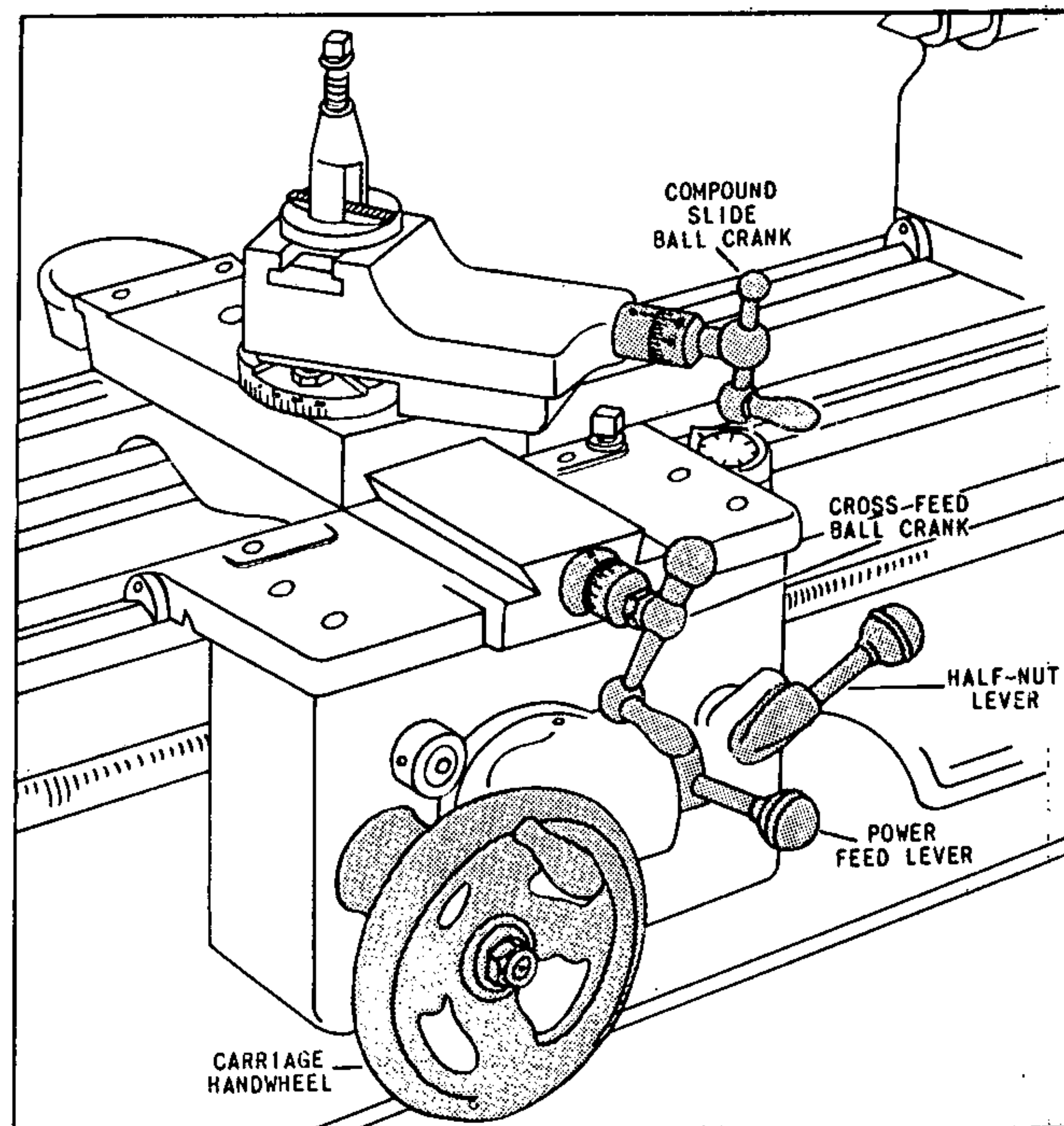
10 Lathe headstock showing location of feed reverse lever.

AUTOMATIC APRON

Figure 11 gives the names and positions of the carriage controls. The carriage handwheel moves the carriage along the

lathe bed. The cross feed and compound slide ball cranks move the carriage slide and tool rest in and out.

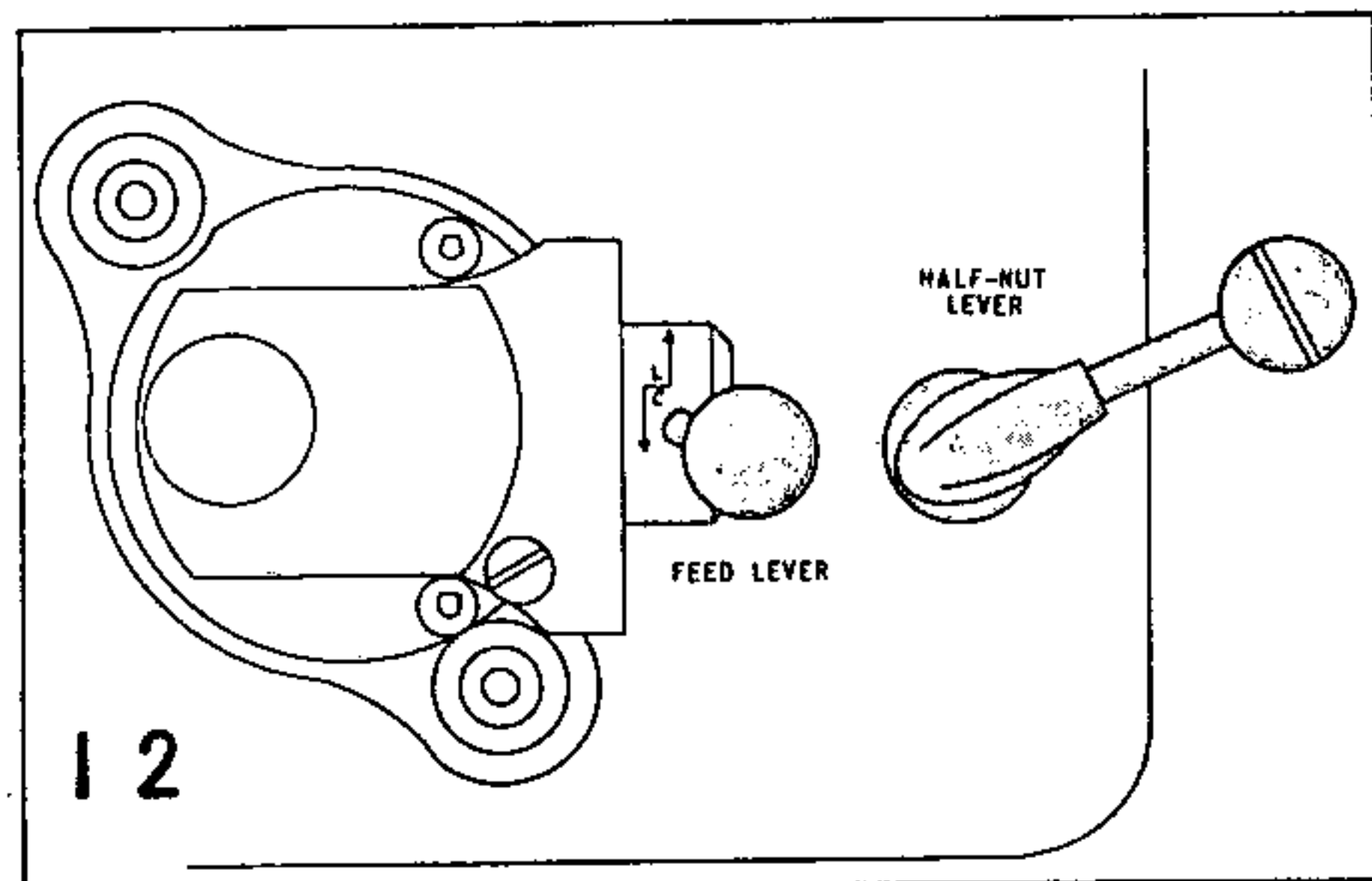
The carriage lock screw is used to lock the carriage to the bed - use it for facing or cut-off operations only.



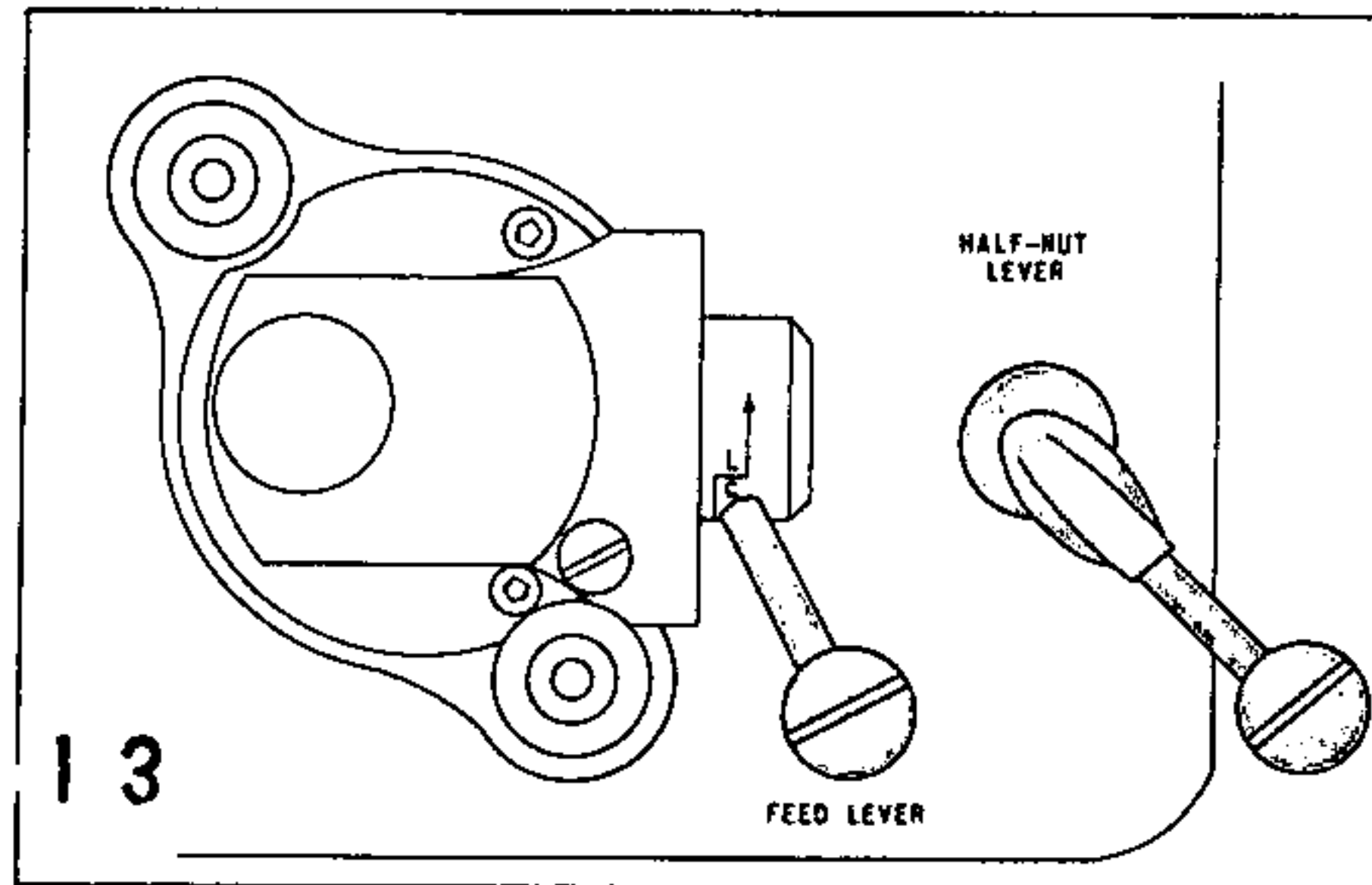
11 Controls on the lathe carriage

IMPORTANT - Use the half-nut lever for threading only - never for feeds. It will prolong the life of the lead screw, and preserve its accuracy for threading operations.

The power feed lever controls the operation of both power longitudinal and power cross feeds - the half-nut lever engages the half-nuts with the lead screw. When one of these levers is en-

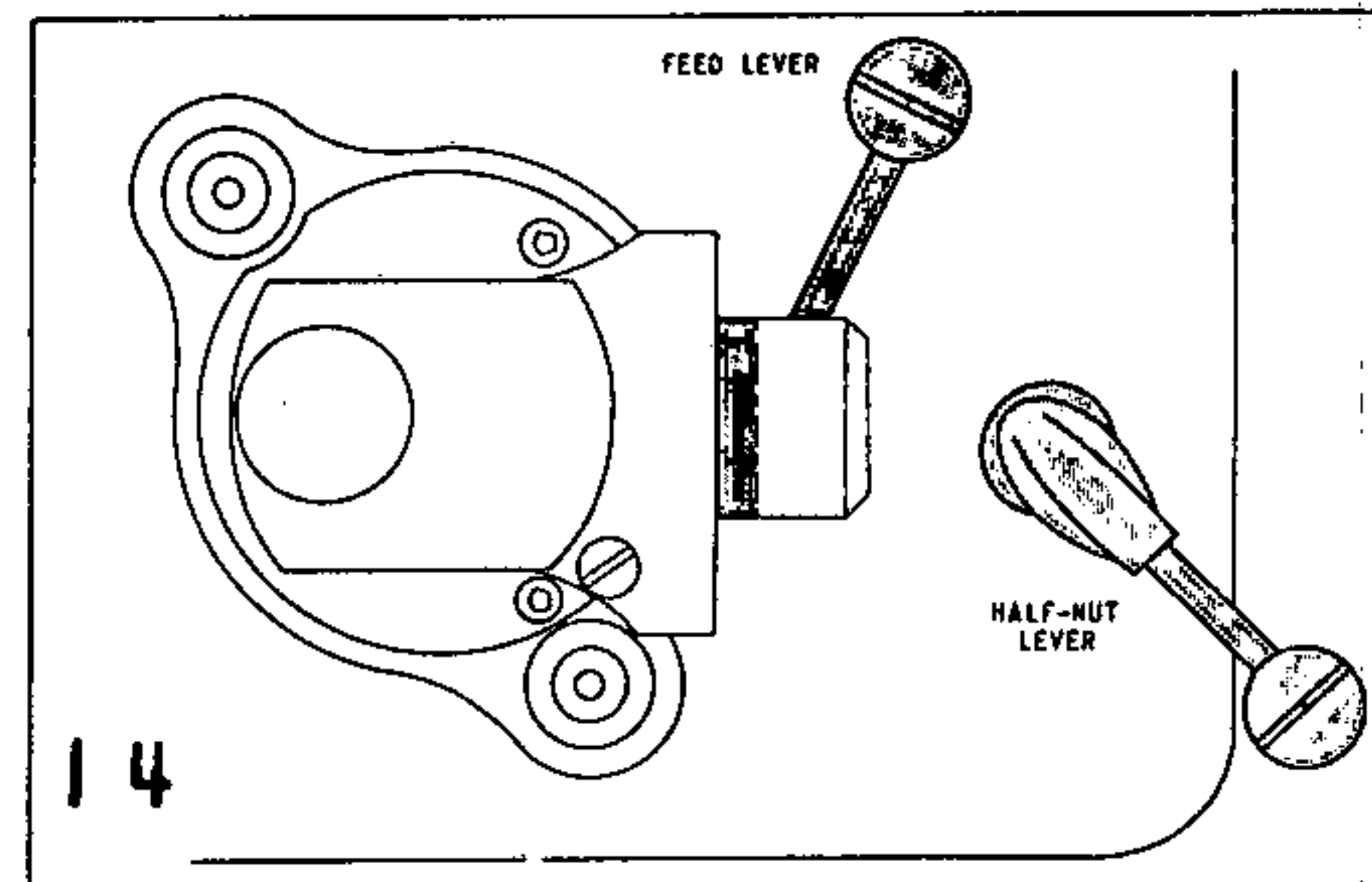


THREADING - Place feed lever at the horizontal position. Shift half-nut lever upwards to engage half-nuts with lead screw. Use the half-nuts for threading only - never for feeds.



POWER CROSS FEED -

To engage power cross feed, place half-nut lever in the down position - the feed handle cannot be moved until half-nut lever is in this position. Move feed handle downward to the vertical position.



LONGITUDINAL FEED -

To engage longitudinal feed, first make sure half-nut lever is in the down position. Shift feed lever sideways to the right about 1/4", and then upwards to the vertical position.

gaged, the other is locked and cannot be moved - DO NOT FORCE. The positions of the levers to obtain a thread or feed are illustrated in Figures 12, 13 and 14. The direction of feed is controlled by the reverse lever on front of headstock.

SELECTION OF THREADS AND FEEDS

Study the chart on the gear box - it lists the threads and feeds available and indicates the position of the controls for thread or feed desired. Figure 15 illustrates and names these controls.

NOTE: The gear chart shows only the carriage longitudinal feeds that are commonly used. Many other feeds can be obtained - these are shown in the table, Figure 16, along with the cross feeds available. To obtain one of these longitudinal or cross feeds, set up the lathe gear box for the equivalent thread as shown in the table. FOR EXAMPLE - To obtain a carriage longitudinal feed of .0033", set up the controls to cut 44 threads.

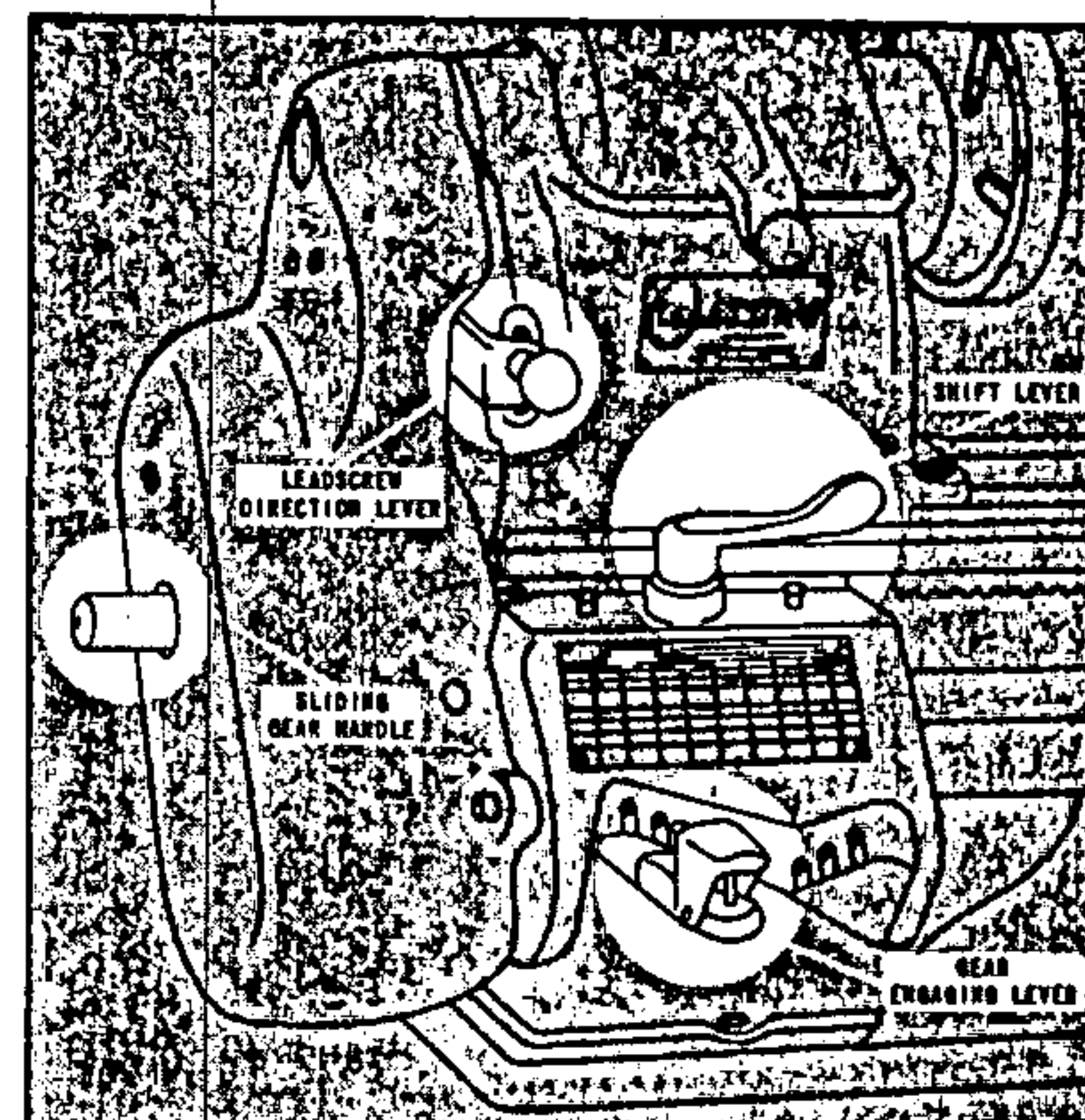
SEQUENCE OF ENGAGING CONTROLS FOR THREADS OR FEEDS ARE OUTLINED BELOW:

1. Disengage the carriage feed controls.
3. Move sliding gear handle "In" or "Out" as indicated on chart for thread or feed desired.
3. Release and lower gear engaging lever.
4. To make sure gears in the quick-change gear box will engage, first start motor. Now move gear engaging lever to the slot directly below thread or feed desired. Raise lever and snap plunger into position.
5. Next, check the chart for position of the shift lever. There are three positions - center, left and right. Shift the lever to position indicated on chart. (It may take a few seconds for the dog gears to engage if lathe is in back gear or turning slowly.) Lathe is now ready to cut thread or feed.

The threading dial is used in thread cutting and indicates the proper time to engage the half-nut lever so that

the cutting tool will enter the same groove of the thread for each cut.

When cutting even numbered threads engage the half-nut lever at any one of the markings on the threading dial for each cut of the thread. When cutting odd numbered threads, engage the half-nut lever for the first cut and all successive cuts at either the 1 or 2 positions on the dial. When cutting half-numbered threads, engage the half-nut lever at the same mark on the threading dial for each cut of the thread.



15 Name and location of controls used to obtain a thread or feed.

CARRIAGE FEED CHART

Thds. Per Inch	Carriage Long Feed	Cross Feed	Thds. Per Inch	Carriage Long Feed	Cross Feed
4	.0367	.00917	32	.0046	.0011
4.5	.0326	.0081	36	.0041	.0010
5	.0293	.0073	40	.0036	.0009
5.5	.0267	.0066	44	.0033	.0008
5.75	.0255	.0063	46	.0031	.00079
6	.0244	.0061	48	.0030	.00076
6.5	.0226	.0056	52	.0028	.00070
6.75	.0218	.0054	54	.0027	.00068
7	.0209	.0052	56	.0026	.00065
8	.0183	.0045	64	.0022	.00057
9	.0163	.0040	72	.0020	.00052
10	.0147	.0036	80	.0018	.00045
11	.0134	.0033	88	.0017	.00041
11.5	.0127	.0032	92	.0016	.00039
12	.0122	.0030	96	.0015	.00038
13	.0113	.0028	104	.0014	.00035
13.5	.0109	.0027	108	.00136	.00034
14	.0105	.0026	112	.0013	.000325
16	.0092	.0023	128	.0011	.00027
18	.0081	.0020	144	.00094	.000235
20	.0073	.0018	160	.00092	.000225
22	.0066	.00166	176	.00083	.00020
23	.0063	.00159	184	.00079	.000197
24	.0061	.00152	192	.00076	.000190
26	.0056	.0014	208	.00070	.000175
27	.0054	.00136	216	.00068	.00017
28	.0052	.0013	224	.00065	.00016

16 Chart listing available threads per inch with equivalent carriage, longitudinal and power cross feeds.

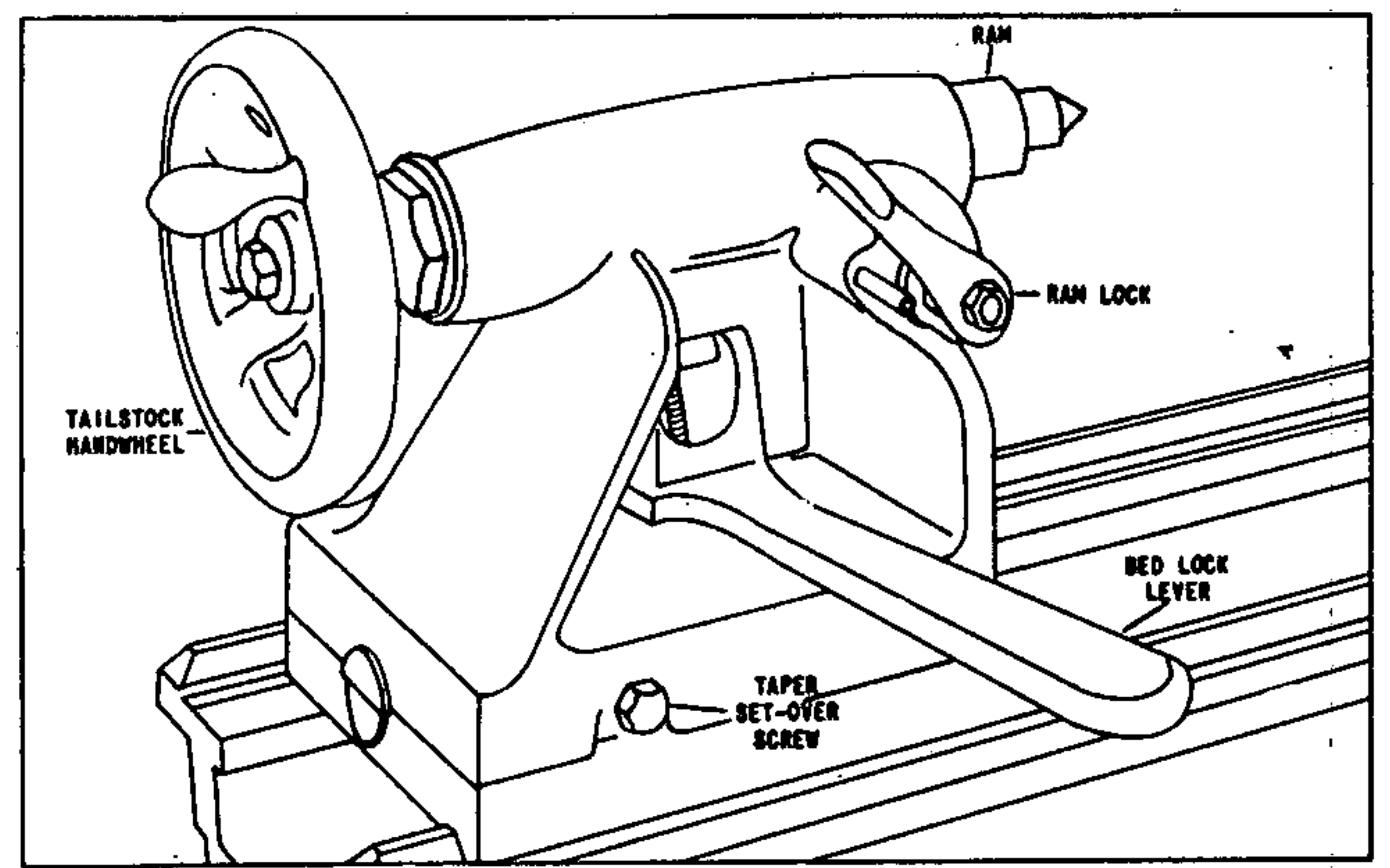
TAILSTOCK, See Figure 17

Tailstock is securely locked to the bed with the lever-controlled bed lock located on the rear of the tailstock. Graduations on the ram simplify accurate boring and drilling. Ram is locked in place with the lock handle located on top of tailstock. Before inserting the center in the tailstock ram, clean both tapers thoroughly with a dry cloth.

Tailstock can be set over 1" for taper turning by first loosening the bed clamp and then adjusting the screws on front and back of tailstock base.

MOUNTING CHUCKS AND FACE PLATES

1. Carefully wipe face of chuck hub and threads (or face plate) clean of dirt and chips.
2. Carefully wipe spindle threads and shoulder clean of any dirt and chips.
3. Oil lathe spindle threads with a light film of clean oil - chuck or face plate will thread more freely on spindle.
4. Tighten belt, or place lathe in back gear to hold spindle firmly in position.
5. Screw chuck or face plate on spindle, turning it rapidly as it nears spindle shoulder so it will seat firmly against spindle shoulder face. Make



17 Controls on the lathe tailstock.

sure threads are not crossed - chuck or face plate should thread on spindle easily.

TO REMOVE CHUCK OR FACE PLATE

1. To remove chuck, rotate chuck until wrench hole is on top. Lock spindle by engaging back gears without pulling out lock pin. Now place chuck wrench in chuck and pull. If chuck doesn't release, tap BASE OF WRENCH lightly with a mallet. Remove chuck carefully so as not to damage spindle threads. Disengage back gears.
2. To remove face plate, lock spindle by engaging back gears without pulling out lock pin, tap slot in face plate with a lead or brass hammer in a counterclockwise direction. Remove face plate carefully so as not to damage spindle threads. Disengage back-gears.

CAUTION - Do not turn power on with the spindle locked - never remove chuck or face plate while lathe is running.

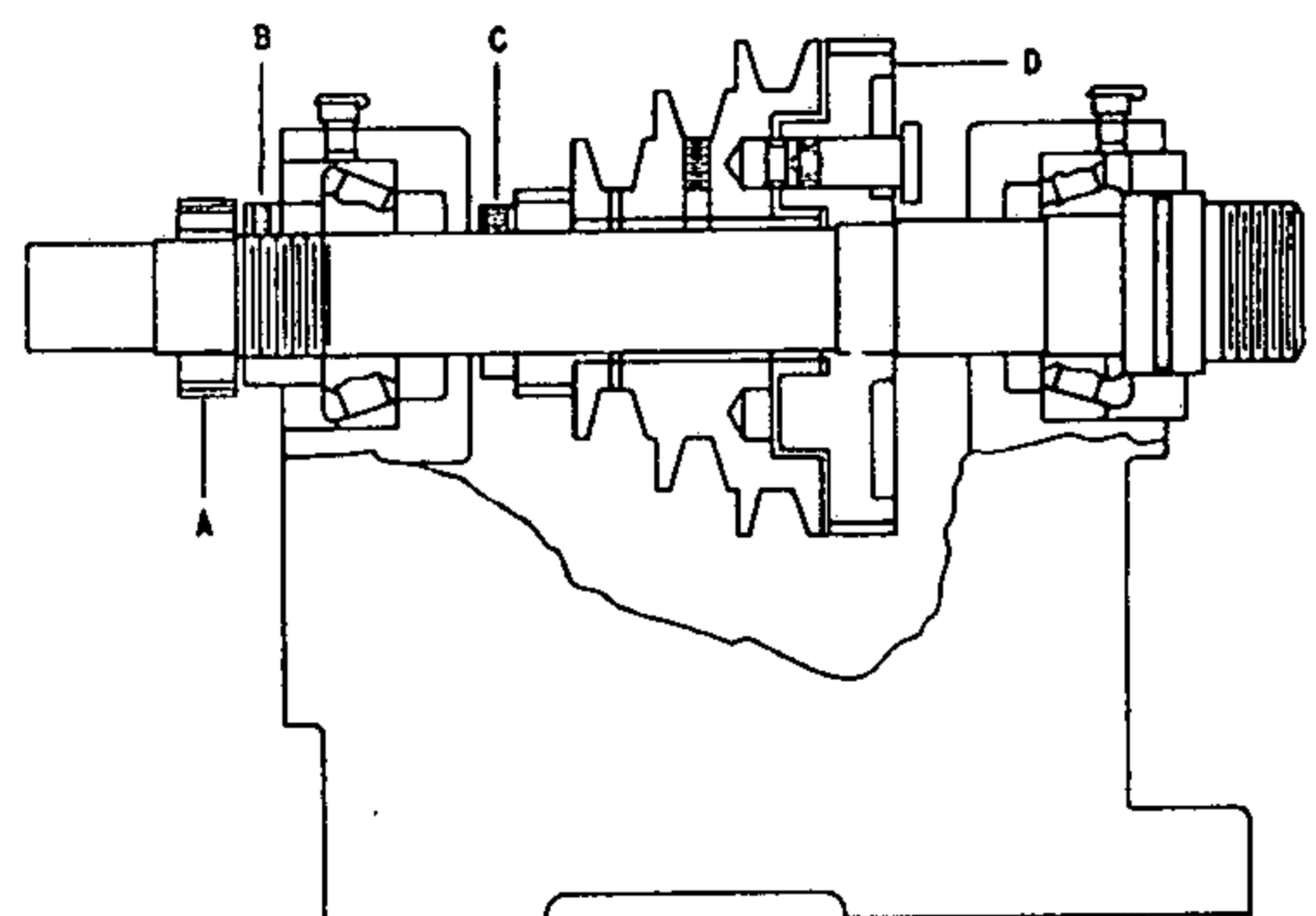
Adjustments

SPINDLE BEARING ADJUSTMENT

If the spindle turns too freely, or if play is noticeable when spindle is pushed back and forth, adjust the bearings as follows:

1. Loosen lock screw in take-up collar "B" (Figure 18) and tighten collar until all spindle play has been removed.
2. To determine correct bearing preload, give spindle pulley a sharp spin with your hand - pulley should rotate about one turn. If it doesn't, adjust collar B, than recheck.

CAUTION - Do not tighten collar too tightly - spindle should rotate freely.



18 Spindle bearing adjustment locations.

BELT ADJUSTMENT ON HORIZONTAL COUNTERSHAFT LATHES

SPINDLE BELT - This belt can be easily adjusted with the four countershaft adjusting screws. Belt should be just tight enough to prevent its slipping when hood is lowered.

IMPORTANT - Do not tighten the four countershaft adjusting screws too tightly - it may compress the outer bearing sleeve and distort the bearing, causing permanent damage. Turn screws up until they are finger tight, then about 1/8" turn more, and lock.

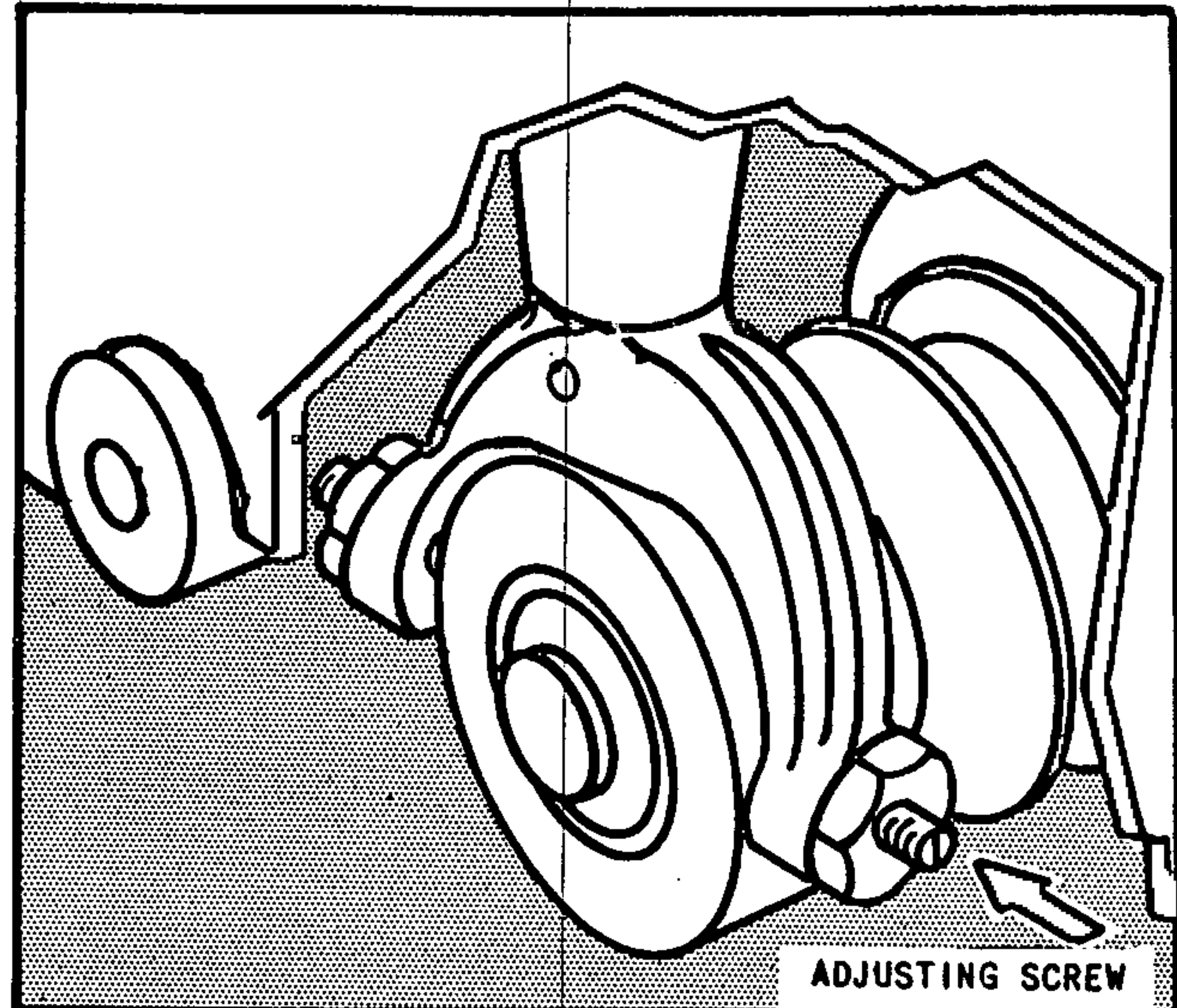
MOTOR BELT - Correct motor belt tension is obtained by adjusting the spring tension nut located on underneath side of motor base - see Figure 7. Belt should be just tight enough to prevent slipping.

HOW TO REMOVE HEADSTOCK SPINDLE, See Figure 18

1. Remove gear train guard.
2. Lower the gear quadrant by loosening the lock screw at bottom of quadrant and remove screw that holds top end of quadrant to headstock.
3. Slide feed reverse lever assembly from headstock to prevent possible damage to the gears when spindle is being removed.
4. Remove spindle gear "A" from spindle by tapping gear with a block of wood and hammer until there's enough clearance between gear and take-up collar to use a pulley puller to remove gear.
5. Remove key from spindle with pliers.
6. Loosen lock screw in take-up collar "B" and remove collar by turning counterclockwise.
7. Loosen set screw in collar "C".
8. Wedge two pieces of wood between headstock and large spindle gear "D" on both sides of spindle. Then, with a wood hammer drive spindle out of the headstock. Remove the rear spindle bearing cone when it is released from the spindle.
9. Remove belt from countershaft by

loosening the two countershaft adjusting screws on right end of countershaft spindle - see Figure 19.

Complete countershaft can be removed by loosening all four countershaft adjusting screws and the lock screw that holds clutch lever in shift yoke.



19 Loosen the rear adjusting screws on each hanger to remove countershaft spindle assembly.

RE-ASSEMBLY

1. Lightly file all burrs, if any, from the spindle.
2. Clean all parts thoroughly, including the bearings.
3. Place new belt, or bearings, in position and replace spindle. **CAUTION** - Use the palm of the hand as much as possible when replacing the spindle - it will avoid damaging the precision surfaces of the spindle nose. Slide large spindle gear against the spindle shoulder. Then slide on the pulley until pulley bushing is against the gear.
4. Tighten collar "C" against small gear with just enough clearance to permit pulley to turn freely but with no play.
5. Carefully tap rear spindle bearing cone in place using a brass rod and hammer.
6. Replace take-up collar "B" and tighten until all play has been removed from the spindle - check by tapping spindle back and forth with the hand. Do not tighten collar too tightly - spindle should rotate freely.

To determine correct bearing preload, give spindle pulley a sharp spin with your hand - pulley should rotate about one turn. If it doesn't, adjust collar B, then recheck.

7. Replace Woodruff key and spindle gear.

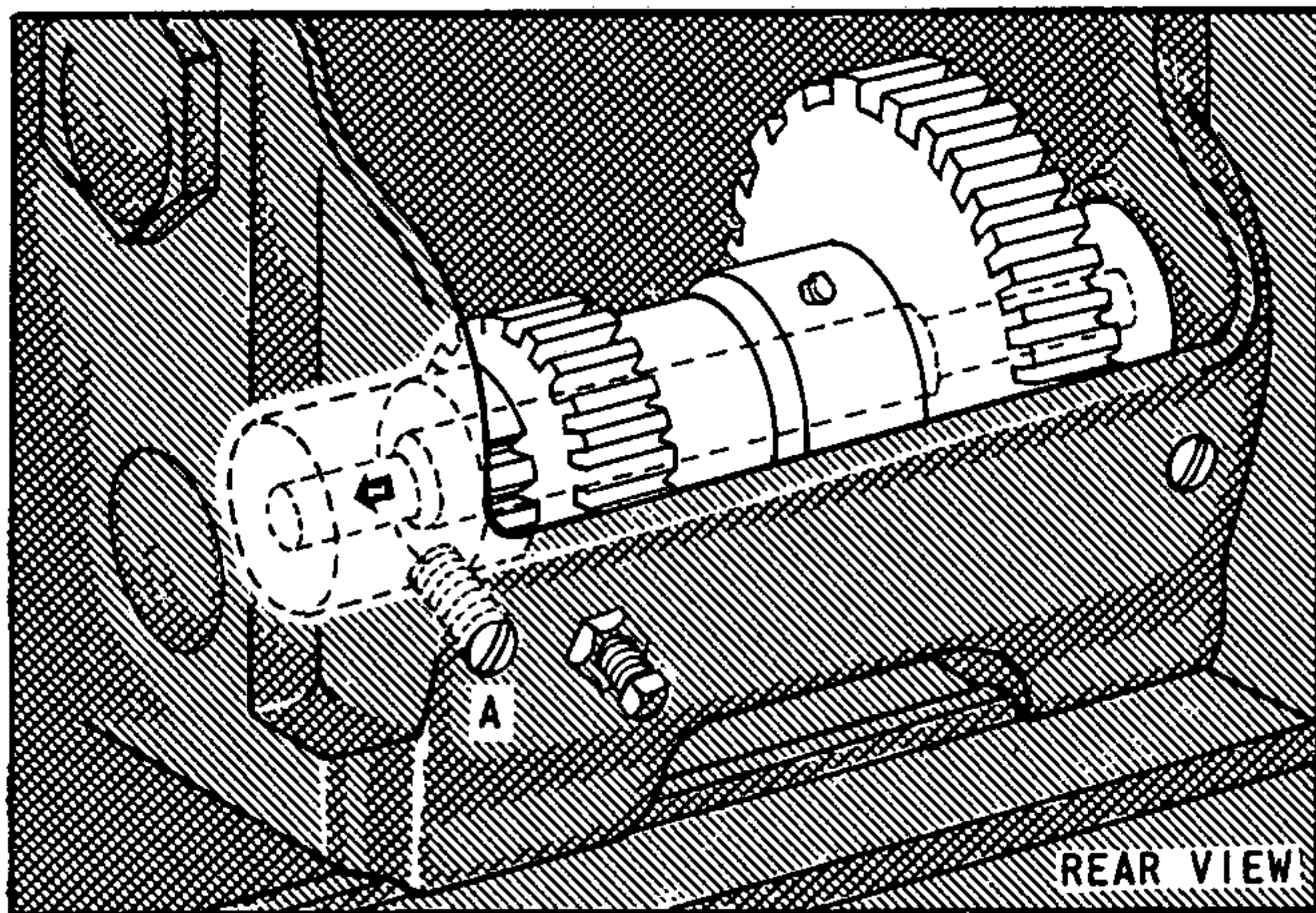
HOW TO REMOVE BACK-GEARS

1. Remove headstock spindle - see instructions "HOW TO REMOVE HEADSTOCK SPINDLE", page 11.

2. Loosen lock screw "A", see Fig. 20.

3. Now drive out back-gear shaft in the direction indicated by the arrow, Figure 20.

4. Remove back-gears from the headstock.



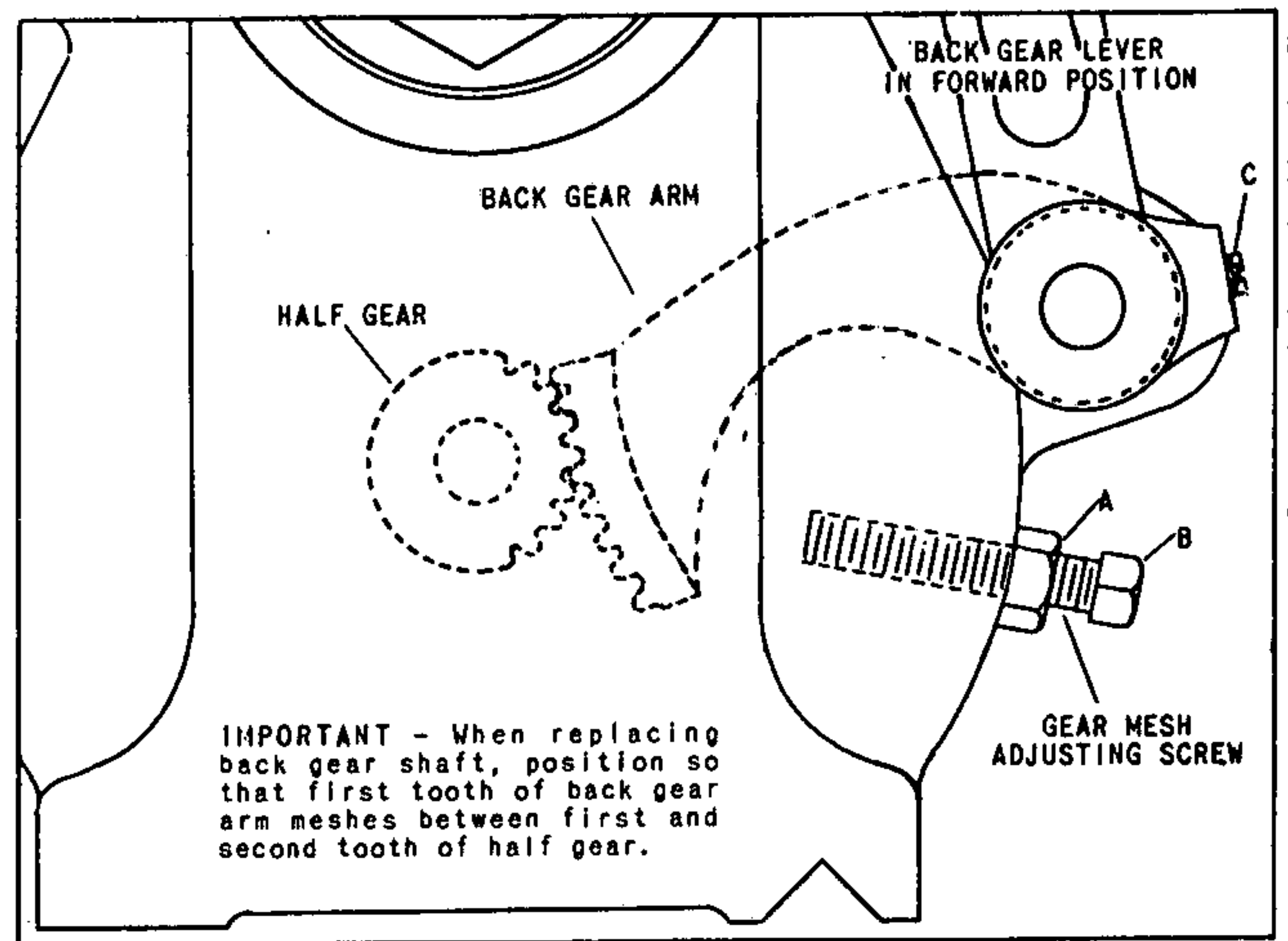
20 Adjusting and removing spindle back-gears.

RE-ASSEMBLING THE BACK-GEARS

1. Move back-gear lever to the forward or back-gear drive position, see Figure 21.

2. Place back-gears in headstock and replace back-gear shaft. The teeth of the half-gear on the shaft must slip into mesh with the teeth on the back-gear arm in only one position - see Figure 21. This is necessary so that the full travel of the back-gears is obtained.

3. Replace sleeve and plug. Lock sleeve in position with lock screw "A", Figure 20.



21 Correct mesh of back-gear arm with half-gear and location of adjustment nuts.

BACK-GEAR ADJUSTMENTS

TO ADJUST GEAR MESH, or gear play, between back-gears and spindle-gears; first place back-gear lever in the forward position. Then loosen lock nut "A", Figure 21, and turn adjusting screw "B" until gears properly mesh. tighten lock nut "A".

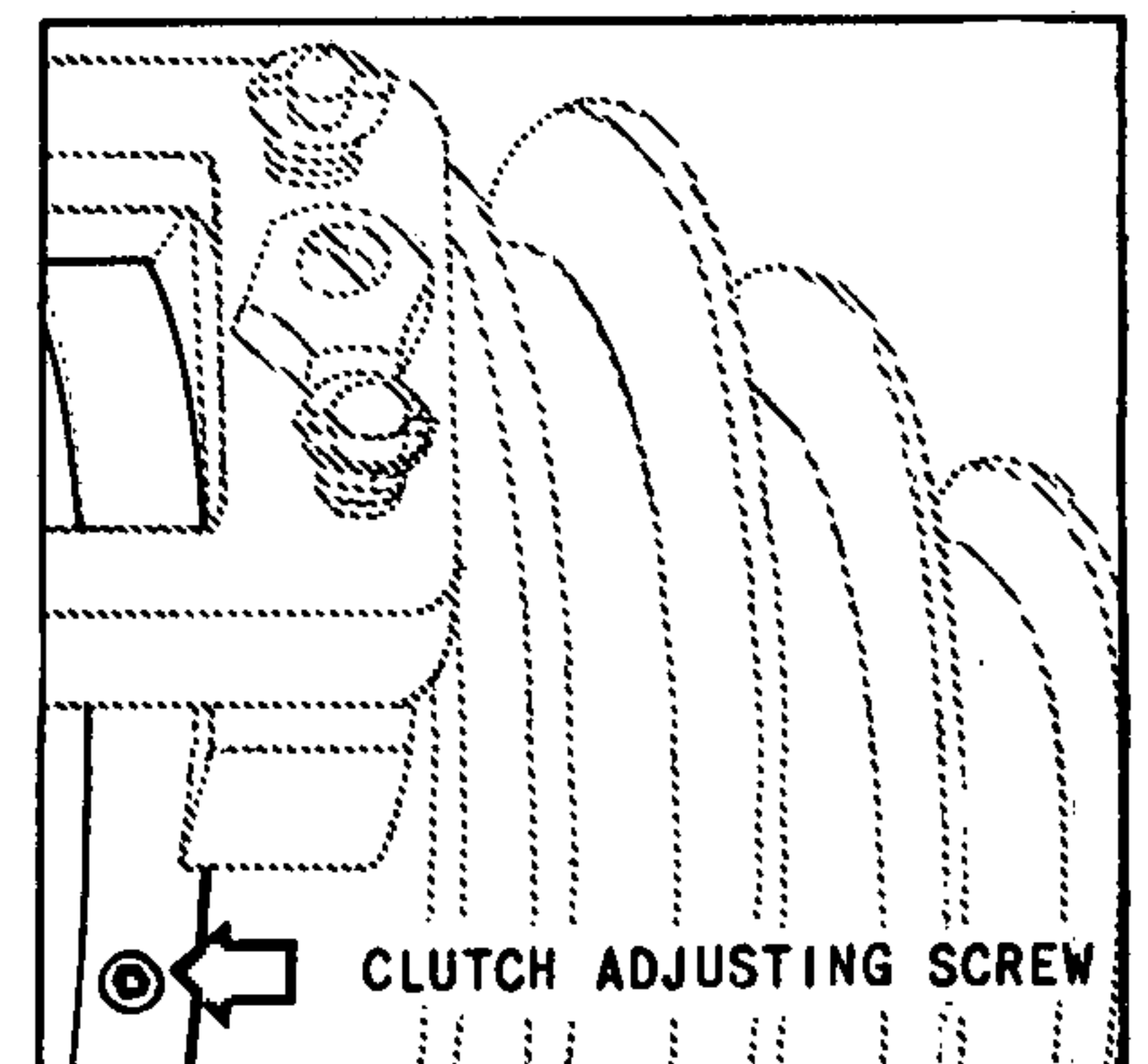
TO PREVENT GEARS FROM JUMPING OUT OF MESH tighten screw "C", see Figure 21.

CLUTCH COUNTERSHAFT ADJUSTMENT

If the countershaft clutch slips when the spindle drive is engaged, adjust as follows:

First remove the lock screw that's just ahead of the clutch adjusting screw - see Figure 22. Then tighten clutch adjusting screw just enough to prevent slippage. DO NOT tighten screw too tight - spindle drive will not disengage when clutch lever is moved to the right if screw is too tight.

22 Tighten clutch-adjusting screw if countershaft clutch slips when spindle drive is engaged.



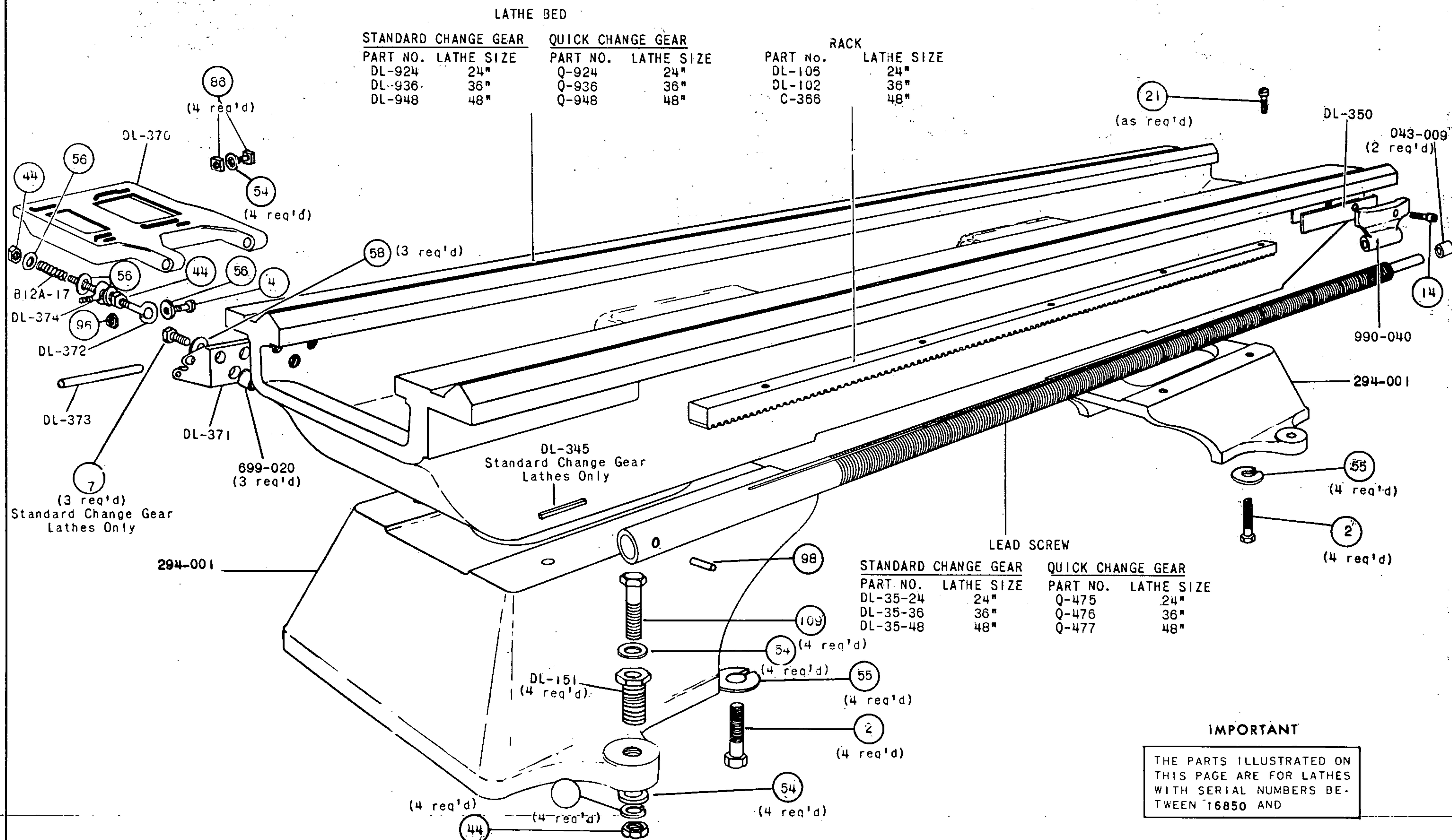
LATHE BED

PART No.	RACK	LATHE SIZE
DL-105		24"
DL-102		36"
C-366		48"

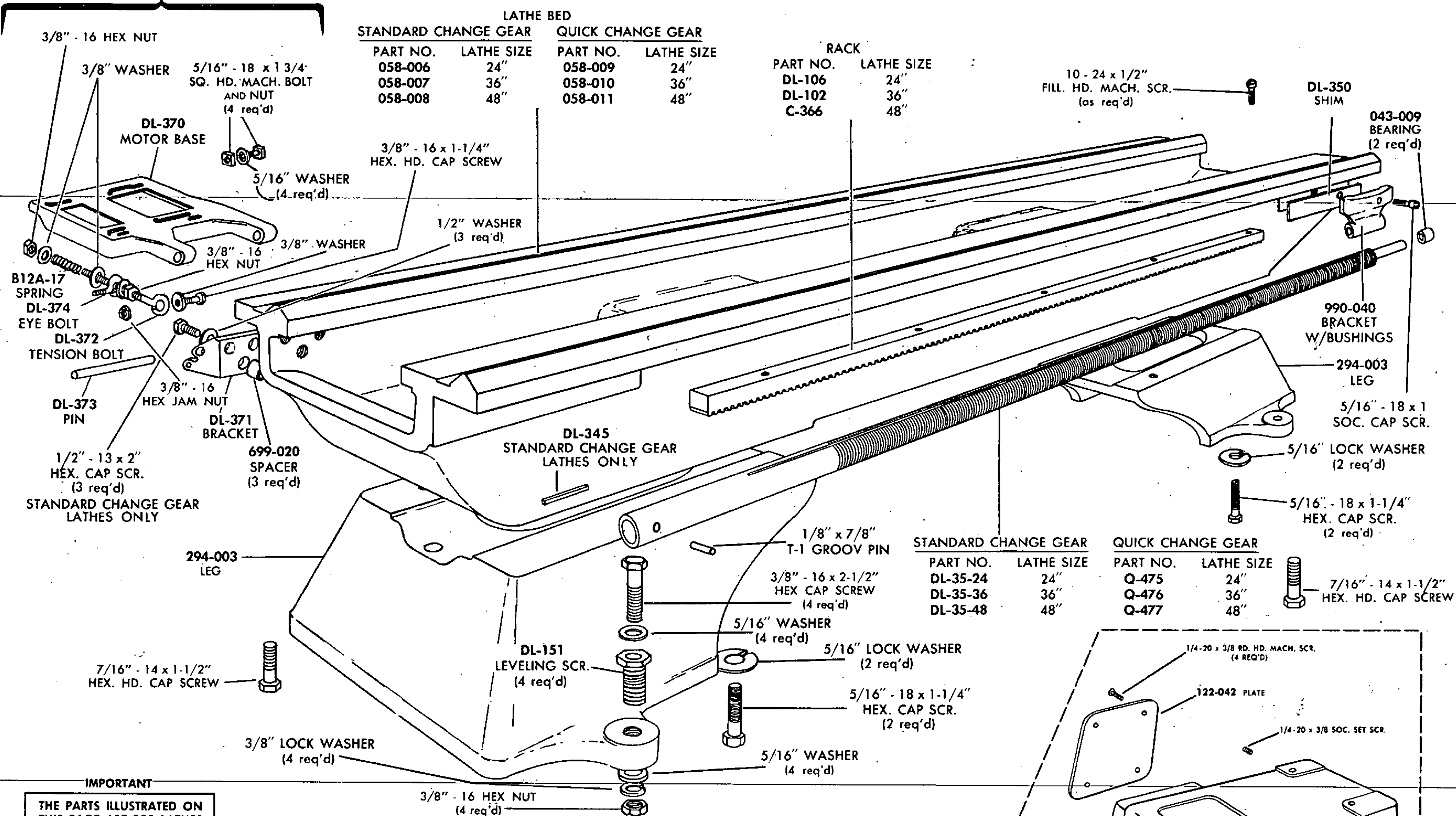


THE PARTS ILLUSTRATED ON THIS PAGE ARE FOR LATHES WITH SERIAL NUMBERS BETWEEN 15000 AND

BED AND LEADSCREW ASSEMBLY



NOTE: FOR USE WITH
HORIZONTAL COUNTER
SHAFT LATHES.



LATHE BED			
STANDARD CHANGE GEAR		QUICK CHANGE GEAR	
PART NO.	LATHE SIZE	PART NO.	LATHE SIZE
058-006	24"	058-009	24"
058-007	36"	058-010	36"
058-008	48"	058-011	48"

RACK	
PART NO.	LATHE SIZE
DL-106	24"
DL-102	36"
C-366	48"

STANDARD CHANGE GEAR	
PART NO.	LATHE SIZE
DL-35-24	24"
DL-35-36	36"
DL-35-48	48"

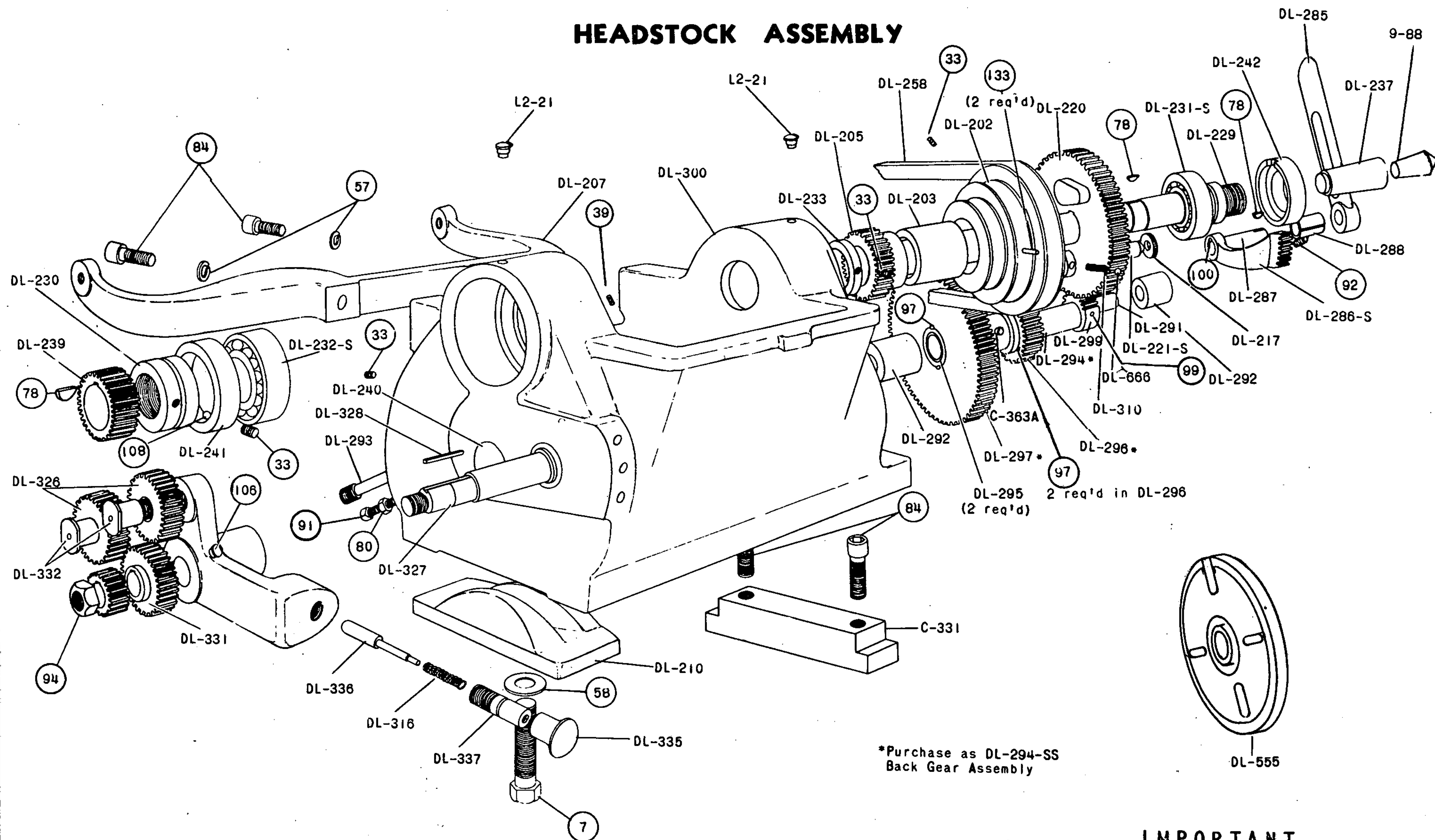
QUICK CHANGE GEAR	
PART NO.	LATHE SIZE
Q-475	24"
Q-476	36"
Q-477	48"

IMPORTANT
THE PARTS ILLUSTRATED ON
THIS PAGE ARE FOR LATHES
WITH SERIAL NUMBERS BE-
TWEEN 17403 AND

NOTE: FOR USE WITH
UNDER-DRIVE LATHES

BED AND LEADSCREW ASSEMBLY

HEADSTOCK ASSEMBLY

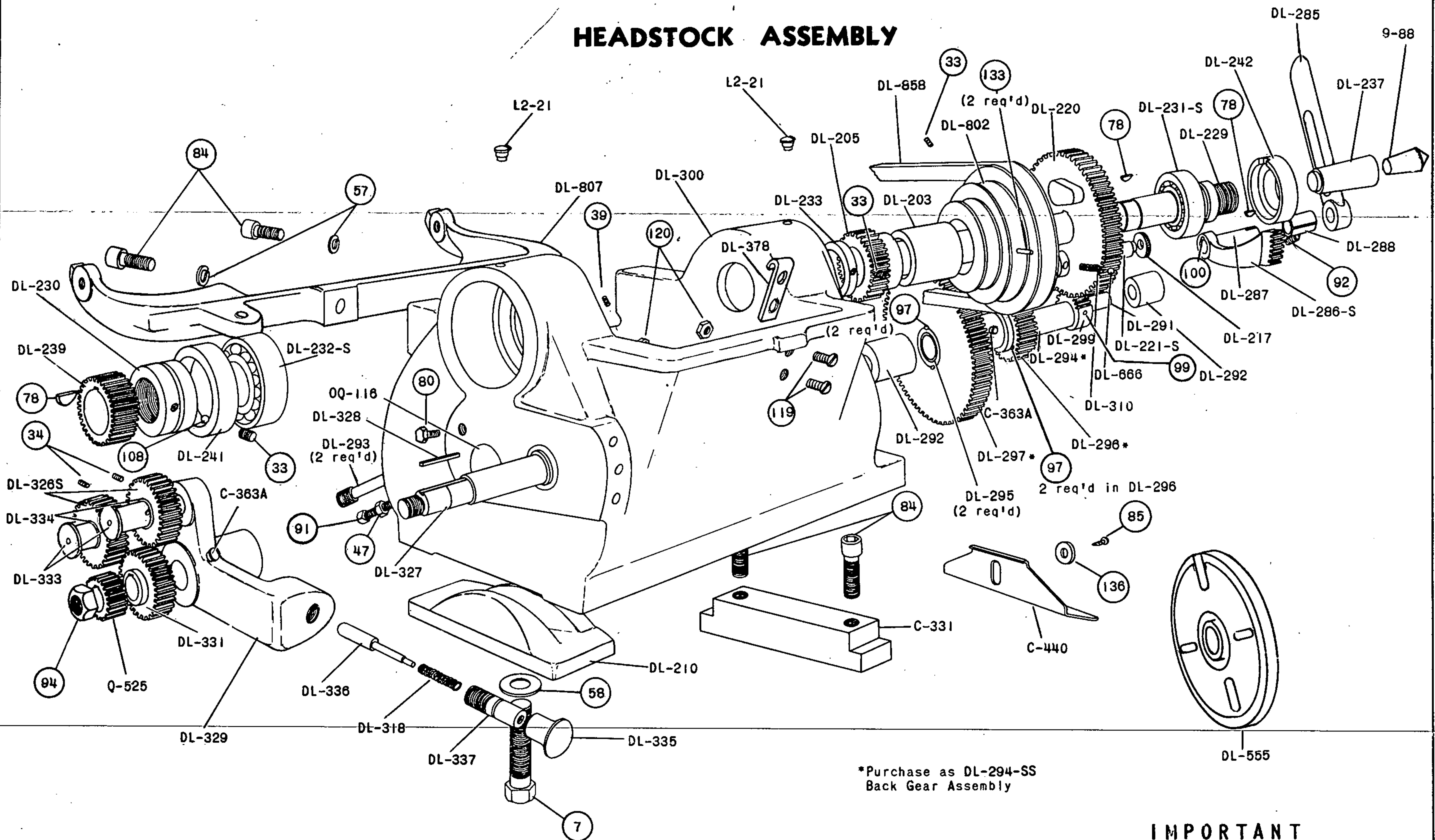


*Purchase as DL-294-SS
Back Gear Assembly

IMPORTANT

THE PARTS ILLUSTRATED
ON THIS PAGE ARE FOR
LATHES WITH SERIAL
NUMBERS BETWEEN 15120
AND 16021.

HEADSTOCK ASSEMBLY

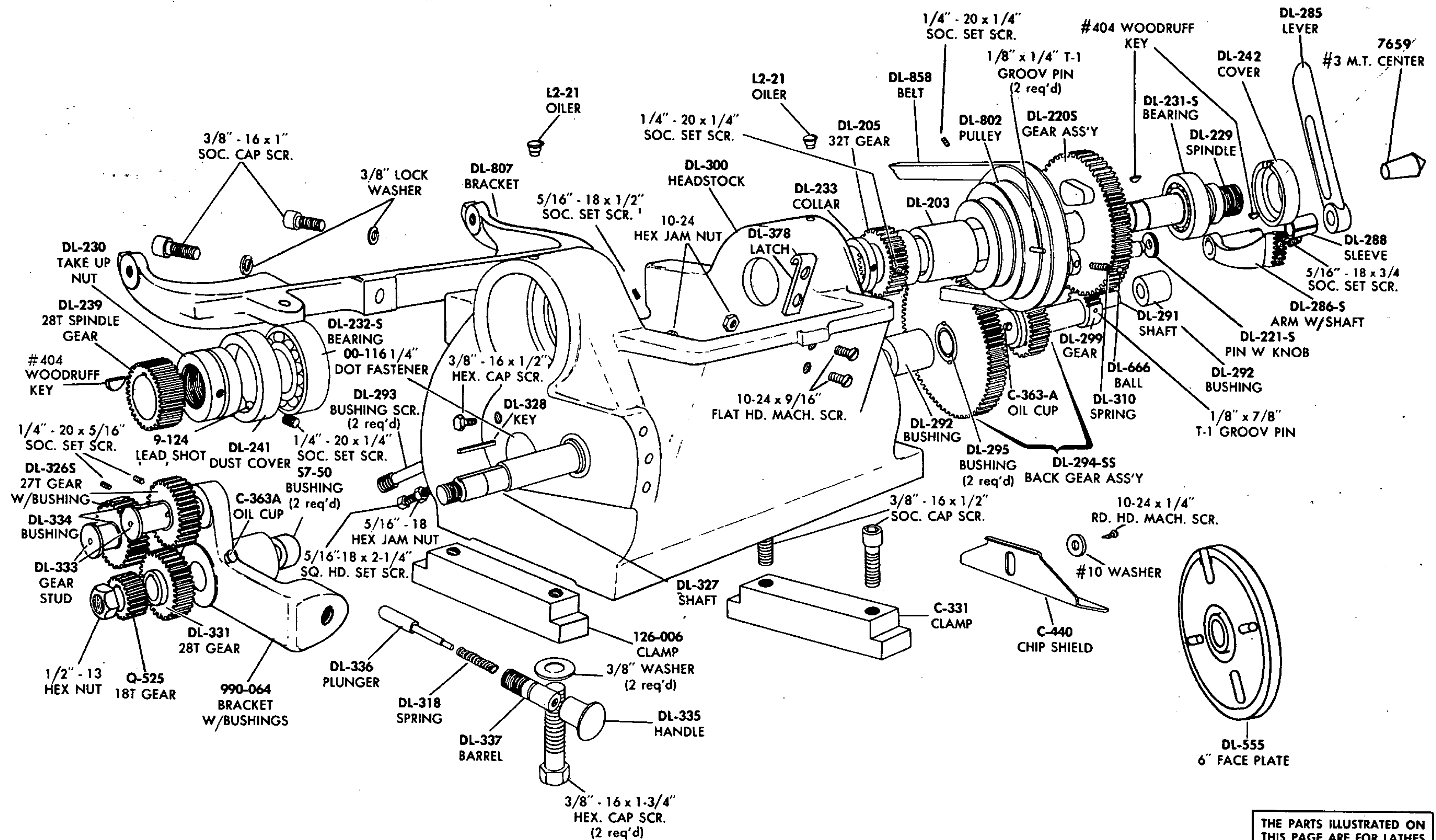


*Purchase as DL-294-SS
Back Gear Assembly

IMPORTANT

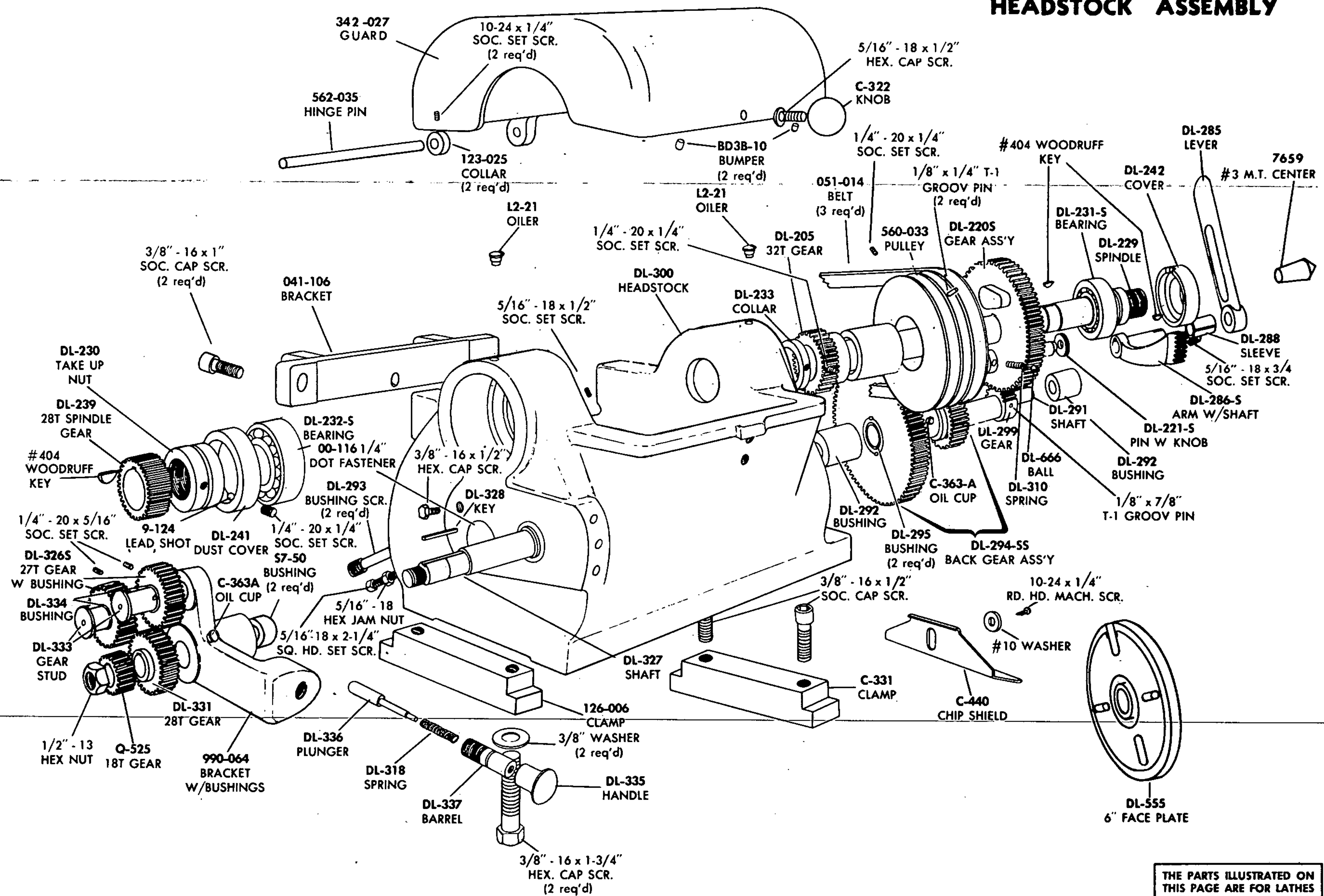
THE PARTS ILLUSTRATED
ON THIS PAGE ARE FOR
LATHES WITH SERIAL
NUMBERS BETWEEN 16022
AND -----

HEADSTOCK ASSEMBLY



THE PARTS ILLUSTRATED ON THIS PAGE ARE FOR LATHES WITH SERIAL NUMBERS BETWEEN 16673 AND

HEADSTOCK ASSEMBLY

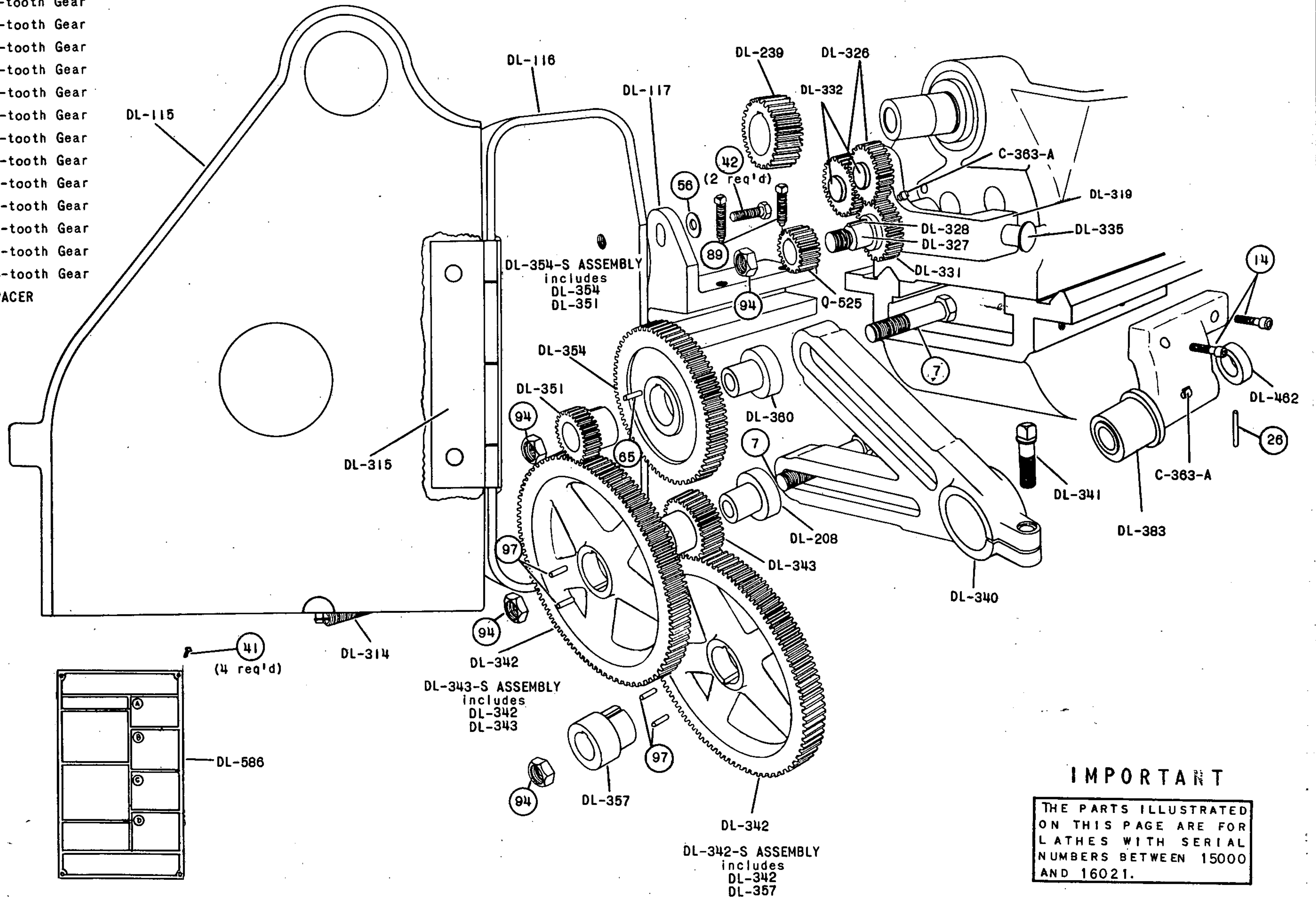


THE PARTS ILLUSTRATED ON THIS PAGE ARE FOR LATHES WITH SERIAL NUMBERS BETWEEN 17434 AND

STANDARD CHANGE GEAR TRAIN ASSEMBLY

CHANGE GEARS and SPACER
(furnished)

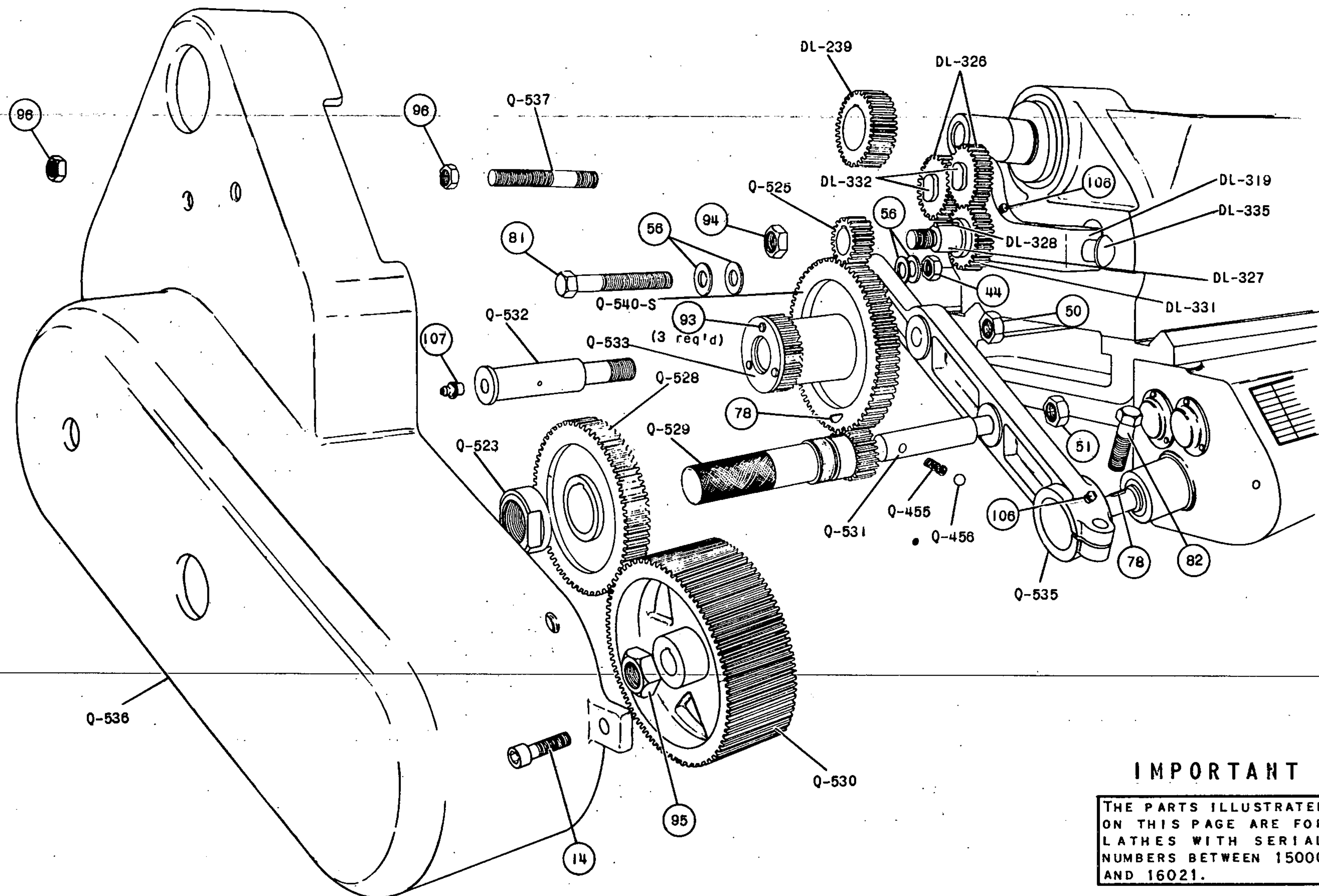
Part No.	Name
DL-330	16-tooth Gear
DL-585	32-tooth Gear
DL-585	32-tooth Gear
DL-590	38-tooth Gear
DL-584	40-tooth Gear
DL-583	44-tooth Gear
DL-582	46-tooth Gear
DL-581	48-tooth Gear
DL-580	52-tooth Gear
DL-591	54-tooth Gear
DL-579	56-tooth Gear
DL-578	60-tooth Gear
DL-577	64-tooth Gear
DL-359	SPACER



IMPORTANT

THE PARTS ILLUSTRATED
ON THIS PAGE ARE FOR
LATHES WITH SERIAL
NUMBERS BETWEEN 15000
AND 16021.

QUICK CHANGE GEAR TRAIN



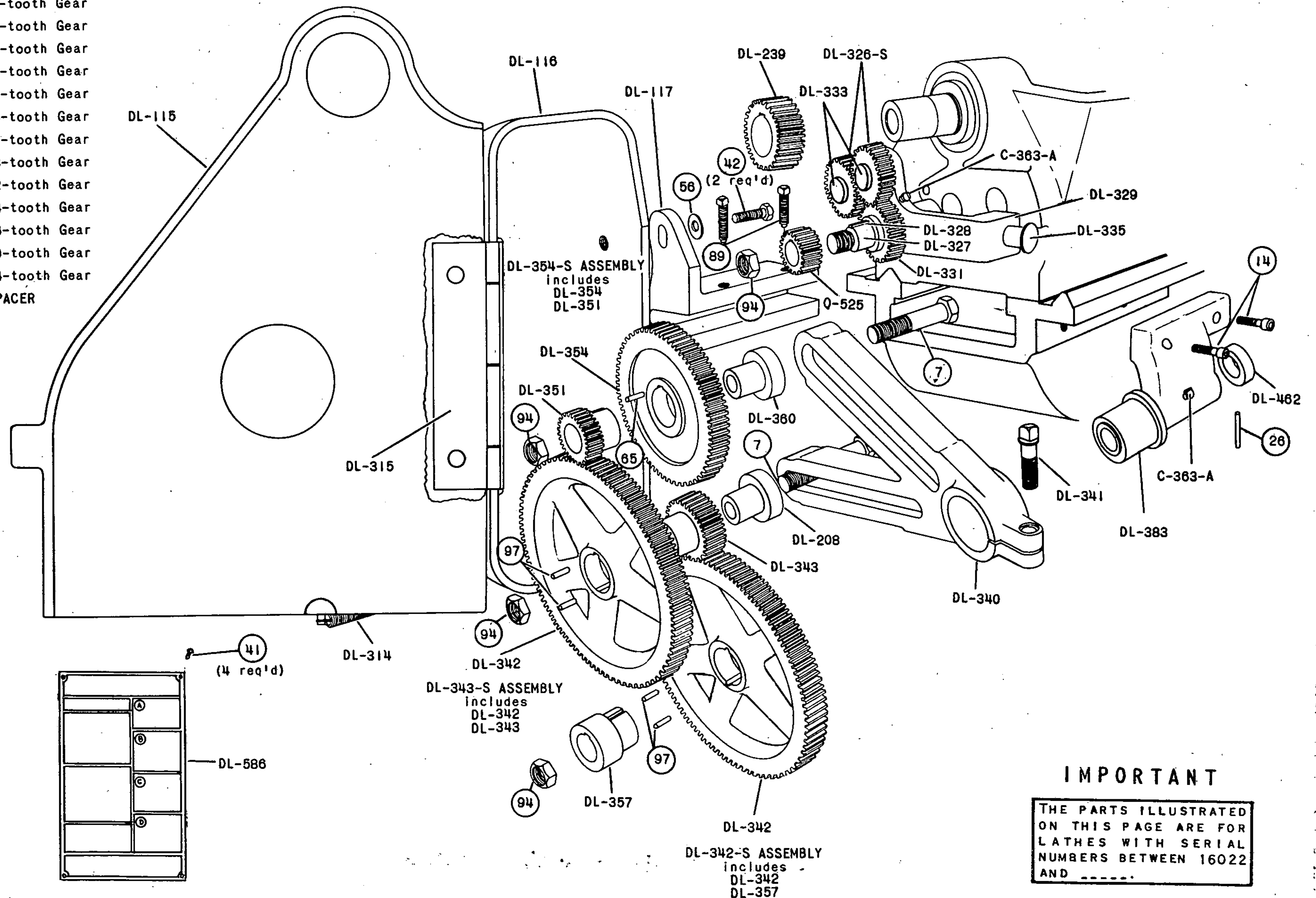
IMPORTANT

THE PARTS ILLUSTRATED
ON THIS PAGE ARE FOR
LATHES WITH SERIAL
NUMBERS BETWEEN 15000
AND 16021.

STANDARD CHANGE GEAR TRAIN ASSEMBLY

CHANGE GEARS and SPACER
(furnished)

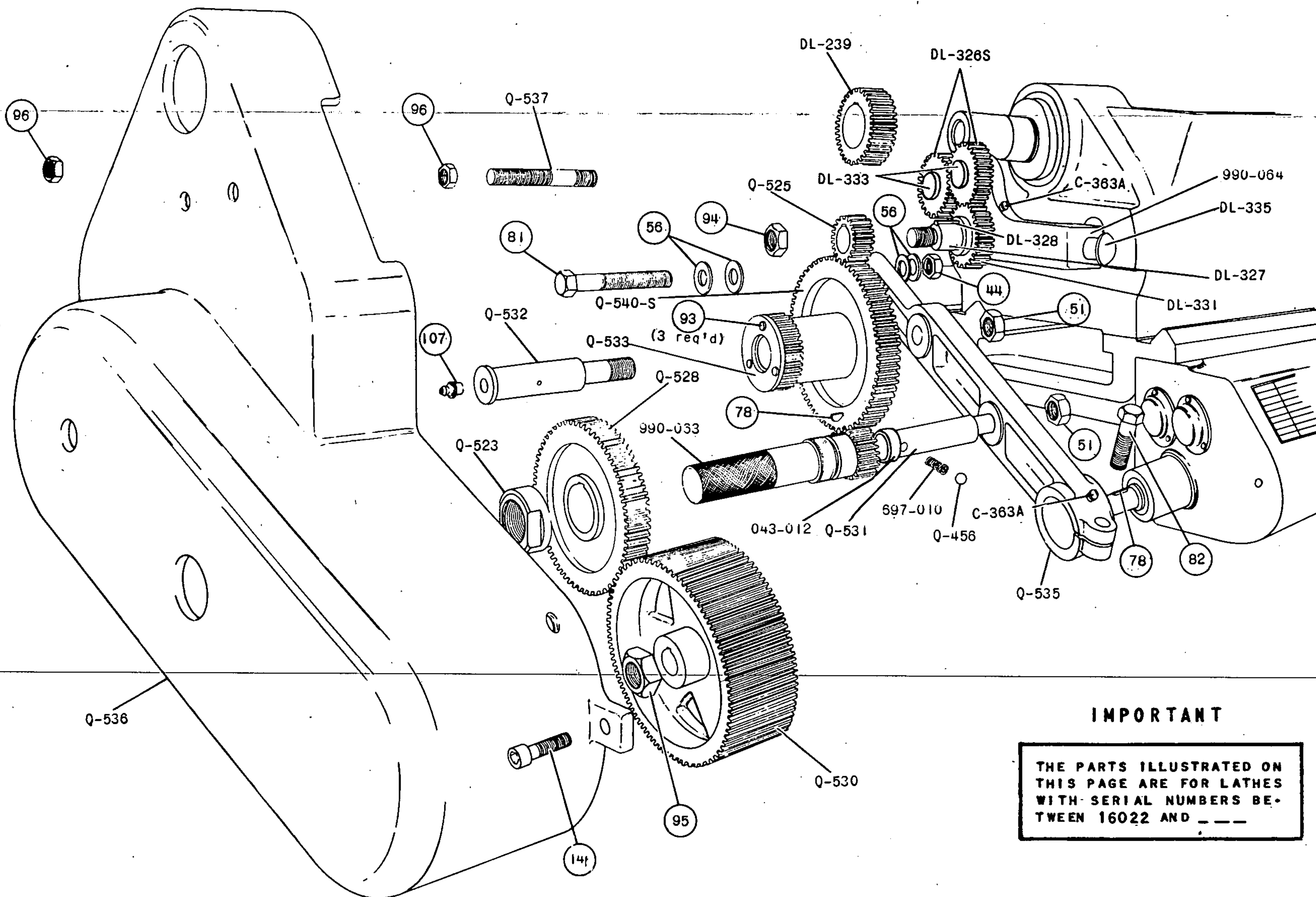
Part No.	Name
DL-330	16-tooth Gear
DL-585	32-tooth Gear
DL-585	32-tooth Gear
DL-590	36-tooth Gear
DL-584	40-tooth Gear
DL-583	44-tooth Gear
DL-582	46-tooth Gear
DL-581	48-tooth Gear
DL-580	52-tooth Gear
DL-591	54-tooth Gear
DL-579	56-tooth Gear
DL-578	60-tooth Gear
DL-577	64-tooth Gear
DL-359	SPACER



IMPORTANT

THE PARTS ILLUSTRATED
ON THIS PAGE ARE FOR
LATHES WITH SERIAL
NUMBERS BETWEEN 16022
AND -----

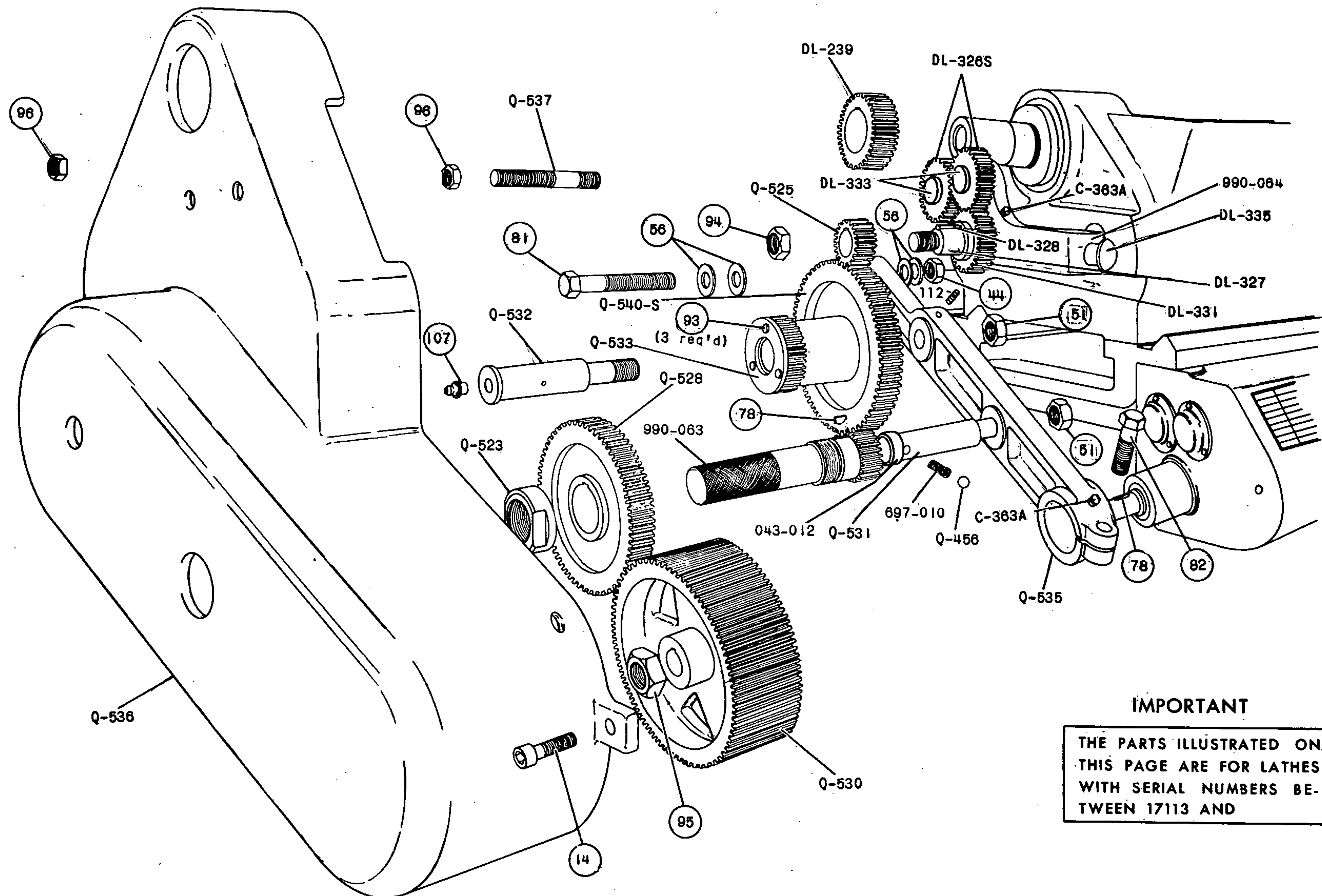
QUICK CHANGE GEAR TRAIN



IMPORTANT

THE PARTS ILLUSTRATED ON THIS PAGE ARE FOR LATHES WITH SERIAL NUMBERS BETWEEN 16022 AND _ _ _

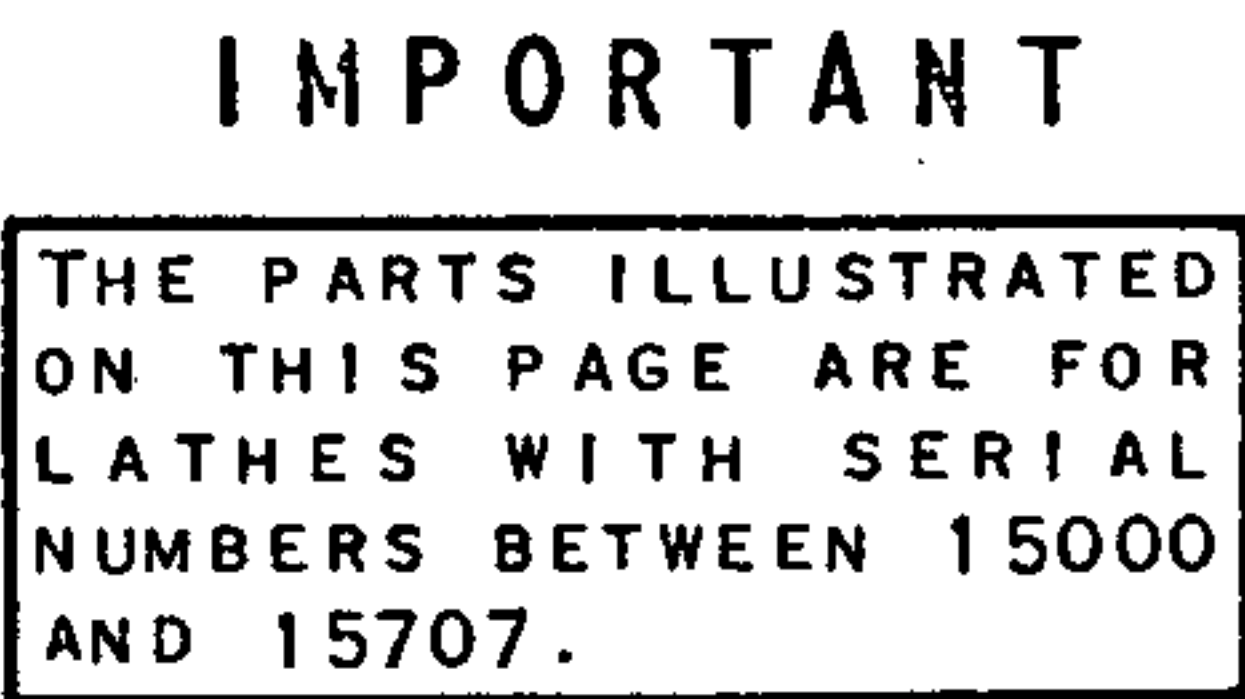
QUICK CHANGE GEAR TRAIN



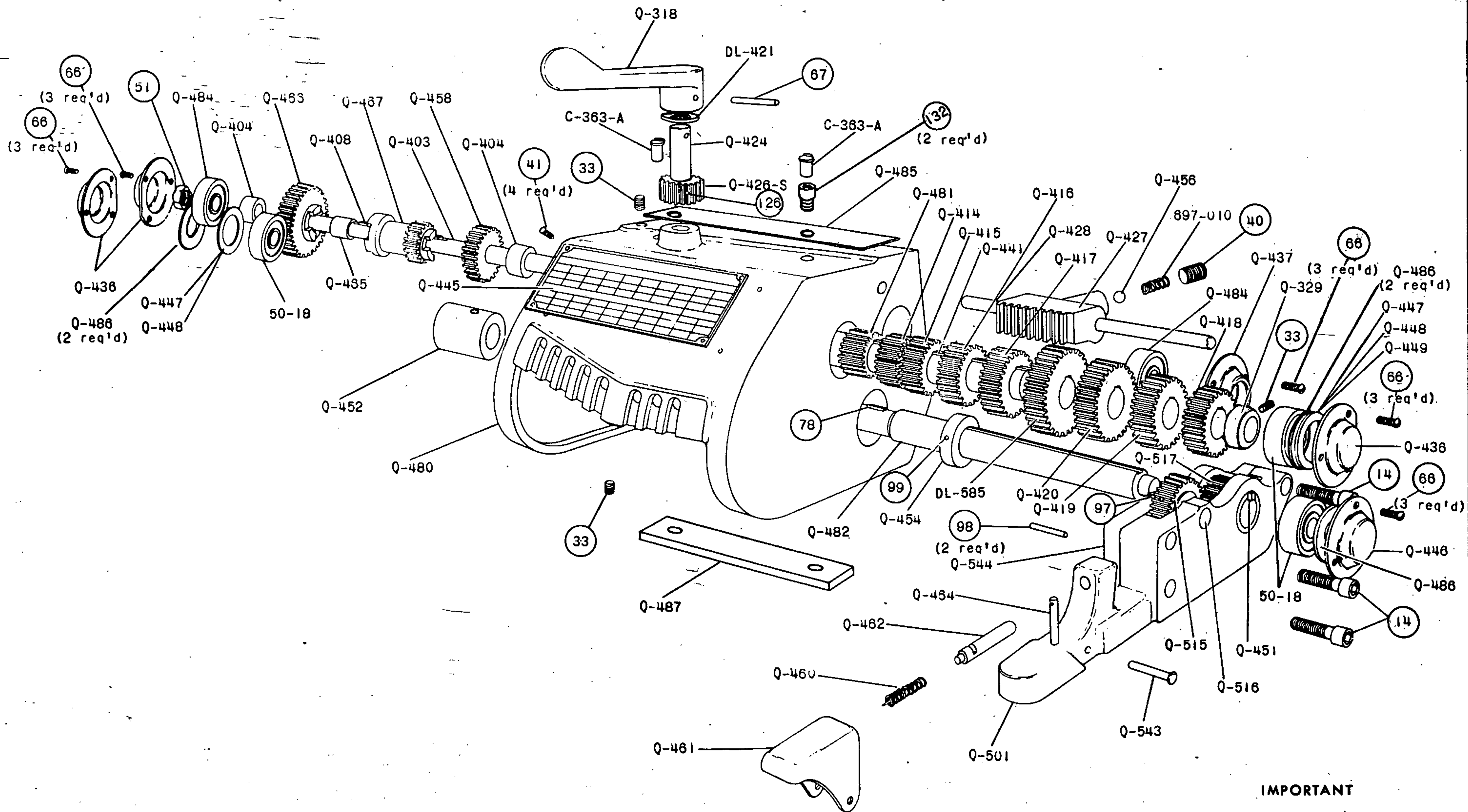
IMPORTANT

THE PARTS ILLUSTRATED ON THIS PAGE ARE FOR LATHES WITH SERIAL NUMBERS BETWEEN 17113 AND

16



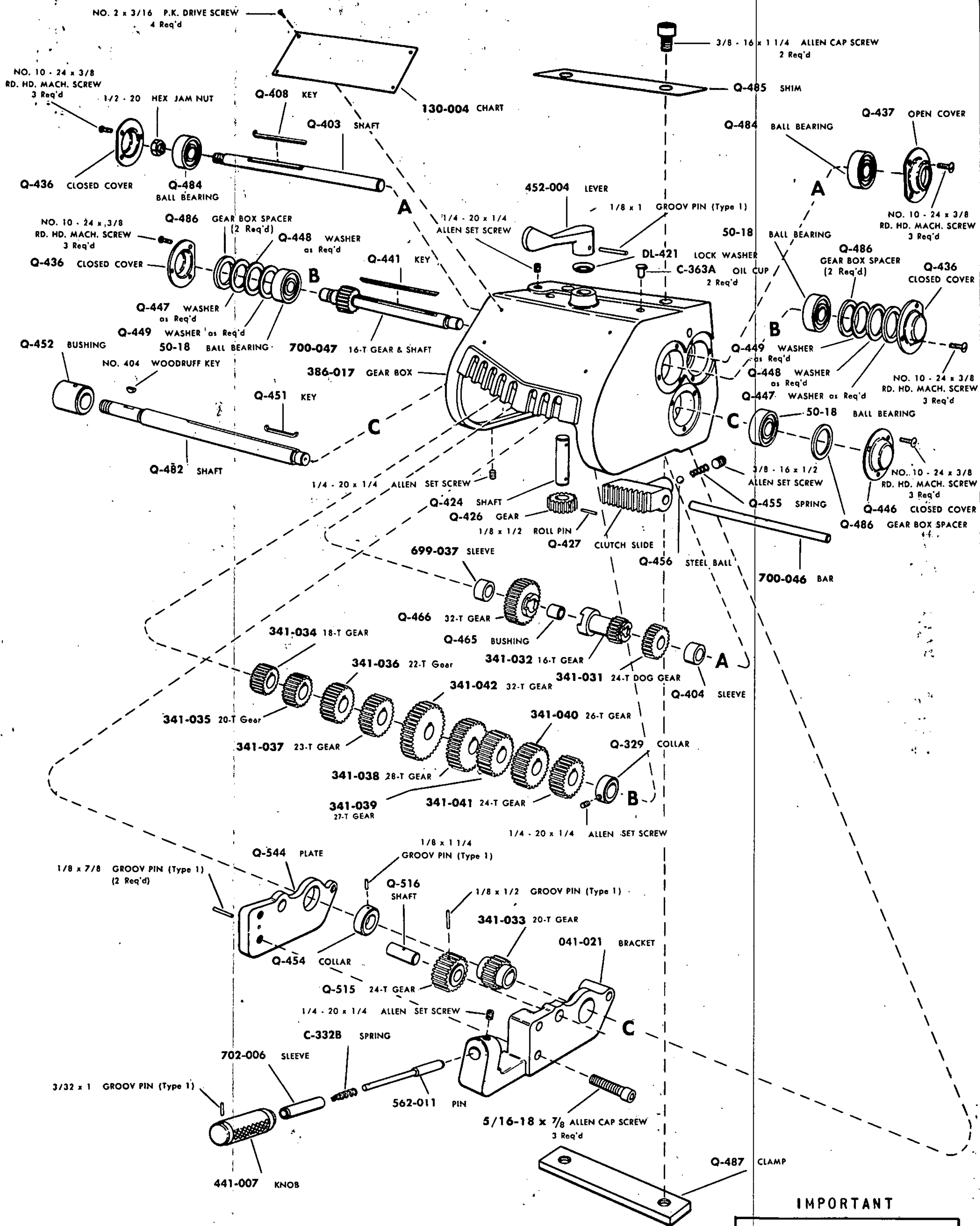
GEAR BOX ASSEMBLY



IMPORTANT

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THIS PAGE ARE FOR LATHES
WITH SERIAL NUMBERS BE-
TWEEN 15708 AND

QUICK-CHANGE GEAR BOX ASSEMBLY



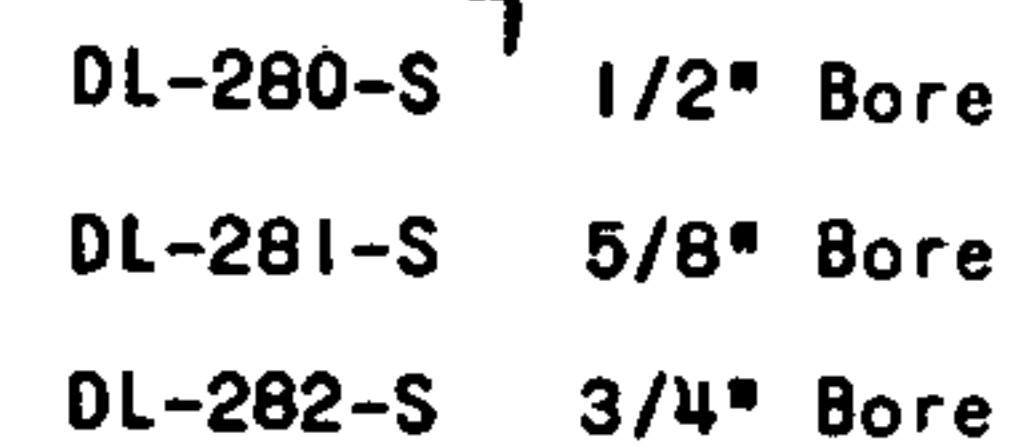
IMPORTANT

THE PARTS ILLUSTRATED ON
THIS PAGE ARE FOR LATHES
WITH SERIAL NUMBERS BE-
TWEEN 17039 AND

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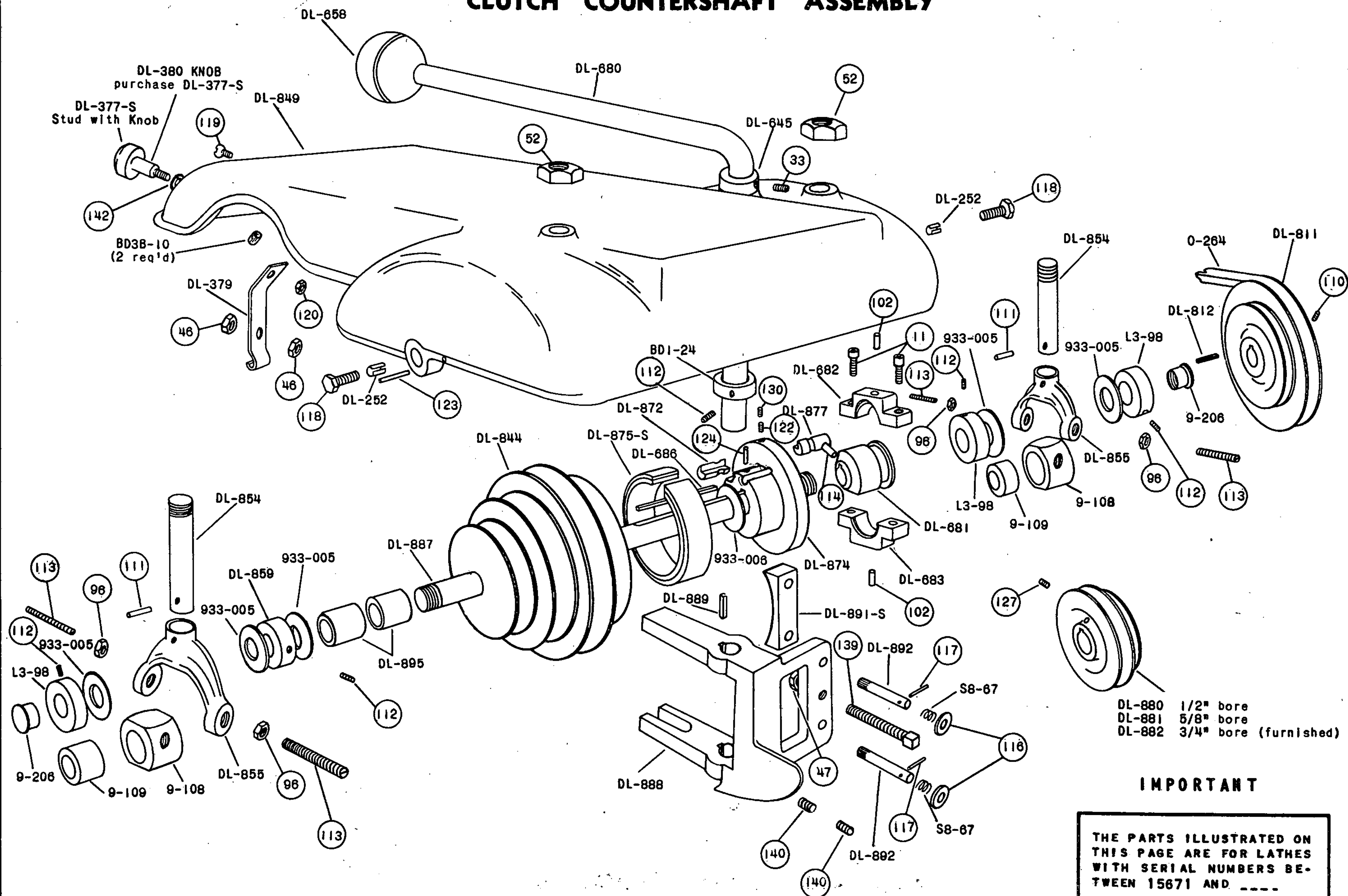
THE PARTS ILLUSTRATED ON THIS PAGE ARE FOR LATHES WITH SERIAL NUMBERS BETWEEN **17382** AND

17



THE PARTS ILLUSTRATED
ON THIS PAGE ARE FOR
LATHES WITH SERIAL
NUMBERS BETWEEN 15000
AND 15670.

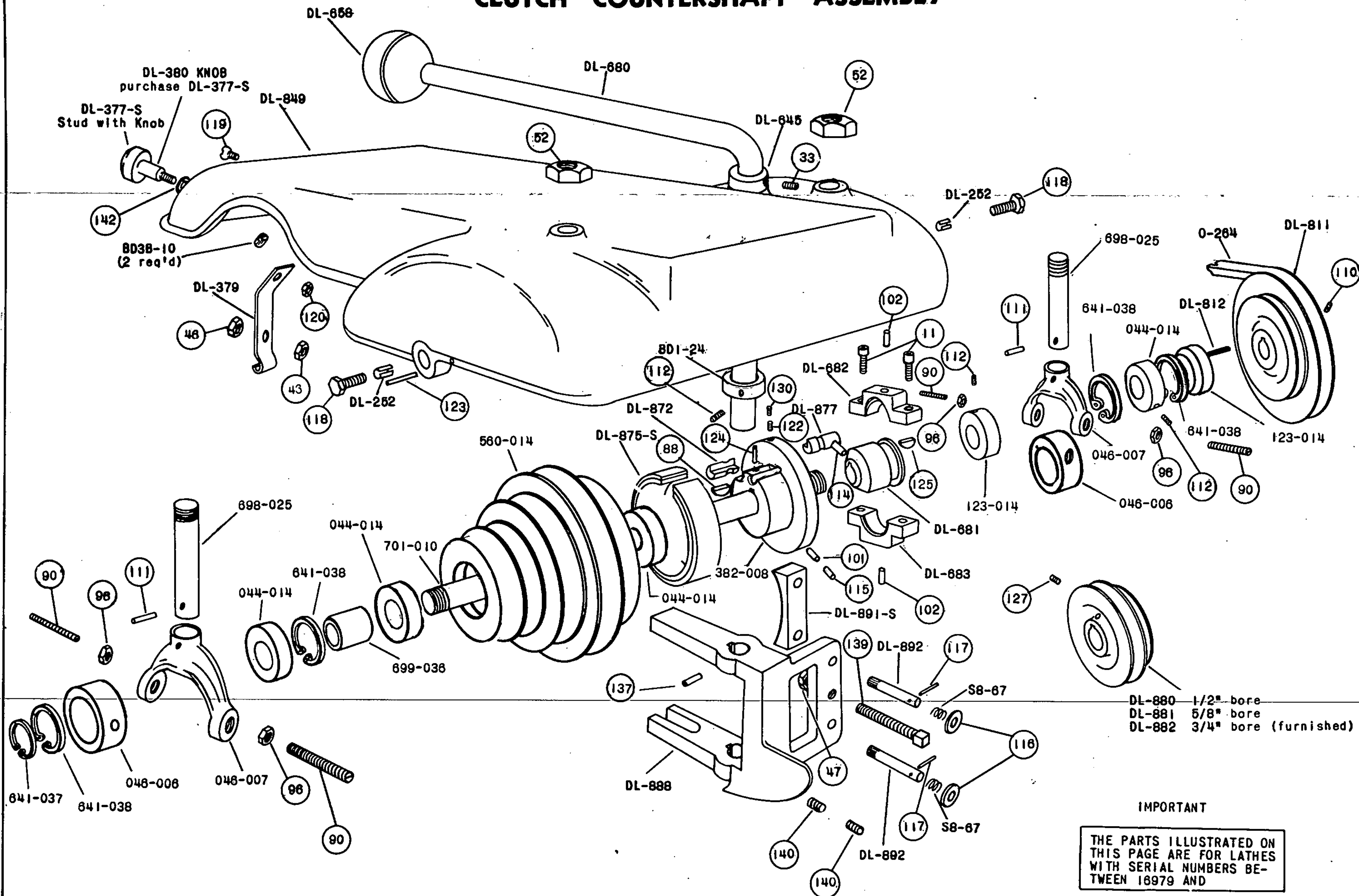
CLUTCH COUNTERSHAFT ASSEMBLY



IMPORTANT

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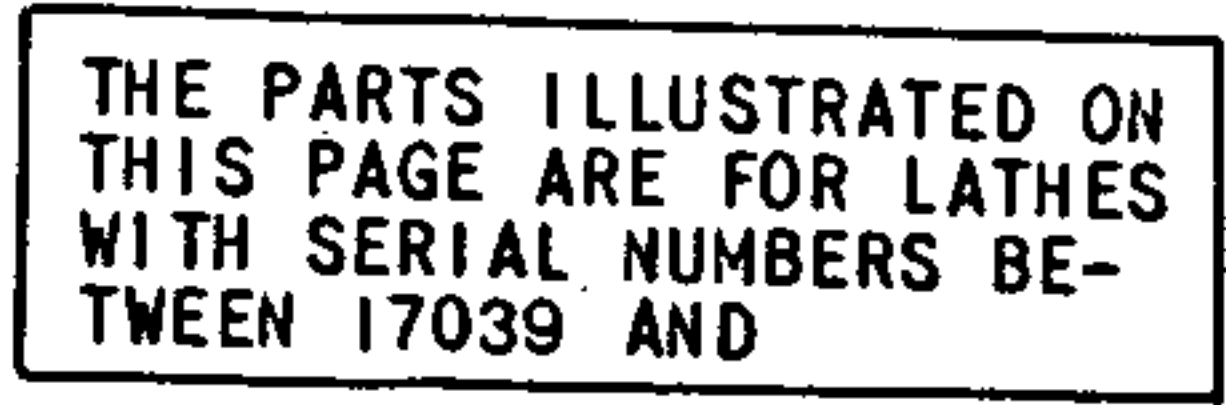
CLUTCH COUNTERSHAFT ASSEMBLY



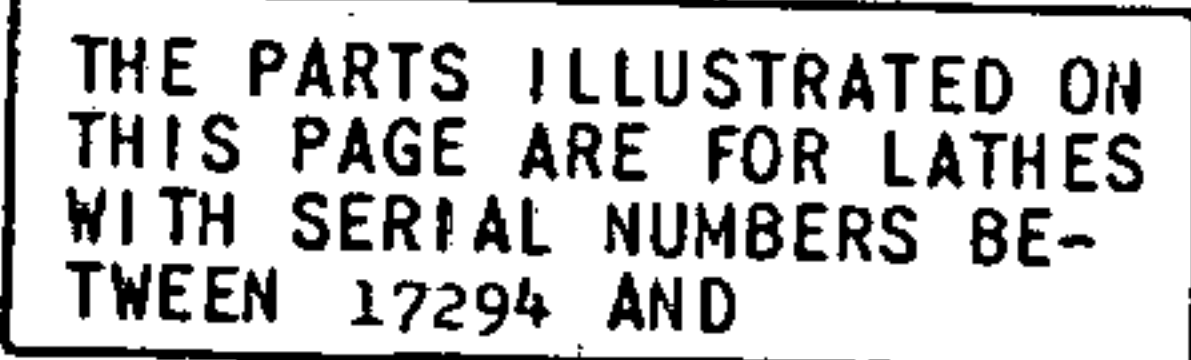
IMPORTANT

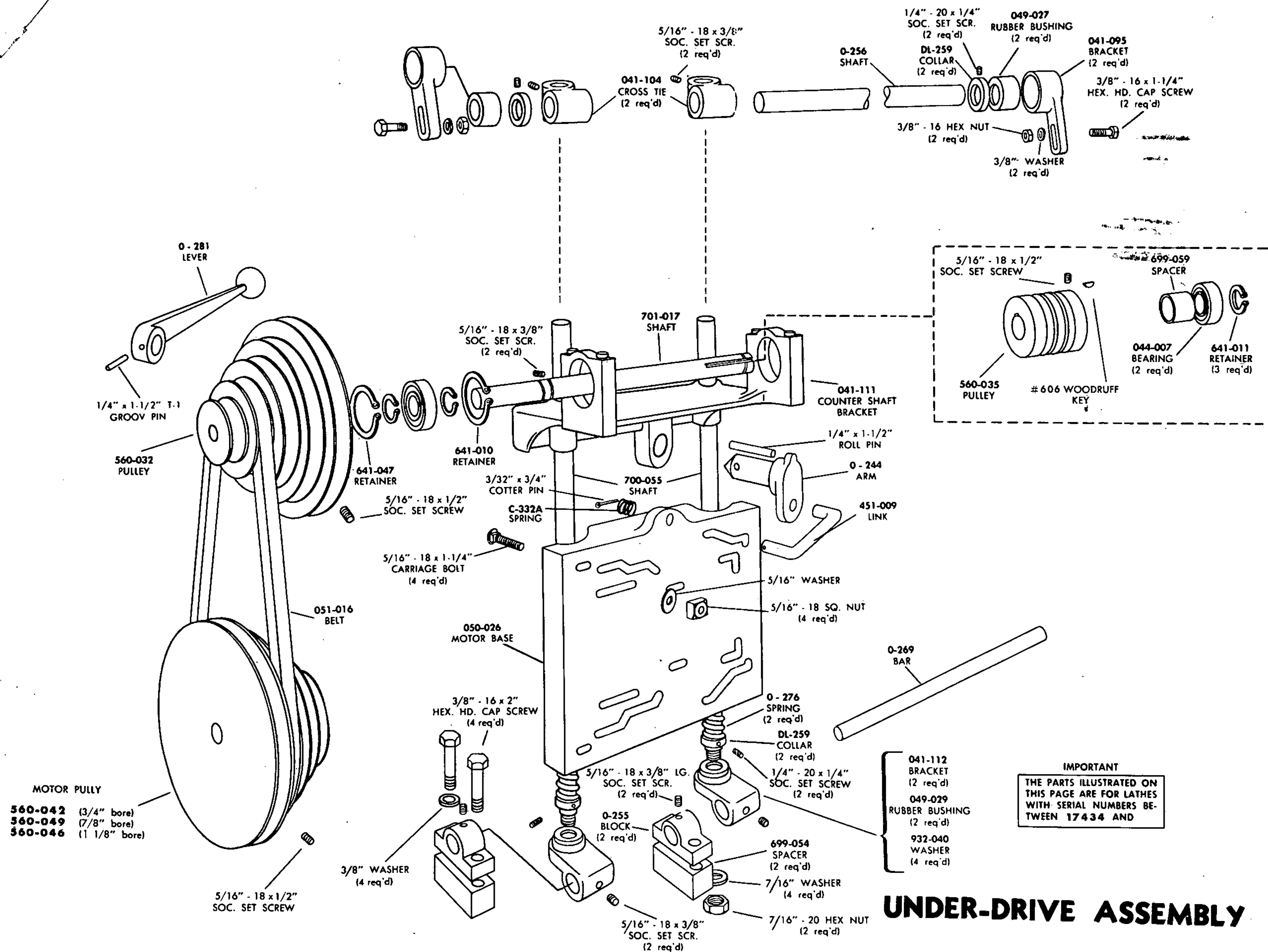
THE PARTS ILLUSTRATED ON
THIS PAGE ARE FOR LATHES
WITH SERIAL NUMBERS BE-
TWEEN 16979 AND

17-C



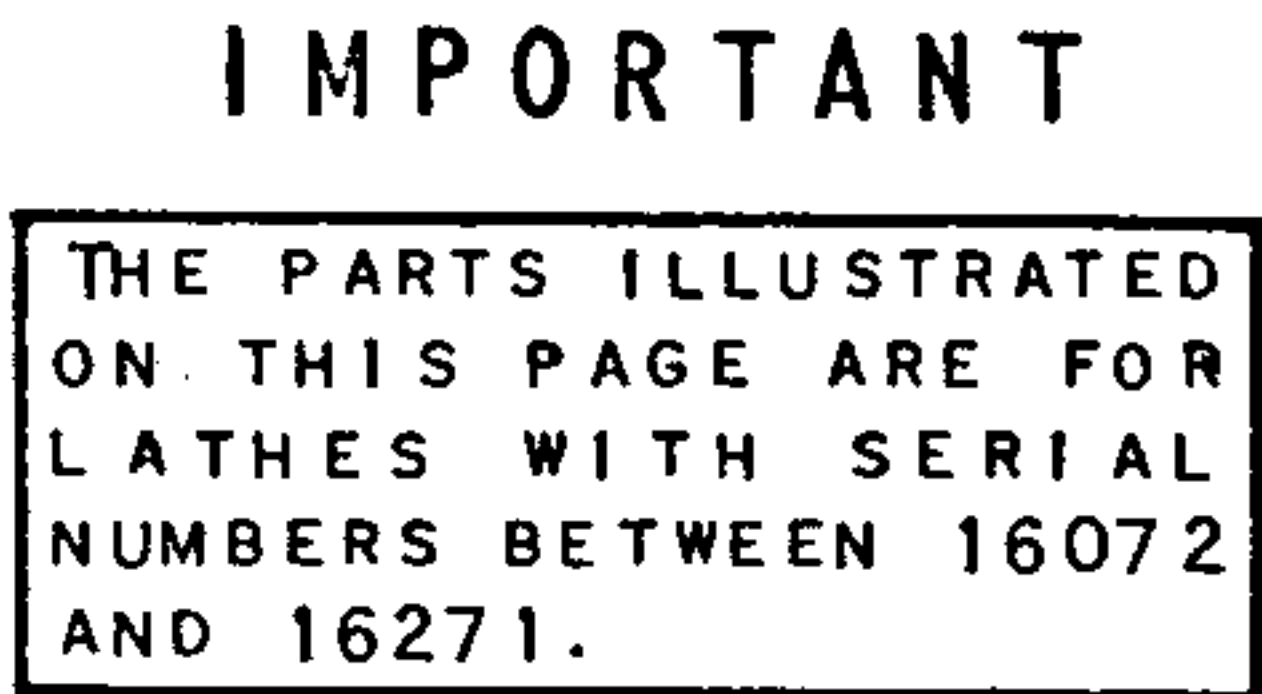
17-D For Horizontal Countershaft Lathes



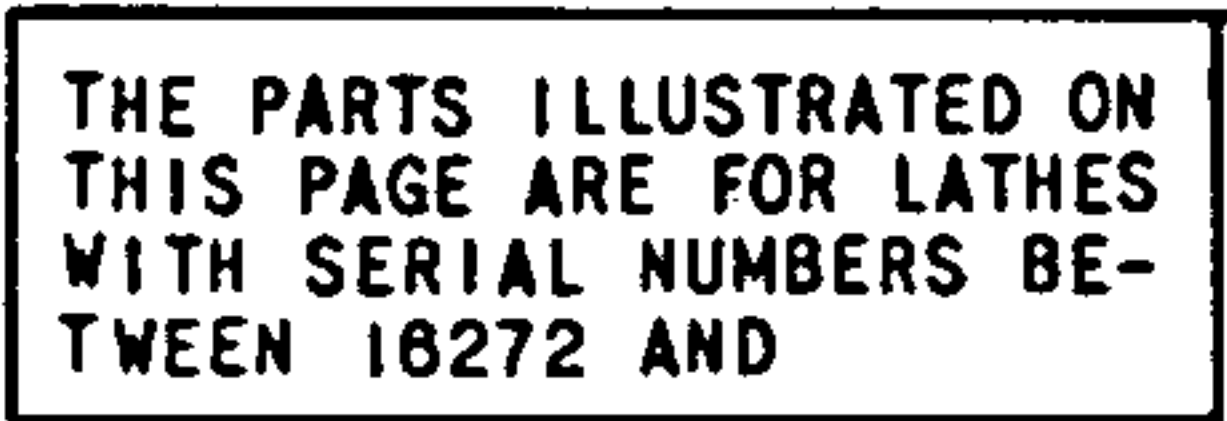


UNDER-DRIVE ASSEMBLY

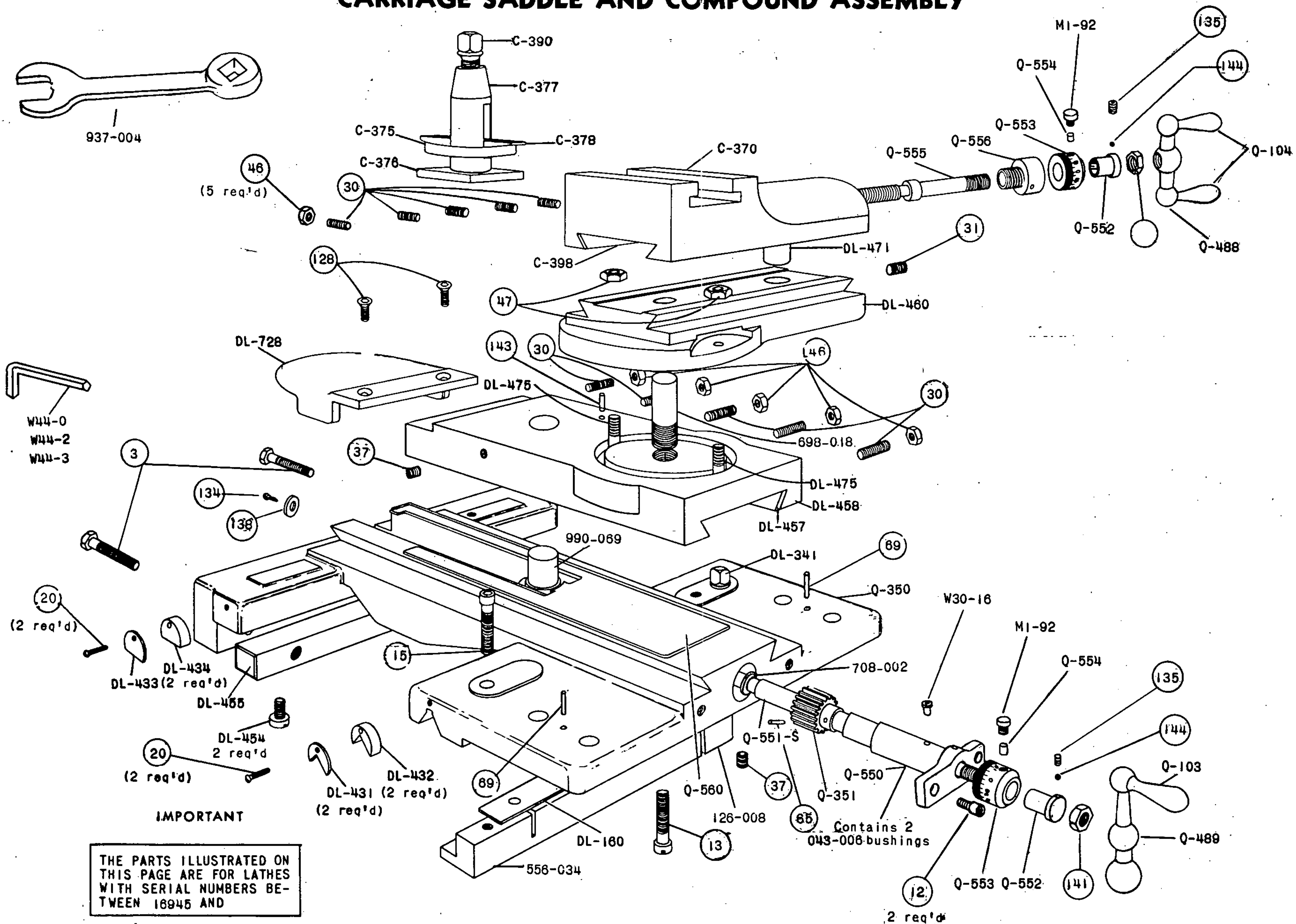
18-A



18-B



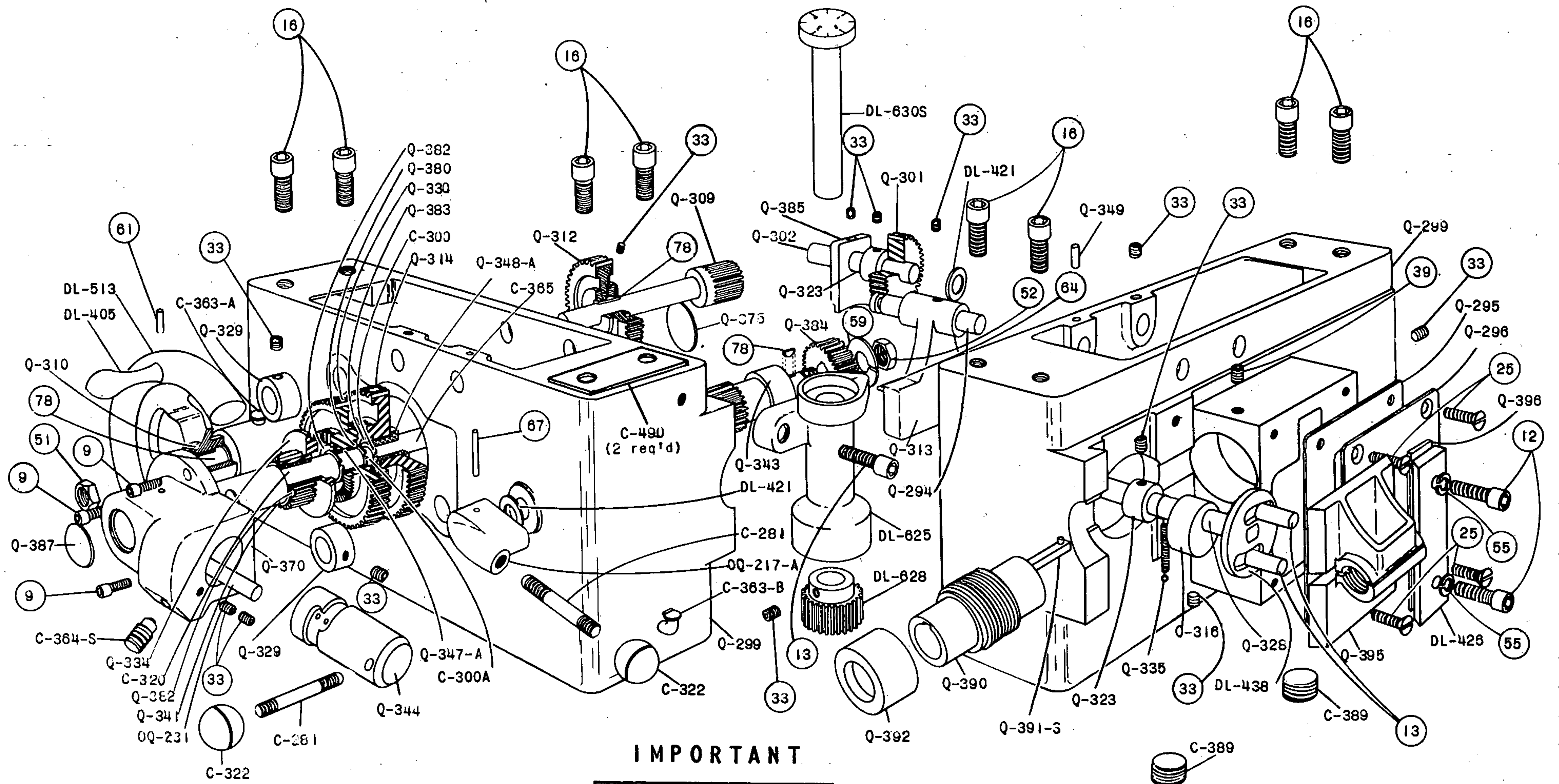
CARRIAGE SADDLE AND COMPOUND ASSEMBLY



IMPORTANT

THE PARTS ILLUSTRATED ON THIS PAGE ARE FOR LATHES WITH SERIAL NUMBERS BETWEEN 18945 AND

APRON ASSEMBLY



FRONT VIEW

IMPORTANT

THE PARTS ILLUSTRATED
ON THIS PAGE ARE FOR
LATHES WITH SERIAL
NUMBERS BETWEEN 15000
AND 15971.

NEAR VIEW

Q-387 Q-341 C-365 Q-382 as req'd Q-334 C-320 C-300-A Q-382 Q-347-A Q-380 C-300-A C-300 Q-383 as req'd Q-314 Q-348-A

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FRONT VIEW

REAR VIEW

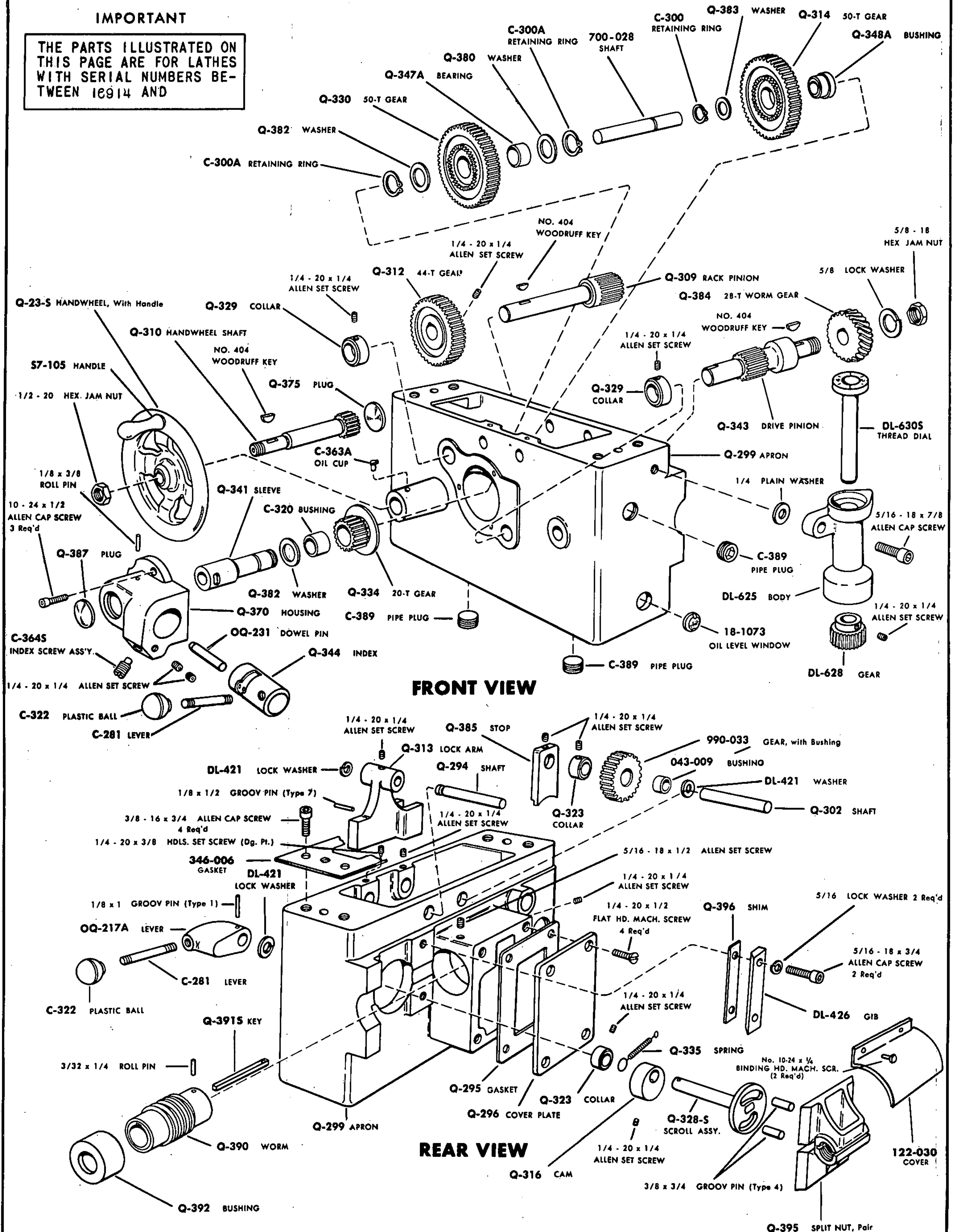
THE PARTS ILLUSTRATED ON
THIS PAGE ARE FOR LATHES
WITH SERIAL NUMBERS BE-
TWEEN 15972 AND

CARRIAGE APRON ASSEMBLY Q-299-S

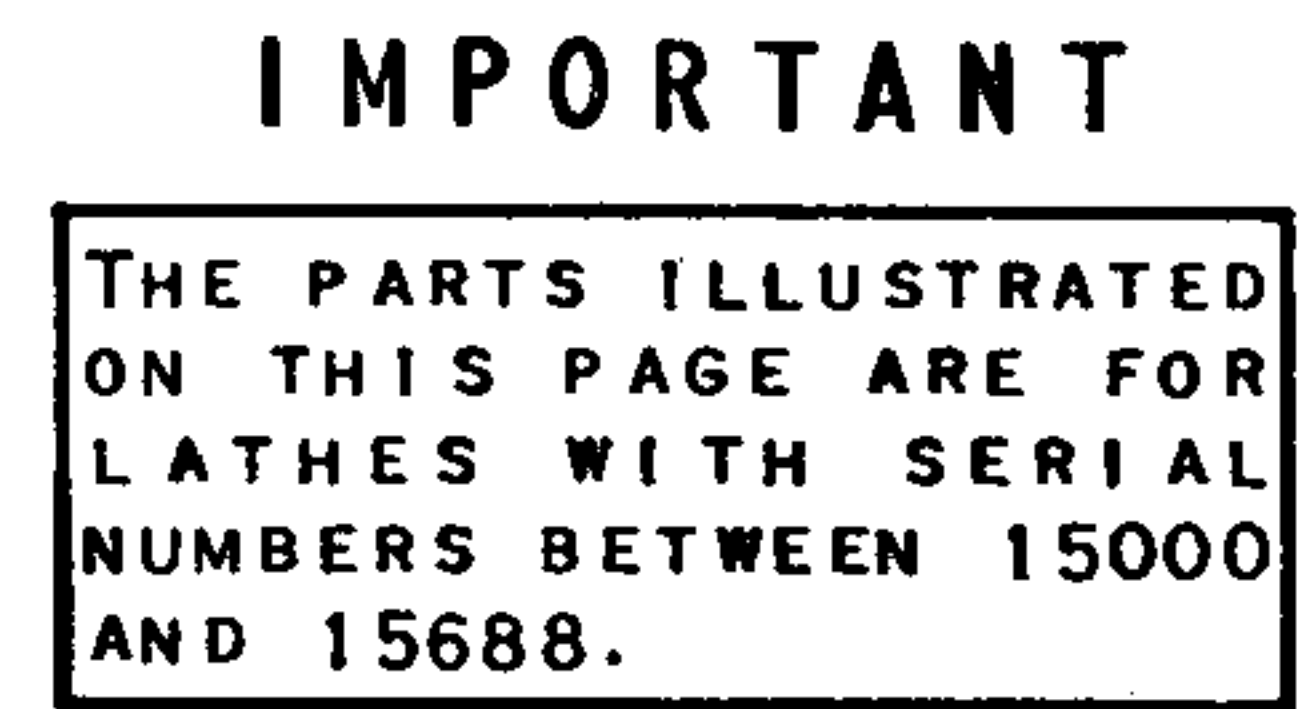
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IMPORTANT

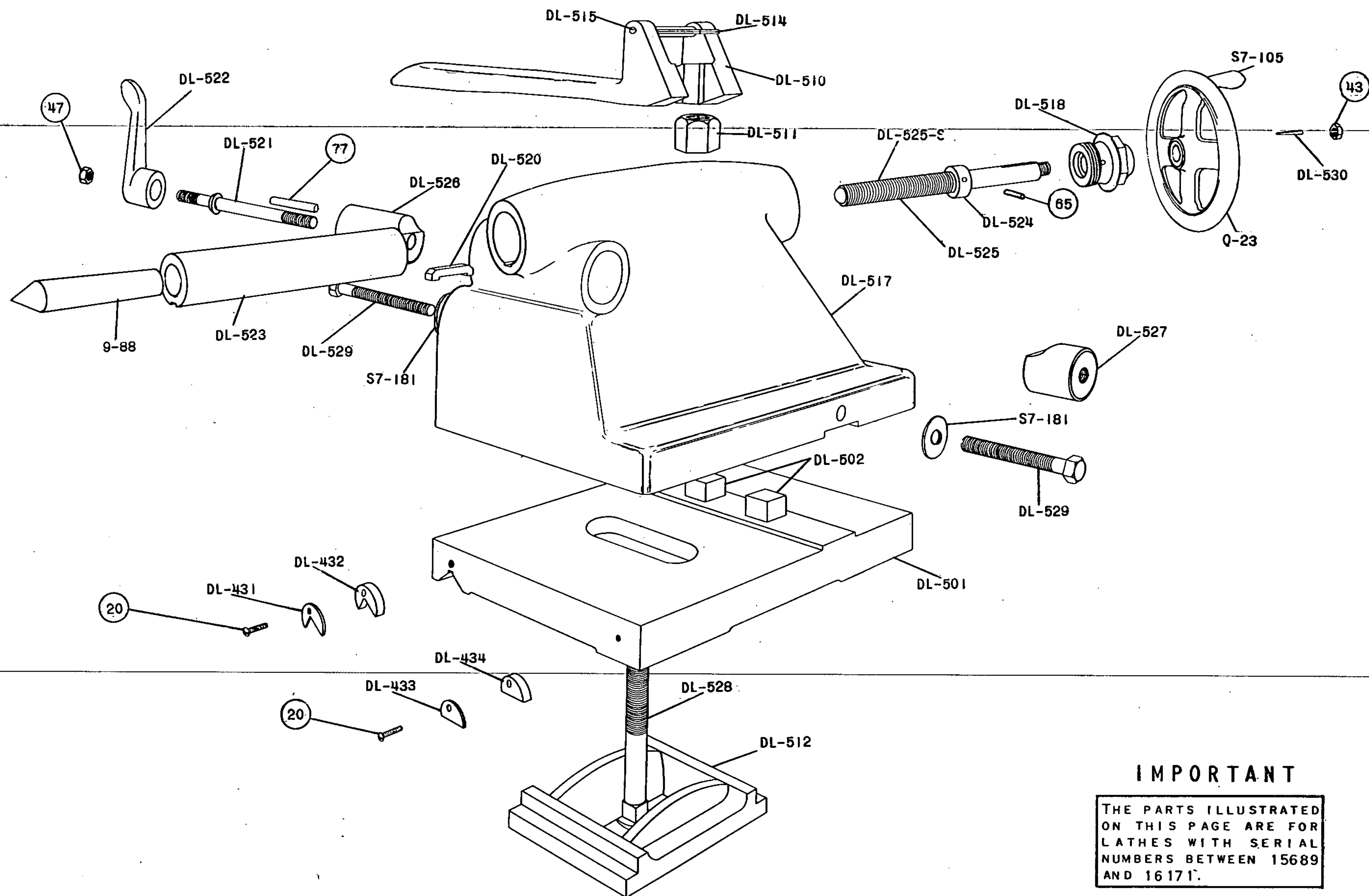
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20



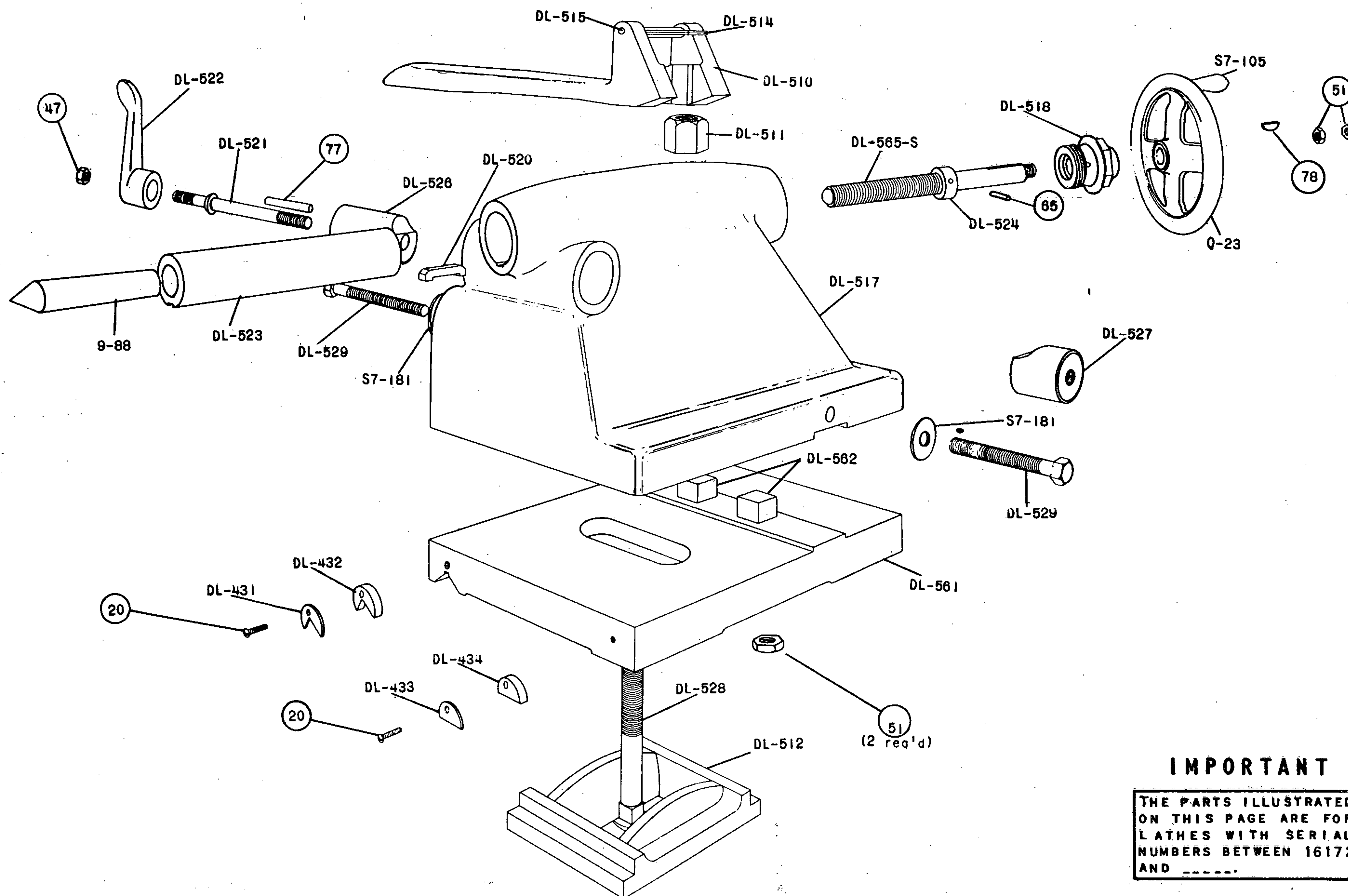
TAILSTOCK ASSEMBLY



IMPORTANT

THE PARTS ILLUSTRATED
ON THIS PAGE ARE FOR
LATHES WITH SERIAL
NUMBERS BETWEEN 15689
AND 16171.

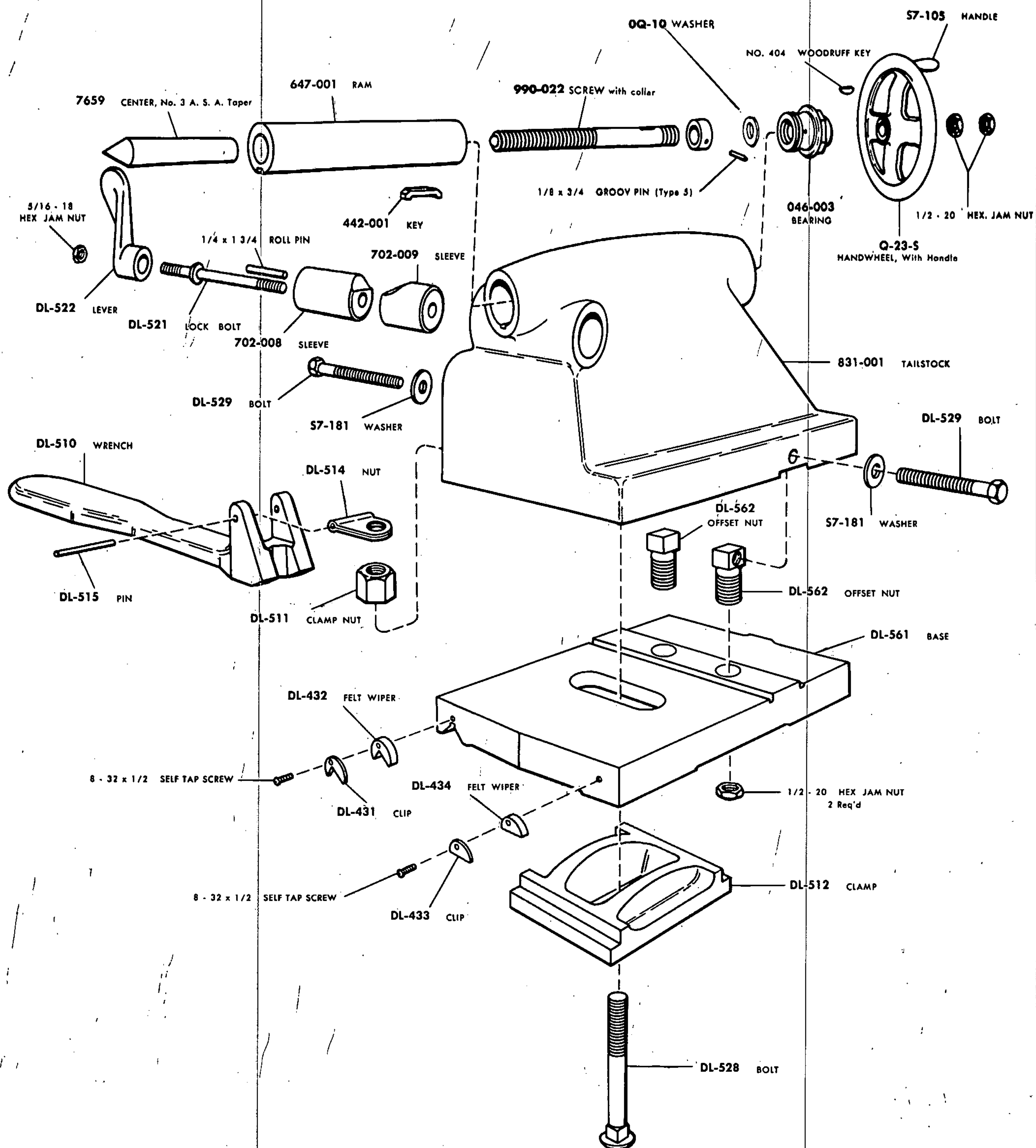
TAILSTOCK ASSEMBLY



IMPORTANT

THE PARTS ILLUSTRATED
ON THIS PAGE ARE FOR
LATHES WITH SERIAL
NUMBERS BETWEEN 16172
AND 16173.

TAILSTOCK ASSEMBLY 990-039



IMPORTANT

THE PARTS ILLUSTRATED ON THIS PAGE ARE FOR LATHES WITH SERIAL NUMBERS BETWEEN 16758 AND -----.

STANDARD PARTS

CODE NO.	DESCRIPTION	CODE NO.	DESCRIPTION
1	5/16"-18 x 3/4" Hex Cap Screw	43	5/16"-18 Hex Full Nut
2	5/16"-18 x 1 1/4" Hex Cap Screw	44	3/8"-16 Hex Full Nut
3	5/16"-18 x 1 1/2" Hex Cap Screw	45	3/8"-24 Hex Full Nut
4	3/8"-16 x 1 1/4" Hex Cap Screw	46	1/4"-20 Hex Jam Nut
5	3/8"-16 x 1 3/4" Hex Cap Screw	47	5/16"-18 Hex Jam Nut
6	1/2"-18 x 1 3/4" Hex Cap Screw	48	7/16"-14 Hex Jam Nut
7	1/2"-18 x 2" Hex Cap Screw	49	1/2"-18 x 2 1/2" Hex Cap Screw
8	No. 10-32 x 3/8" Socket Cap Screw	50	1/2"-18 Hex Jam Nut
9	No. 10-24 x 1/2" Socket Cap Screw	51	1/2"-20 Hex Jam Nut
10	1/4"-20 x 7/16" Socket Cap Screw	52	5/8"-18 Hex Jam Nut
11	1/4"-20 x 3/4" Socket Cap Screw	53	5/16"-18 Square Nut
12	5/16"-18 x 3/4" Socket Cap Screw	54	5/16" Plain Washer
13	5/16"-18 x 7/8" Socket Cap Screw	55	5/16" Lock Washer
14	5/16"-18 x 1" Socket Cap Screw	56	3/8" Plain Washer
15	5/16"-18 x 1 1/4" Socket Cap Screw	57	3/8" Lock Washer
16	3/8"-16 x 3/4" Socket Cap Screw	58	1/2" Plain Washer
17	3/8"-16 x 1" Socket Cap Screw	59	5/8" Lock Washer
18	3/8"-16 x 1 5/8" Socket Cap Screw	60	3/32" x 3/4" Cotter Pin
19	5/16"-18 x 7/8" Fillister Hd. Cap Screw	61	1/8" x 3/8" T-1 Groov Pin
20	No. 8-32 x 1/2" Self Tapping Screw	62	No 10 - 24 x 1/4" Rd. Hd. Mach. Screw
21	No. 10-24 x 1/2" Fillister Hd. Mach. Screw	63	1/8" x 1/2" T-4 Groov Pin
22	No. 6-32 x 1/4" Parker Kalon Self Tapping Scr	64	1/8" x 1/2" T-7 Groov Pin
23	1/8" x 3/8" Roll Pin	65	1/8" x 3/4" T-5 Groov Pin
24	No. 10-24 x 1/2" Flat Hd. Mach. Screw	66	No. 10 - 24 x 3/8" Rd. Hd. Mach. Screw
25	1/4"-20 x 1/2" Flat Hd. Mach. Screw	67	1/8" x 1" T-1 Groov Pin
26	3/16" x 1 3/8" T-1 Groov Pin	68	3/16" x 5/8" T-1 Groov Pin
27	5/16"-18 x 1 1/4" Carriage Bolt	69	1/4" x 3/4" T-2 Groov Pin
28	No. 10-24 x 3/16" Headless Set Screw	70	1/4" x 7/8" T-1 Groov Pin
29	1/4"-20 x 1/2" Headless Set Screw	71	1/4" x 1 1/4" Groov Pin T-1
30	1/4"-20 x 3/4" Headless Set Screw (cone pt.)	72	1/4" x 1 1/2" T-1 Groov Pin
31	5/16"-18 x 1/2" Headless Set Screw (cone pt.)	73	3/8" x 3/4" T-4 Groov Pin
32	3/8"-16 x 1 1/4" Sq. Hd. Set Screw	74	1/4 - 28 x 3/16 Socket Set Screw
33	1/4"-20 x 1/4" Socket Set Screw	75	No. 8-32 x 5/8 Rd. Hd. Mach. Screw
34	1/4"-20 x 5/16" Socket Set Screw	76	1/4" x 1 1/2" Roll Pin
35	1/4"-20 x 1 5/16" Socket Set Screw	77	1/4" x 1 3/4" Roll Pin
36	5/16"-18 x 1/4" Socket Set Screw	78	No. 404 Woodruff Key
37	5/16"-18 x 5/16" Socket Set Screw	79	No. 61 (3/16" x 5/8") Woodruff Key
38	5/16"-18 x 3/8" Socket Set Screw	80	3/8"-16 x 1/2" Hex Cap Screw
39	5/16"-18 x 1/2" Socket Set Screw	81	3/8"-16 x 2 1/4" Hex Cap Screw
40	3/8"-16 x 1/2" Socket Set Screw	82	7/16"-14 x 2" Hex Cap Screw
41	No. 2 x 3/16" Parker Kalon Drive Screw	83	1/2"-18 x 1 1/4" Hex Cap Screw
42	3/8"-16 x 1" Hex Cap Screw	84	3/8"-16 x 1 1/2" Socket Cap Screw
		85	No. 10-24 x 3/8" Self Tapping Screw

CODE NO.	DESCRIPTION	CODE NO.	DESCRIPTION
86	5/16"-18 x 1 3/4" Sq. Hd. Mach. Bolt	129	1/4"-20 x 5/16" Socket Set Screw
87	3/8"-16 x 3" Sq. Hd. Mach. Bolt	130	10-32 x 1/4" Socket Set Screw (flat pt.)
88	No. 404 Woodruff Key	131	1/4"-20 x 3/8" Headless Set Screw (dog pt.)
89	5/16"-18 x 1" Sq. Head Set Screw	132	3/8"-16 x 1 1/4" Socket Cap Screw
90	3/8"-16x1 1/2" Hdless Set Scr (Flat Pt)	133	1/8" x 1/4" T-1 Groov Pin
91	5/16"-18 x 2 1/4" Sq. Head Set Screw	134	No. 8-32 x 3/8" Self Tapping Screw
92	5/16"-18 x 3/4" Socket Set Screw	135	No. 8-32 x 3/16" Socket Set Screw
93	No. 6 x 5/16" Parker Kalon Drive Screw	136	No. 10 Plain Washer
94	1/2"-13 Hex Full Nut	137	3/16"-1 1/8" Roll Pin
95	5/8"-18 Hex Full Nut	138	No. 8 Plain Washer
96	3/8"-16 Hex Jam Nut	139	5/16"-18 x 1 1/4" Sq. Hd. Set Screw (dog pt.)
97	1/8" x 1/2" T-1 Groov Pin	140	5/16"-18 x 3/8" Socket Set Screw (flat pt.)
98	1/8" x 7/8" T-1 Groov Pin	141	3/8"-24 Hex Jam Nut
99	1/8" x 1 1/4" T-1 Groov Pin	142	1/4" Lock Washer
100	3/16" x 1/2" T-4 Groov Pin	143	1/8" x 13/16" T-1 Groov Pin
101	3/8"-16x3/8" Socket Set Scr (Cone Pt)	144	(.120 dia.) Lead Shot
102	1/4" x 5/8" T-4 Groov Pin	145	3/8"-16 x 1 3/4" Socket Cap Screw
103	5/16" x 1 3/4" T-1 Groov Pin	146	No. 6-32 x 3/16" Rd. Hd. Mach. Screw
104	3/32" x 1/4" Roll Pin	147	No. 8-32 x 1/2" Rd. Hd. Mach. Screw
105	3/32" x 5/16" Roll Pin	148	10-24 x 1/4" Binding Hd. Mach. Scr.
106	1/4" Oiler (Purchase C-363-A)	149	1/8" x 7/16" T-7 Groov Pin
107	5/16" Zerk Fitting	150	1/8" x 3/4" T-1 Groov Pin
108	(.180 dia.) Lead Shot	151	No. 4 x 3/16" Parker Kalon Drive Screw
109	3/8"-16 x 2 1/2" Hex Cap Screw	152	1/2"-20 Hex Full Nut
110	5/16"-18 x 7/16" Socket Set Screw	153	1/4 x 1/4 Roll Pin
111	3/16" x 1 1/4" T-1 Groov Pin	154	1/4"-28 x 1/4" Socket Set Screw
112	1/4"-20 x 3/8" Socket Set Screw	155	1/4"-20 x 5/8" Headless Set Screw
113	3/8"-16 x 1 3/4" Headless Set Screw (flat pt.)		
114	1/4" x 1 1/4" T-2 Groov Pin		
115	3/8"-16x3/8" Socket Set Scr (Flat Pt)		
116	1/4" Plain Washer		
117	3/32" x 3/8" T-1 Groov Pin		
118	3/8"-16 x 1 1/2" Hex Cap Screw		
119	10-24 x 9/16" Flat Hd. Mach. Screw		
120	10-24 Hex Jam Nut		
121	5/16"-18 x 1 1/4" Headless Set Screw		
122	10-32 x 3/8" Socket Set Screw (dog pt.)		
123	3/16" x 3/4" T-1 Groov Pin		
124	3/16" x 5/8" T-7 Groov Pin		
125	No. 606 Woodruff Key		
126	1/8" x 1/2" Roll Pin		
127	5/16"-18 x 3/8" Socket Set Screw		
128	1/4"-20 x 3/4" Socket Cap Screw (flat hd.)		